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(54) **MODIFIED STRAINS FOR IMPROVED
SECRETION OF RECOMBINANT PROTEINS**

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Related U.S. Application Data

(63) Continuation of application No. 18/178,275, filed on Mar. 3, 2023, now Pat. No. 12,286,640, which is a continuation of application No. 16/415,605, filed on May 17, 2019, now Pat. No. 11,634,729.

(57) **ABSTRACT**

Disclosed herein are modified strains for improving secretion of recombinantly expressed products secreted from a host organism with improved growth and productivity characteristics, as well as methods of using the modified strains.

Specification includes a Sequence Listing.

Complex Membership		<i>S. cerevisiae</i>	<i>P. pastoris</i>	% ID (blastp)
SEC (heptameric)	SEC61 (trimeric)	SSH1	PAS_chr1-4_0629	39%
		SSS1	PAS_chr1-1_0023	52%
		SBH2*	PAS_chr2-2_0210	57%
	SEC63 complex	SEC61	PAS_chr1-3_0202	70%
		SEC62	PAS_chr3_1014	35%
		SEC63	PAS_chr4_0395	38%
		SEC66	PAS_chr2-1_0433	41%
		SEC72	PAS_chr2-1_0448	29%

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SEC (heptameric)	SEC61 (trimeric)	SSH1	PAS_chr1-4_0629	39%
		SSS1	PAS_chr1-1_0023	52%
		SBH2*	PAS_chr2-2_0210	57%
	SEC63 complex	SEC61	PAS_chr1-3_0202	70%
		SEC62	PAS_chr3_1014	35%
		SEC63	PAS_chr4_0395	38%
		SEC66	PAS_chr2-1_0433	41%
		SEC72	PAS_chr2-1_0448	29%

Figure 1

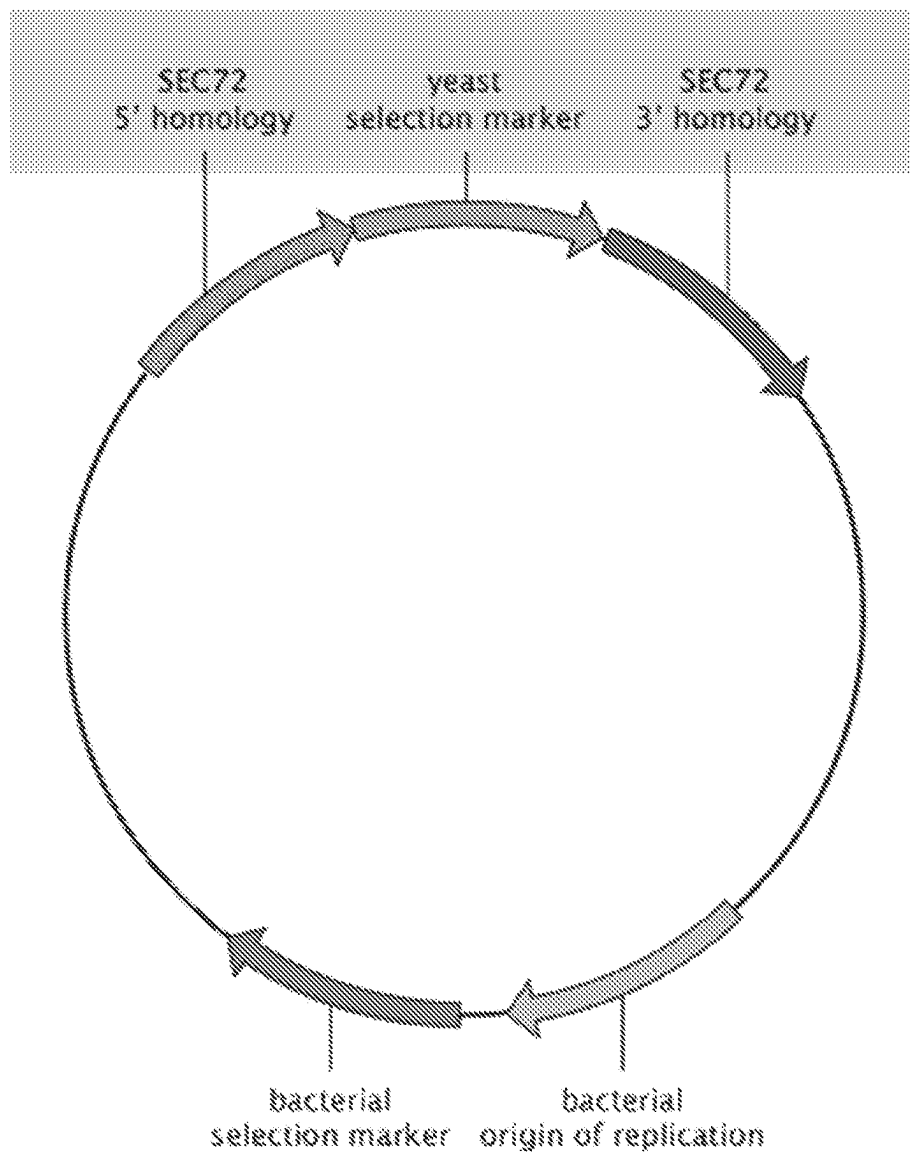


Figure 2

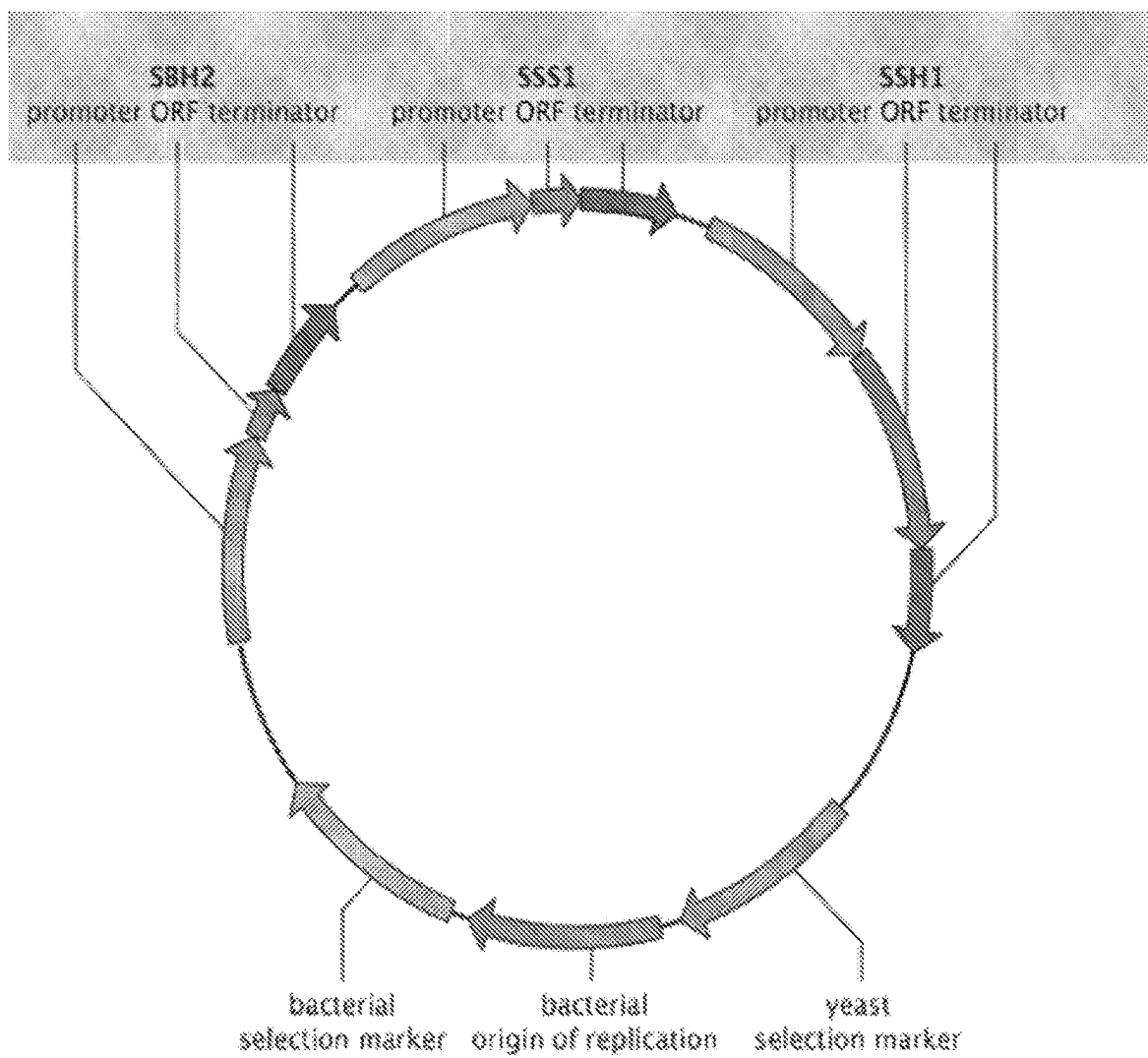


Figure 3

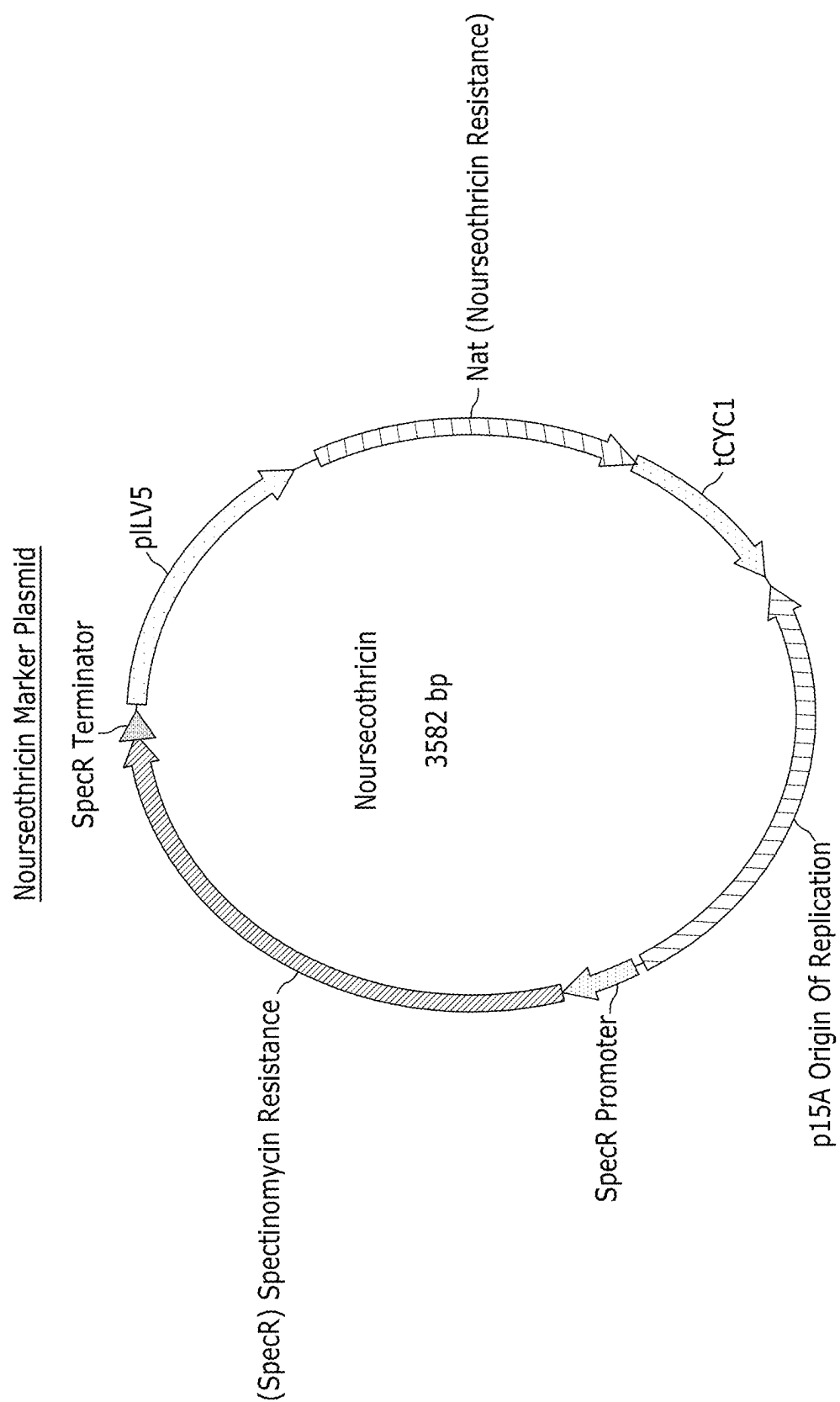


Figure 4A

Homology Arm Insertion into Nourseothricin Marker Plasmid



Figure 4B



Figure 4C

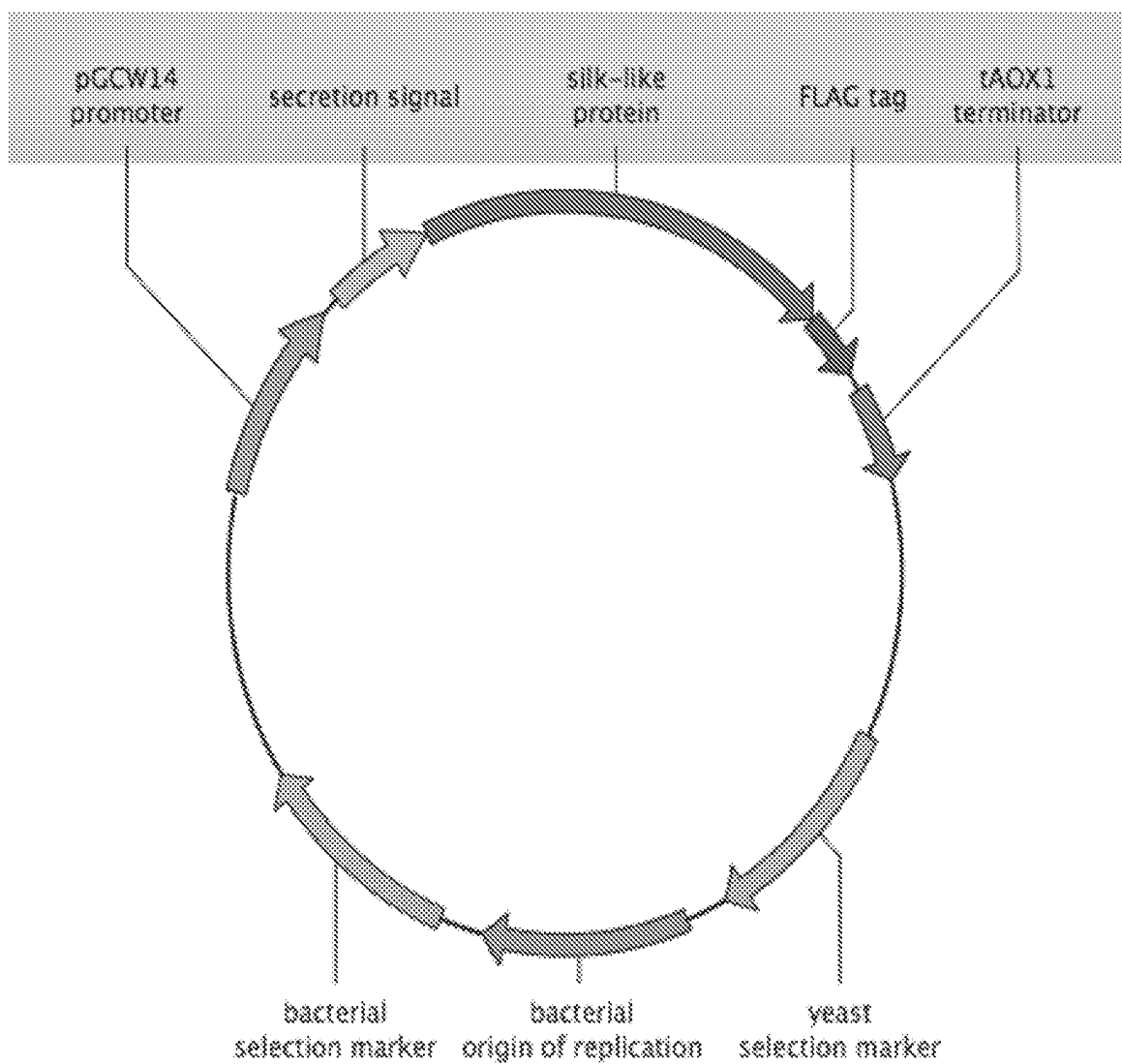


Figure 5

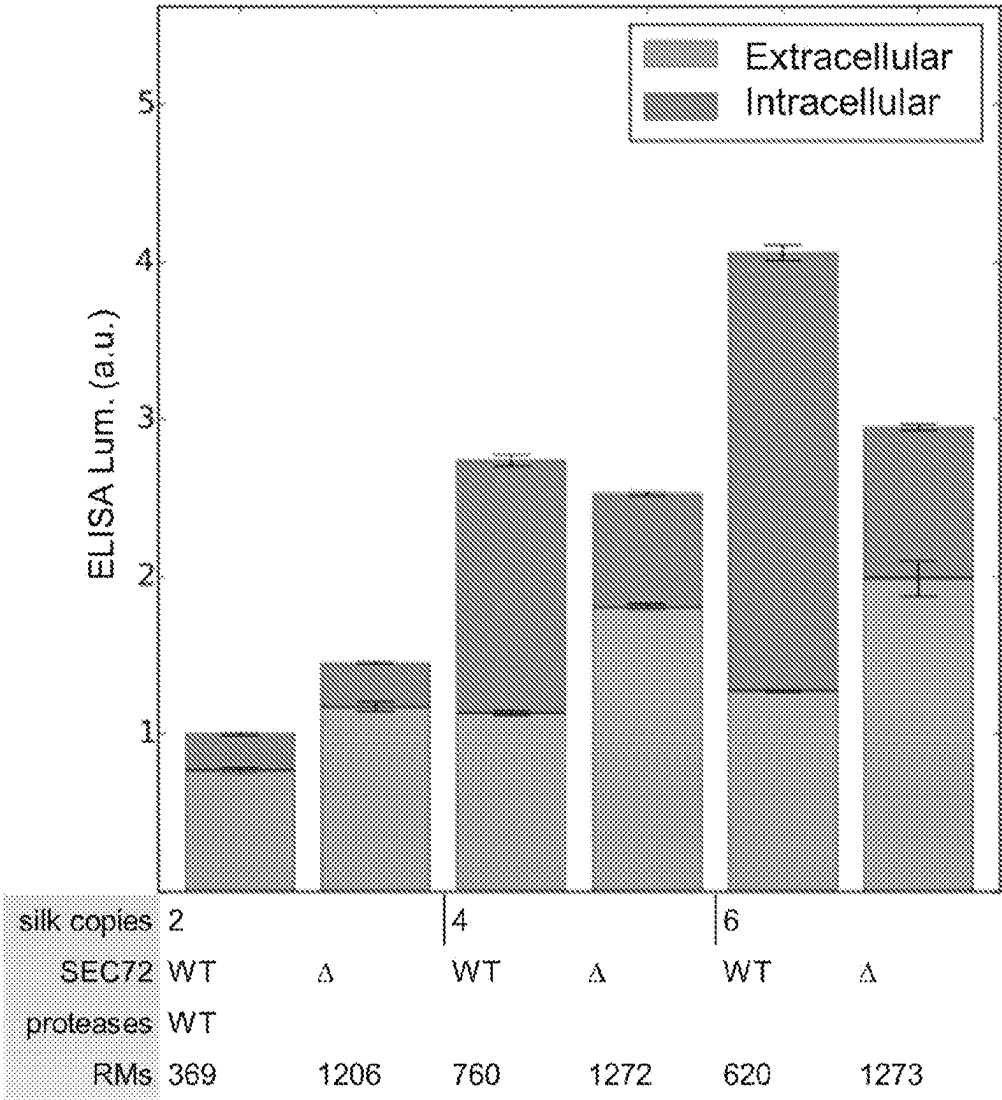


Figure 6

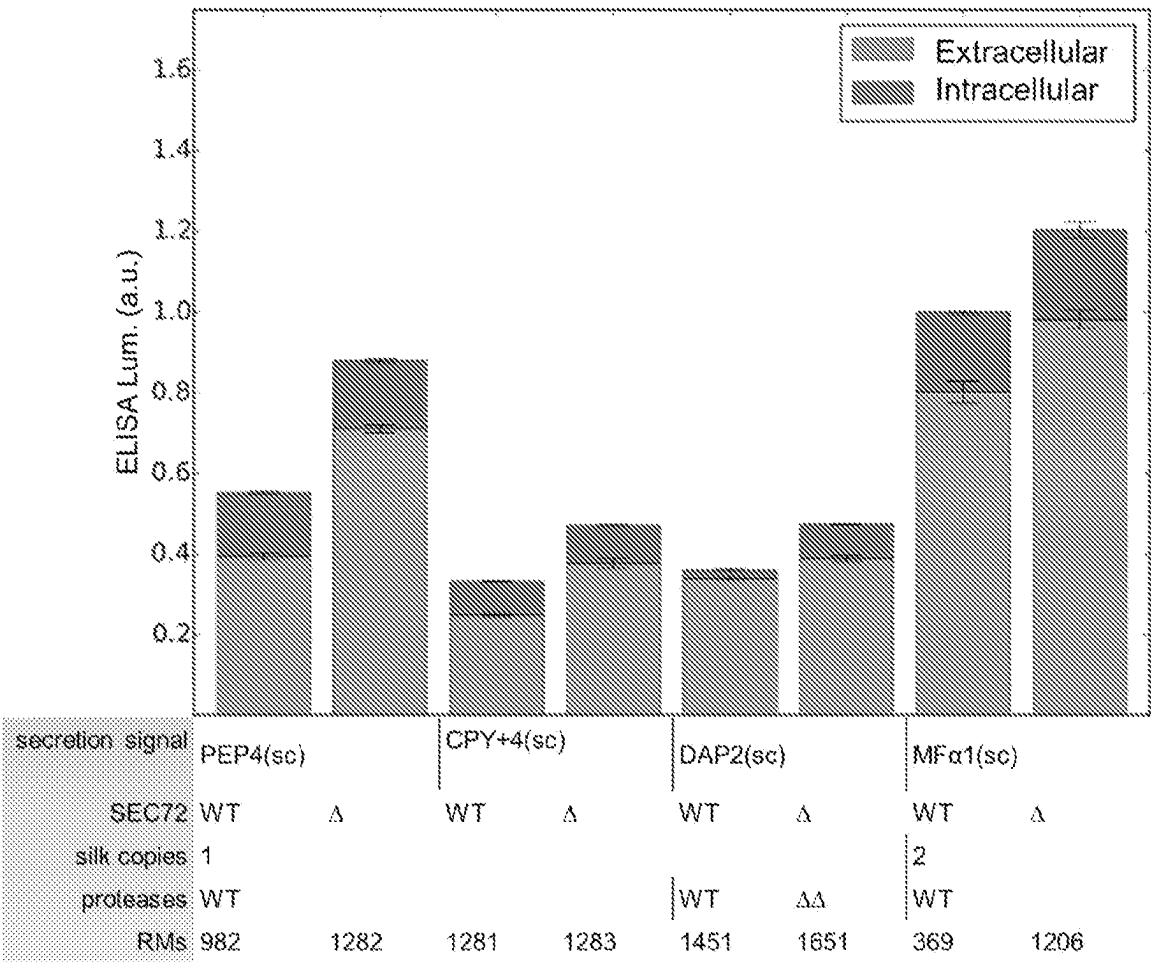


Figure 7

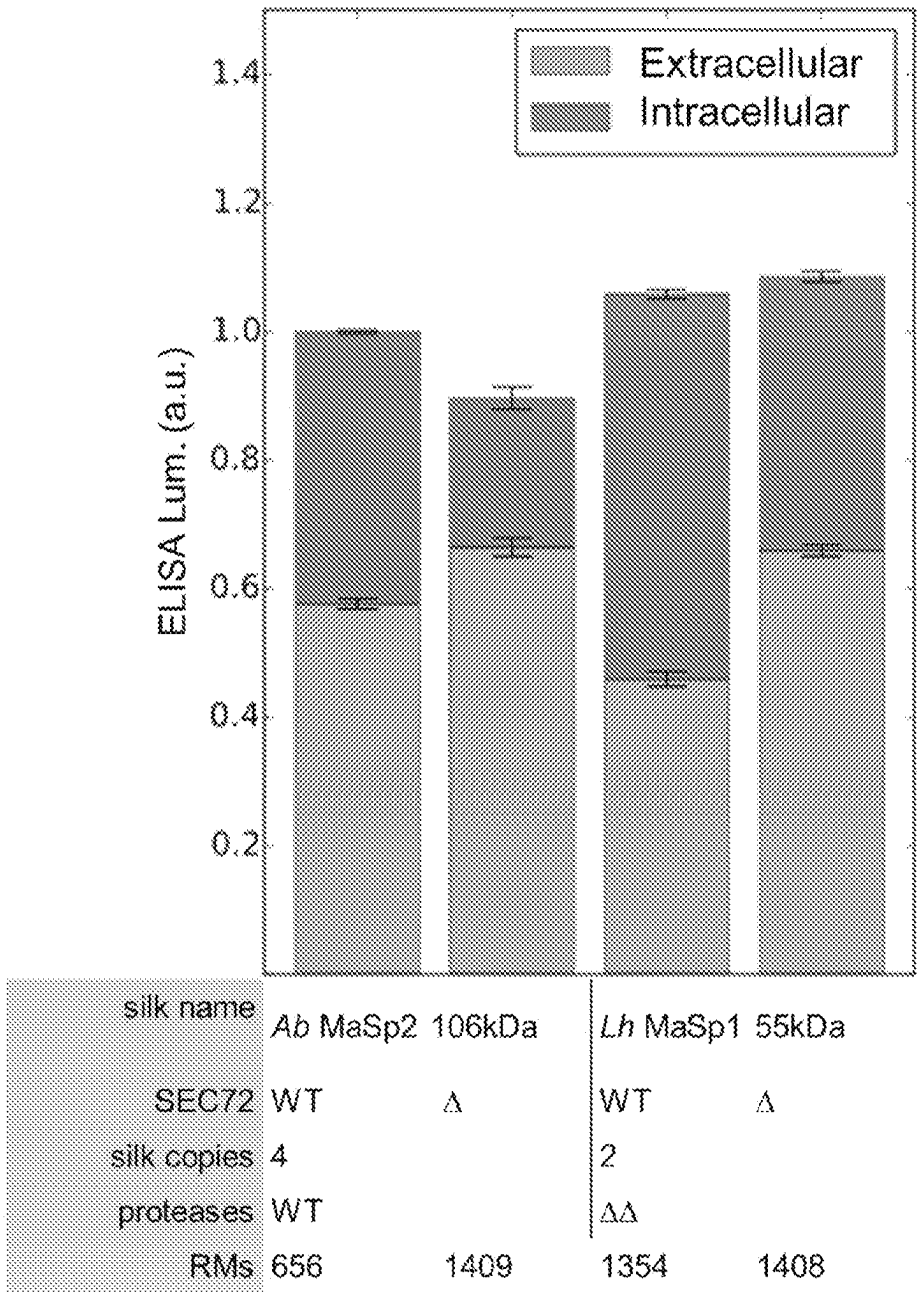


Figure 8

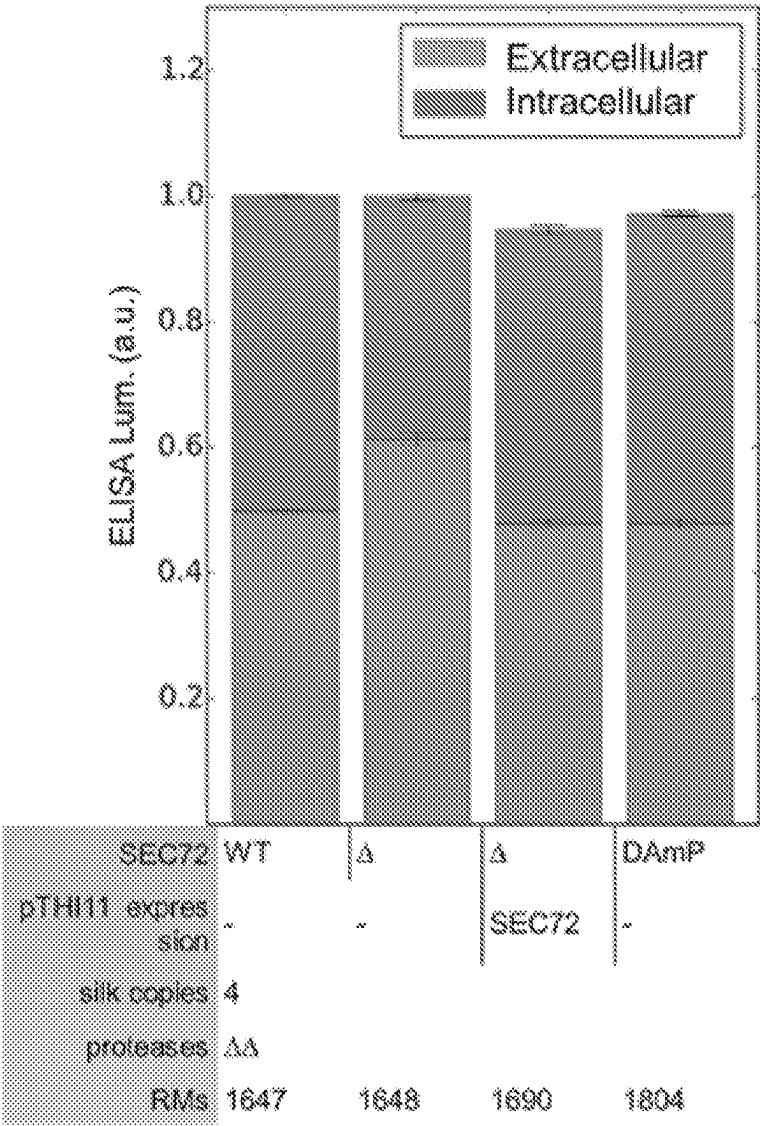


Figure 9A

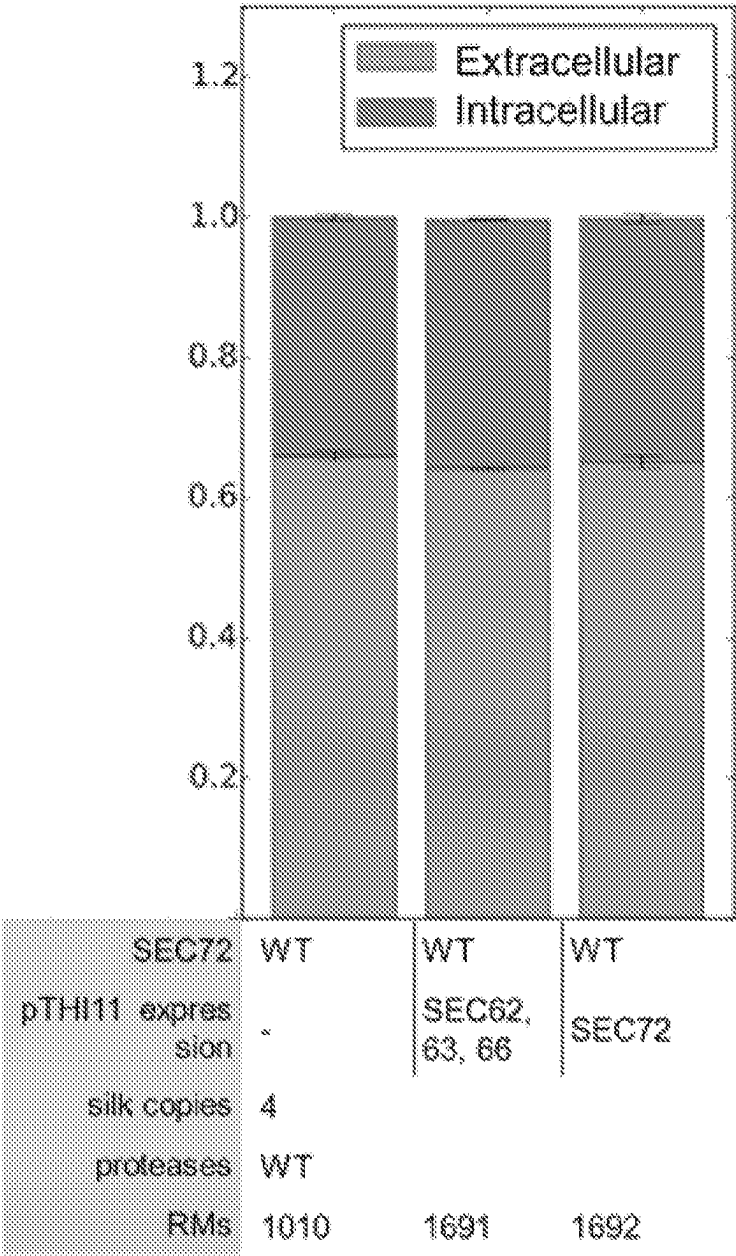


Figure 9B

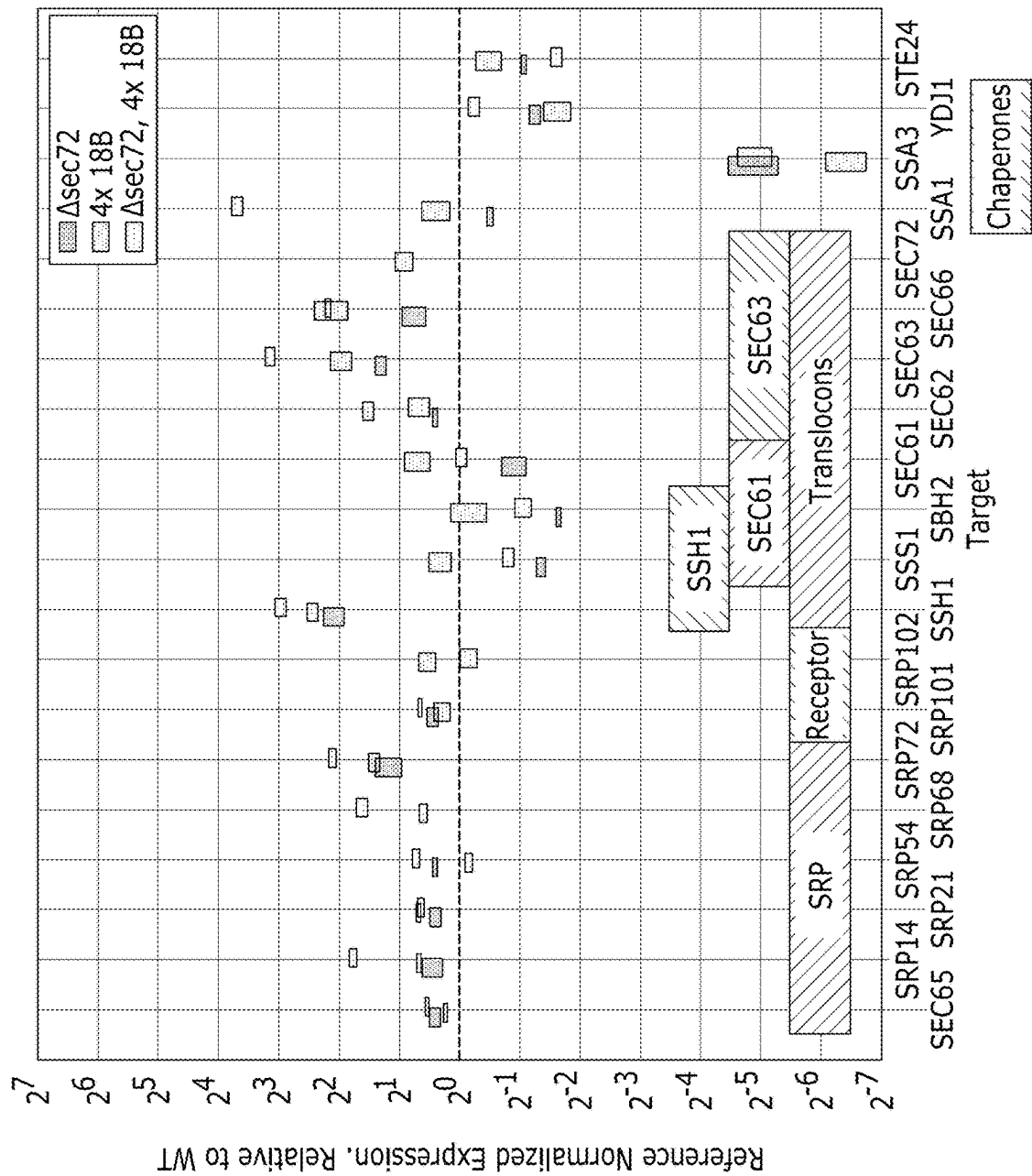


Figure 10A

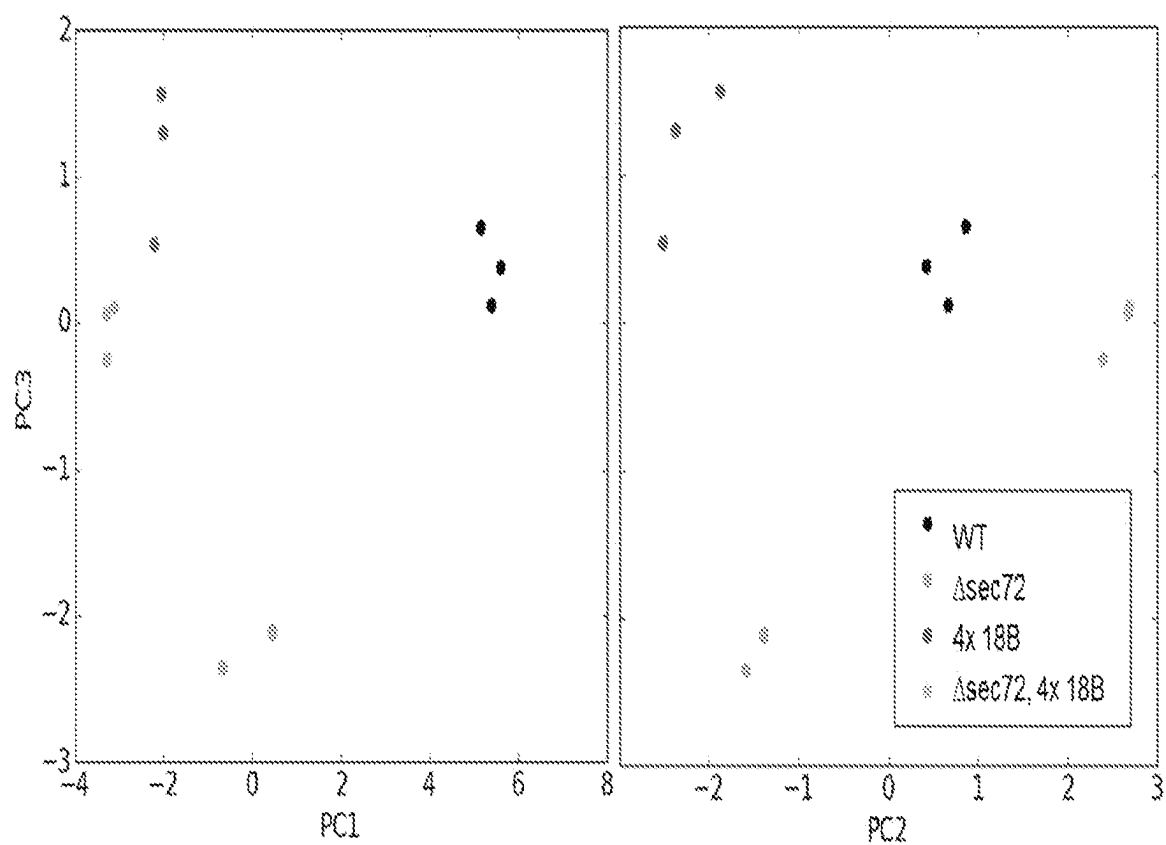


Figure 10B

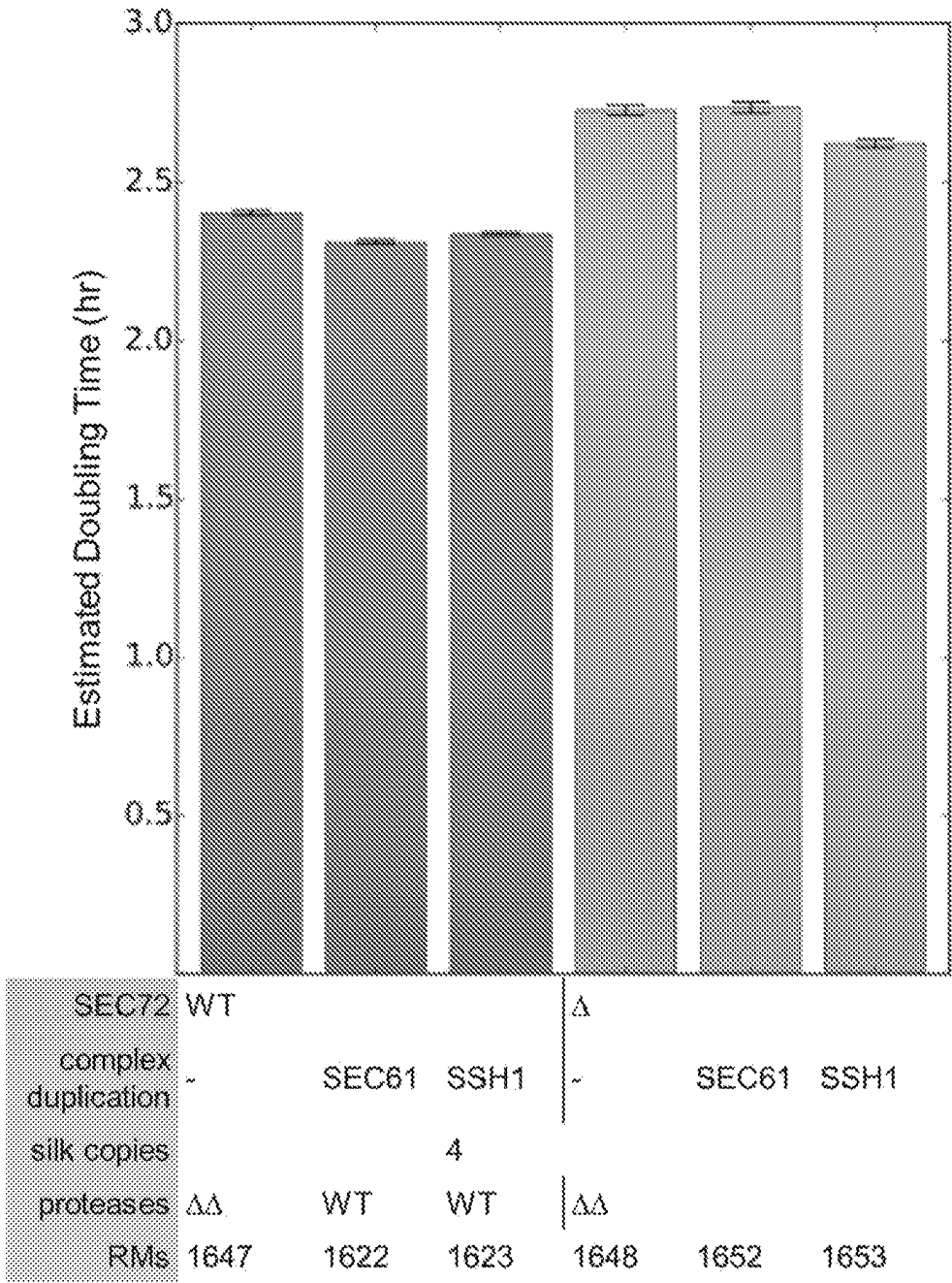


Figure 11

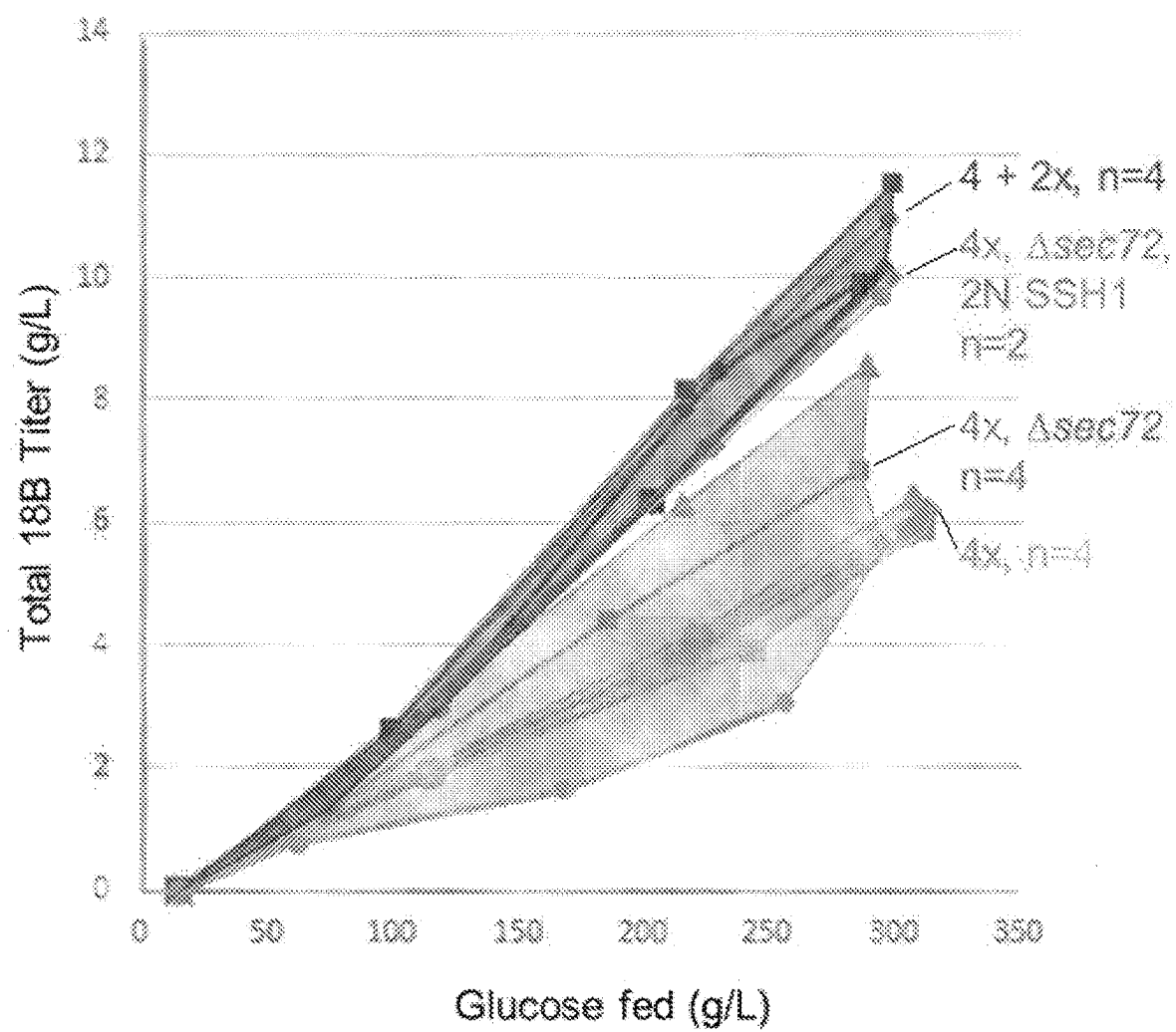


Figure 12A

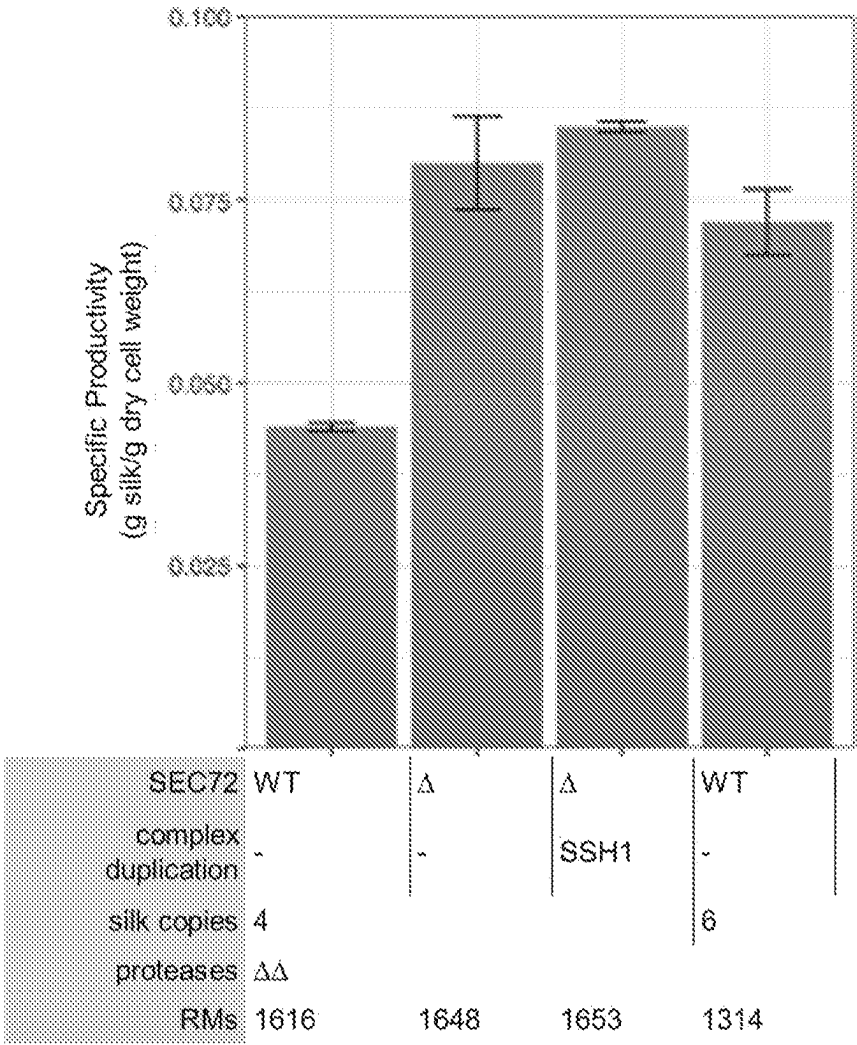


Figure 12B

MODIFIED STRAINS FOR IMPROVED SECRETION OF RECOMBINANT PROTEINS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/178,275, filed Mar. 3, 2023, which is a continuation of U.S. patent application Ser. No. 16/415,605, filed May 17, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/673,001, filed May 17, 2018, the contents of each of which are incorporated by reference in its entirety.

INCORPORATION BY REFERENCE OF MATERIAL SUBMITTED ELECTRONICALLY

[0002] A sequence listing, which is a part of the present disclosure, is submitted concurrently with the specification. The name of the file containing the sequence listing is "50019B_SubSeqListing.xml". The sequence listing was created on May 7, 2025, and is 315,392 bytes in size. The subject matter of the sequence listing is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

[0003] The present disclosure relates to methods of strain optimization to enhance secretion of proteins or metabolites from cells. The present disclosure also relates to compositions resulting from those methods. In particular, the disclosure relates to yeast cells selected or genetically engineered to enhance secretion of recombinant proteins expressed by the yeast cells, while minimizing or improving growth yields, and to methods of cultivating yeast cells for the production of useful compounds.

BACKGROUND OF THE INVENTION

[0004] The methylotrophic yeast *Pichia pastoris* is widely used in the production of recombinant proteins. *P. pastoris* grows to high cell density, provides tightly controlled methanol-inducible trans gene expression and is capable of secreting recombinant proteins into a defined media.

[0005] However, much of the expressed recombinant protein remains located intracellularly, which can make collection difficult and can negatively impact cell growth and longevity. Furthermore, recombinantly expressed proteins may be degraded in the cell before they are secreted, resulting in a mixture of proteins that includes fragments of recombinantly expressed proteins and a decreased yield of full-length recombinant proteins. What is needed, therefore, are tools and engineered strains to enhance secretion in *P. pastoris*, while mitigating any loss of or improving recombinant protein productivity, including maintaining growth characteristics, as many modifications can negatively impact the natural functioning of the cell.

[0006] What is needed, therefore, are modified organisms and methods of using these organisms to produce and secrete recombinant proteins while minimizing any negative effect on cell growth and productivity.

SUMMARY OF THE INVENTION

[0007] In some embodiments, provided herein is a *Pichia pastoris* microorganism, in which the activity of SEC72 has been eliminated or the sec72 gene has been deleted, and

wherein said microorganism expresses a recombinant protein. In some embodiments, the microorganism further compresses a recombinantly expressed SSH1 translocon complex.

[0008] In some embodiments, the SEC72 comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 1. In some embodiments, the SEC72 comprises SEQ ID NO: 1. In some embodiments, the SEC72 is encoded by a sec72 gene. In some embodiments, the sec72 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 2. In some embodiments, the sec72 gene comprises at least 15, 20, 25, 30, 40, or 50 contiguous nucleotides of SEQ ID NO: 2. In some embodiments, the sec72 gene comprises SEQ ID NO: 2. In some embodiments, the sec72 gene is at locus PAS_chr2-1_0448 of said microorganism.

[0009] In some embodiments, the SSH1 translocon complex comprises a first polypeptide sequence at least 95% identical to SEQ ID NO: 4, a second polypeptide sequence at least 95% identical to SEQ ID NO: 6, and a third polypeptide sequence at least 95% identical to SEQ ID NO: 8. In some embodiments, the SSH1 translocon complex comprises a first polypeptide comprising SEQ ID NO: 4, a second polypeptide comprising SEQ ID NO: 6, and a third polypeptide comprising SEQ ID NO: 8.

[0010] In some embodiments, the microorganism further comprises a recombinantly expressed translocon complex. In some embodiments, the translocon complex is expressed from a recombinant SSH1 gene, a recombinant SSS1 gene, and a recombinant SBH2 gene.

[0011] In some embodiments, the SSH1 gene comprises SEQ ID NO: 3. In some embodiments, the SSH1 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 3. In some embodiments, the SSH1 gene comprises at least 15, 20, 25, 30, 40, or 50 contiguous nucleotides of SEQ ID NO: 3.

[0012] In some embodiments, the SSS1 gene comprises SEQ ID NO: 5. In some embodiments, the SSS1 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 5. In some embodiments, the SSS1 gene comprises at least 15, 20, 25, 30, 40, or 50 contiguous nucleotides of SEQ ID NO: 5.

[0013] In some embodiments, the SBH2 gene comprises SEQ ID NO: 7. In some embodiments, the SBH2 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 7. In some embodiments, the SBH2 gene comprises at least 15, 20, 25, 30, 40, or 50 contiguous nucleotides of SEQ ID NO: 7.

[0014] In some embodiments, expression of said recombinant SSH1 gene to increase levels of said SSH1 translocon in said microorganism above that expressed by the native organism improves the growth rate and/or fermentation performance of said microorganism.

[0015] In some embodiments, the translocon complex comprises an SSH1 protein, an SSS1 protein, and an SBH2 protein. In some embodiments, the SSH1 protein comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 4. In some embodiments, the SSS1 protein comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 6. In some embodiments, the SBH2 protein comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 8. In some embodiments, the SBH2 protein comprises SEQ ID NO: 8.

[0016] In some embodiments, the activity of a YPS1-1 protease and a YPS1-2 protease in said microorganism has been attenuated or eliminated.

[0017] In some embodiments, the YPS1-1 protease comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 10. In some embodiments, the YPS1-1 protease comprises SEQ ID NO: 10. In some embodiments, the YPS1-1 protease is encoded by a YPS1-1 gene. In some embodiments, the YPS1-1 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 9. In some embodiments, the YPS1-1 gene comprises at least 15, 20, 25, 30, 40, or 50 contiguous nucleotides of SEQ ID NO: 9. In some embodiments, the YPS1-1 gene comprises SEQ ID NO: 9. In some embodiments, the YPS1-1 gene is at locus PAS_chr4_0584 of said microorganism.

[0018] In some embodiments, the YPS1-2 protease comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 12. In some embodiments, the YPS1-2 protease comprises SEQ ID NO: 12. In some embodiments, the YPS1-2 protease is encoded by a YPS1-2 gene. In some embodiments, the YPS1-2 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 11. In some embodiments, the YPS1-2 gene comprises at least 15, 20, 25, 30, 40, or 50 contiguous nucleotides of SEQ ID NO: 11. In some embodiments, the YPS1-2 gene comprises SEQ ID NO: 11. In some embodiments, the YPS1-2 gene is at locus PAS_chr3_1157 of said microorganism.

[0019] In some embodiments, the YPS1-1 gene or said YPS1-2 gene, or both, of said microorganism has been mutated or knocked out. In some embodiments, the activity of one or more additional proteases of said microorganism has been attenuated or eliminated.

[0020] In some embodiments, the recombinant protein expressed by said microorganism comprises at least one block polypeptide sequence from a silk protein. In some embodiments, the recombinant protein comprises a silk-like polypeptide. In some embodiments, the silk-like polypeptide comprises one or more repeat sequences $\{GGY-[GPG-X_1]_{n1}-GPS-(A)_{n2}\}_{n3}$, wherein $X_1=SGGQQ$ or $GAGQQ$ or $GQGPPY$ or $AGQQ$ or SQ ; $n1$ is from 4 to 8; $n2$ is from 6 to 20; and $n3$ is from 2 to 20. In some embodiments, the silk-like polypeptide comprises a polypeptide sequence encoded by SEQ ID NO: 21.

[0021] In some embodiments, the recombinant protein comprises a secretion signal peptide. In some embodiments, the secretion signal peptide is selected from the group consisting of: a PEP4 signal sequence, a CPY+4 signal sequence, a DAP2 signal sequence, and a MFa1 signal sequence. In some embodiments, the secretion signal peptide is selected from the group consisting of: SEQ ID NO: 84, SEQ ID NO: 86, SEQ ID NO: 88, and SEQ ID NO: 90.

[0022] Also provided herein, according to some embodiments, is a *Pichia pastoris* microorganism, wherein the activity of SEC72 of said microorganism has been eliminated or an sec72 gene comprising SEQ ID NO: 1 has been knocked out, wherein the microorganism comprises a recombinantly expressed SSH1 gene comprising SEQ ID NO: 3, a recombinantly expressed SSS1 gene comprising SEQ ID NO: 5, and a recombinantly expressed SBH2 gene comprising SEQ ID NO: 7, and wherein said microorganism comprises a silk-like polypeptide comprising a polypeptide sequence encoded by SEQ ID NO: 21.

[0023] Also provided herein, according to some embodiments, is a cell culture comprising a recombinant microorganism described herein.

[0024] Also provided, herein, according to some embodiments, is a cell culture comprising a recombinant microorganism as described herein, wherein said cell culture has an improved strain growth rate and fermentation performance under standard cell culture conditions as compared to a cell culture that does not comprise a recombinantly expressed SSH1 translocon complex.

[0025] Also provided, herein, according to some embodiments, is a cell culture comprising a recombinant microorganism as described herein, wherein said cell culture has an improved yield or specific productivity of said recombinant protein under standard cell culture conditions as compared to a cell culture of otherwise identical microorganisms that comprises a functional sec72 gene and does not comprise a recombinantly expressed SSH1 translocon complex, wherein each microorganism has the same number of copies of recombinant silk polypeptide genes.

[0026] Also provided herein, according to some embodiments, is a method of producing a recombinant protein, the method comprising: culturing the recombinant microorganism described herein in a culture medium under conditions suitable for expression of the recombinantly expressed protein; and isolating the recombinant protein from the microorganism or the culture medium.

[0027] In some embodiments, the recombinant protein is secreted from said microorganism, and isolating said recombinant protein comprises collecting a culture medium comprising said secreted recombinant protein.

[0028] In some embodiments, the microorganism has an increased yield or specific productivity of said recombinant protein as compared to an otherwise identical microorganism wherein said sec72 gene is not deleted.

[0029] In some embodiments, the microorganism has an increased yield or specific productivity of said recombinant protein as compared to an otherwise identical microorganism not comprising said recombinantly expressed SSH1 translocon complex, and wherein said sec72 gene is not deleted.

[0030] Also provided herein, according to some embodiments, is a method of modifying *Pichia pastoris* to improve the secretion of a recombinantly expressed protein, said method comprising knocking out a gene encoding an SEC72 protein. In some embodiments, the method of modifying *Pichia pastoris* to improve the secretion of a recombinantly expressed protein further comprises transforming said *Pichia pastoris* with a vector comprising genes encoding a recombinantly expressed SSH1 translocon complex.

[0031] In some embodiments, the recombinantly expressed protein comprises a silk-like polypeptide. In some embodiments, the silk-like polypeptide comprises one or more repeat sequences $\{GGY-[GPG-X_1]_{n1}-GPS-(A)_{n2}\}_{n3}$, wherein $X_1=SGGQQ$ or $GAGQQ$ or $GQGPPY$ or $AGQQ$ or SQ ; $n1$ is from 4 to 8; $n2$ is from 6 to 20; and $n3$ is from 2 to 20. In some embodiments, the recombinantly expressed protein comprises a polypeptide sequence encoded by SEQ ID NO: 21.

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] The foregoing and other objects, features and advantages will be apparent from the following description of particular embodiments of the invention, as illustrated in

the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead placed upon illustrating the principles of various embodiments of the invention.

[0033] FIG. 1 depicts groupings of genes in secretion complexes SEC, SEC61, SEC63, and SSH1. Also shown is a value of the homology between *S. cerevisiae* and *P. pastoris* translocon complex proteins.

[0034] FIG. 2 is a plasmid map of a vector for SEC72 deletion comprising a yeast selection marker flanked by SEC72 homology arms.

[0035] FIG. 3 is a plasmid map of a vector for overexpression of genes encoding proteins comprising the SSH1 complex. It includes a promoter, an open reading frame (ORF), and a terminator for SBH2, SSS1, and SSH1, as well as a yeast selection marker.

[0036] FIG. 4A is a plasmid map of a vector comprising a nourseothricin marker used with homology arms for targeted protease gene deletion. FIG. 4B and FIG. 4C are cassettes for protease knockout with homology arms targeting the desired protease gene flanking a nourseothricin resistance marker.

[0037] FIG. 5 is a plasmid map of a vector comprising an open reading frame (ORF) for expressing a silk-like polypeptide comprising a secretion signal and a FLAG tag. The ORF is operably linked to a pGCW14 promoter and a tAOX1 terminator. The vector also comprises selection markers to select successfully transformed cells.

[0038] FIG. 6 shows secretion of recombinant 18B silk-like polypeptide expressed by wild-type (WT) and Δ sec72 (Δ) strains expressing 2 $\times$, 4 $\times$, or 6 $\times$ copies of 18B, as measured by an enzyme-linked immunosorbent assay (i.e., ELISA).

[0039] FIG. 7 shows secretion of recombinant protein comprising PEP4 (sc), CPY4 (sc), DAP2 (sc), or Mfa1 (sc) signal peptides as expressed by wild-type (WT) and Δ sec72 (4) strains and measured by ELISA. (sc) indicates the signal peptide derives from *S. cerevisiae*.

[0040] FIG. 8 shows secretion of long silk polypeptides Argiope bruennichi Major ampullate spidron 2 (i.e., Ab MaSp2) (106 kDa), and *Latrodectus hesperus* Major ampullate spidron 1 (i.e., Lh MaSp1 (55 kDa)) as expressed by wild-type (WT) and Δ sec72 (Δ) strains and measured by ELISA.

[0041] FIG. 9A shows secretion of recombinant 18B silk-like polypeptide expressed by wild-type (WT), Δ sec72 (4) strains, strains overexpressing sec72 under the control of pTHI11, and strains where SEC72 was knocked down by DAmP (Decreased Abundance by mRNA Perturbation), as measured by ELISA. FIG. 9B shows secretion of recombinant 18B silk-like polypeptide expressed by wild-type (WT), strains overexpressing non-SEC72 proteins from the SEC63 complex (i.e., sec62, sec63 and sec66) and strains overexpressing sec72, as measured by ELISA.

[0042] FIG. 10A shows reference-normalized expression in Δ sec72 strains as a fold-change from the WT strain. Shading indicates range and center bar represents median among n=3 biological replicates (except n=2 for Δ sec72). FIG. 10B shows a projection of centered, log-scaled, reference normalized expression data onto principal component space, left=PC1 vs. 3, right=PC2 vs. 3. Replicates are colored by strain genotype, showing low within-genotype variance compared variance across genotypes.

[0043] FIG. 11 shows growth rates observed for WT and Δ sec72 strains, and comparison of each with growth rates for strains modified to overexpress the SSH1 translocon complex or SEC61.

[0044] FIG. 12A shows the production of 18B as a function of glucose (yield) for wild-type (WT) strains expressing 4 (4 $\times$) or 6 (4+2 $\times$) copies of 18B, as compared to a Δ sec72 strain (4 $\times$, Δ sec72) and a Δ sec72 strain overexpressing the SSH1 translocon complex (4 $\times$, Δ sec72, 2N SSH1). FIG. 12B shows the specific productivity (product-per biomass) of each strain.

DETAILED DESCRIPTION

[0045] The details of various embodiments of the invention are set forth in the description below. Other features, objects, and advantages of the invention will be apparent from the description and the drawings, and from the claims.

Definitions

[0046] Unless otherwise defined herein, scientific and technical terms used in connection with the present invention shall have the meanings that are commonly understood by those of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include the plural and plural terms shall include the singular. The terms “a” and “an” includes plural references unless the context dictates otherwise. Generally, nomenclatures used in connection with, and techniques of, biochemistry, enzymology, molecular and cellular biology, microbiology, genetics and protein and nucleic acid chemistry and hybridization described herein are those well-known and commonly used in the art.

[0047] The following terms, unless otherwise indicated, shall be understood to have the following meanings:

[0048] The term “polynucleotide” or “nucleic acid molecule” refers to a polymeric form of nucleotides of at least 10 bases in length. The term includes DNA molecules (e.g., cDNA or genomic or synthetic DNA) and RNA molecules (e.g., mRNA or synthetic RNA), as well as analogs of DNA or RNA containing non-natural nucleotide analogs, non-native internucleoside bonds, or both. The nucleic acid can be in any topological conformation. For instance, the nucleic acid can be single-stranded, double-stranded, triple-stranded, quadruplexed, partially double-stranded, branched, hairpinned, circular, or in a padlocked conformation.

[0049] Unless otherwise indicated, and as an example for all sequences described herein under the general format “SEQ ID NO:”, “nucleic acid comprising SEQ ID NO:1” refers to a nucleic acid, at least a portion of which has either (i) the sequence of SEQ ID NO:1, or (ii) a sequence complementary to SEQ ID NO:1. The choice between the two is dictated by the context. For instance, if the nucleic acid is used as a probe, the choice between the two is dictated by the requirement that the probe be complementary to the desired target.

[0050] The term “recombinant” refers to a biomolecule, e.g., a gene or protein, that (1) has been removed from its naturally occurring environment, (2) is not associated with all or a portion of a polynucleotide in which the gene is found in nature, (3) is operatively linked to a polynucleotide which it is not linked to in nature, or (4) does not occur in nature. The term “recombinant” can be used in reference to cloned DNA isolates, chemically synthesized polynucle-

otide analogs, or polynucleotide analogs that are biologically synthesized by heterologous systems, as well as proteins and/or mRNAs encoded by such nucleic acids.

[0051] An endogenous nucleic acid sequence in the genome of an organism (or the encoded protein product of that sequence) is deemed “recombinant” herein if a heterologous sequence is placed adjacent to the endogenous nucleic acid sequence, such that the expression of this endogenous nucleic acid sequence is altered. In this context, a heterologous sequence is a sequence that is not naturally adjacent to the endogenous nucleic acid sequence, whether or not the heterologous sequence is itself endogenous (originating from the same host cell or progeny thereof) or exogenous (originating from a different host cell or progeny thereof). By way of example, a promoter sequence can be substituted (e.g., by homologous recombination) for the native promoter of a gene in the genome of a host cell, such that this gene has an altered expression pattern. This gene would now become “recombinant” because it is separated from at least some of the sequences that naturally flank it.

[0052] A nucleic acid is also considered “recombinant” if it contains any modifications that do not naturally occur to the corresponding nucleic acid in a genome. For instance, an endogenous coding sequence is considered “recombinant” if it contains an insertion, deletion or a point mutation introduced artificially, e.g., by human intervention. A “recombinant nucleic acid” also includes a nucleic acid integrated into a host cell chromosome at a heterologous site and a nucleic acid construct present as an episome.

[0053] As used herein, the phrase “degenerate variant” of a reference nucleic acid sequence encompasses nucleic acid sequences that can be translated, according to the standard genetic code, to provide an amino acid sequence identical to that translated from the reference nucleic acid sequence. The term “degenerate oligonucleotide” or “degenerate primer” is used to signify an oligonucleotide capable of hybridizing with target nucleic acid sequences that are not necessarily identical in sequence but that are homologous to one another within one or more particular segments.

[0054] The term “percent sequence identity” or “identical” in the context of nucleic acid sequences refers to the residues in the two sequences which are the same when aligned for maximum correspondence. The length of sequence identity comparison may be over a stretch of at least about nine nucleotides, usually at least about 20 nucleotides, more usually at least about 24 nucleotides, typically at least about 28 nucleotides, more typically at least about 32 nucleotides, and preferably at least about 36 or more nucleotides. There are a number of different algorithms known in the art which can be used to measure nucleotide sequence identity. For instance, polynucleotide sequences can be compared using FASTA, Gap or Bestfit, which are programs in Wisconsin Package Version 10.0, Genetics Computer Group (GCG), Madison, Wis. FASTA provides alignments and percent sequence identity of the regions of the best overlap between the query and search sequences. Pearson, *Methods Enzymol.* 183:63-98 (1990) (hereby incorporated by reference in its entirety). For instance, percent sequence identity between nucleic acid sequences can be determined using FASTA with its default parameters (a word size of 6 and the NOPAM factor for the scoring matrix) or using Gap with its default parameters as provided in GCG Version 6.1, herein incorporated by reference. Alternatively, sequences can be compared using the computer program, BLAST (Altschul et al.,

J. Mol. Biol. 215:403-410 (1990); Gish and States, *Nature Genet.* 3:266-272 (1993); Madden et al., *Meth. Enzymol.* 266:131-141 (1996); Altschul et al., *Nucleic Acids Res.* 25:3389-3402 (1997); Zhang and Madden, *Genome Res.* 7:649-656 (1997)), especially blastp or tblastn (Altschul et al., *Nucleic Acids Res.* 25:3389-3402 (1997)).

[0055] The term “substantial homology” or “substantial similarity,” when referring to a nucleic acid or fragment thereof, indicates that, when optimally aligned with appropriate nucleotide insertions or deletions with another nucleic acid (or its complementary strand), there is nucleotide sequence identity in at least about 75%, 80%, 85%, preferably at least about 90%, and more preferably at least about 95%, 96%, 97%, 98% or 99% of the nucleotide bases, as measured by any well-known algorithm of sequence identity, such as FASTA, BLAST or Gap, as discussed above.

[0056] The nucleic acids (also referred to as polynucleotides) of this present invention may include both sense and antisense strands of RNA, cDNA, genomic DNA, and synthetic forms and mixed polymers of the above. They may be modified chemically or biochemically or may contain non-natural or derivatized nucleotide bases, as will be readily appreciated by those of skill in the art. Such modifications include, for example, labels, methylation, substitution of one or more of the naturally occurring nucleotides with an analog, internucleotide modifications such as uncharged linkages (e.g., methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, etc.), charged linkages (e.g., phosphorothioates, phosphorodithioates, etc.), pendent moieties (e.g., polypeptides), intercalators (e.g., acridine, psoralen, etc.), chelators, alkylators, and modified linkages (e.g., alpha anomeric nucleic acids, etc.) Also included are synthetic molecules that mimic polynucleotides in their ability to bind to a designated sequence via hydrogen bonding and other chemical interactions. Such molecules are known in the art and include, for example, those in which peptide linkages substitute for phosphate linkages in the backbone of the molecule. Other modifications can include, for example, analogs in which the ribose ring contains a bridging moiety or other structure such as the modifications found in “locked” nucleic acids.

[0057] The term “mutated” when applied to nucleic acid sequences means that nucleotides in a nucleic acid sequence may be inserted, deleted or changed compared to a reference nucleic acid sequence. A single alteration may be made at a locus (a point mutation) or multiple nucleotides may be inserted, deleted or changed at a single locus. In addition, one or more alterations may be made at any number of loci within a nucleic acid sequence. A nucleic acid sequence may be mutated by any method known in the art including but not limited to mutagenesis techniques such as “error-prone PCR” (a process for performing PCR under conditions where the copying fidelity of the DNA polymerase is low, such that a high rate of point mutations is obtained along the entire length of the PCR product; see, e.g., Leung et al., *Technique*, 1:11-15 (1989) and Caldwell and Joyce, *PCR Methods Applic.* 2:28-33 (1992)); and “oligonucleotide-directed mutagenesis” (a process which enables the generation of site-specific mutations in one cloned DNA segment of interest; see, e.g., Reidhaar-Olson and Sauer, *Science* 241:53-57 (1988)).

[0058] The term “attenuate” as used herein generally refers to a functional deletion, including a mutation, partial or complete deletion, insertion, or other variation made to a

gene sequence or a sequence controlling the transcription of a gene sequence, which reduces or inhibits production of the gene product, or renders the gene product non-functional. In some instances a functional deletion is described as a knockout mutation. Attenuation also includes amino acid sequence changes by altering the nucleic acid sequence, placing the gene under the control of a less active promoter, down-regulation, expressing interfering RNA, ribozymes or antisense sequences that target the gene of interest, or through any other technique known in the art. In one example, the sensitivity of a particular enzyme to feedback inhibition or inhibition caused by a composition that is not a product or a reactant (non-pathway specific feedback) is lessened such that the enzyme activity is not impacted by the presence of a compound. In other instances, an enzyme that has been altered to be less active can be referred to as attenuated.

[0059] The term “deletion” as used herein refers to the removal of one or more nucleotides from a nucleic acid molecule or one or more amino acids from a protein, the regions on either side being joined together.

[0060] The term “knock-out” as used herein is intended to refer to a gene whose level of expression or activity has been reduced to zero. In some examples, a gene is knocked-out via deletion of some or all of its coding sequence. In other examples, a gene is knocked-out via introduction of one or more nucleotides into its open reading frame, which results in translation of a non-sense or otherwise non-functional protein product.

[0061] The term “vector” as used herein is intended to refer to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a “plasmid,” which generally refers to a circular double stranded DNA loop into which additional DNA segments may be ligated, but also includes linear double-stranded molecules such as those resulting from amplification by the polymerase chain reaction (PCR) or from treatment of a circular plasmid with a restriction enzyme. Other vectors include cosmids, bacterial artificial chromosomes (BAC) and yeast artificial chromosomes (YAC). Another type of vector is a viral vector, wherein additional DNA segments may be ligated into the viral genome (discussed in more detail below). Certain vectors are capable of autonomous replication in a host cell into which they are introduced (e.g., vectors having an origin of replication which functions in the host cell). Other vectors can be integrated into the genome of a host cell upon introduction into the host cell, and are thereby replicated along with the host genome. Moreover, certain preferred vectors are capable of directing the expression of genes to which they are operatively linked. Such vectors are referred to herein as “recombinant expression vectors” (or simply “expression vectors”).

[0062] “Operatively linked” or “operably linked” expression control sequences refers to a linkage in which the expression control sequence is contiguous with the gene of interest to control the gene of interest, as well as expression control sequences that act in trans or at a distance to control the gene of interest.

[0063] The term “expression control sequence” refers to polynucleotide sequences which are necessary to affect the expression of coding sequences to which they are operatively linked. Expression control sequences are sequences which control the transcription, post-transcriptional events and translation of nucleic acid sequences. Expression con-

trol sequences include appropriate transcription initiation, termination, promoter and enhancer sequences; efficient RNA processing signals such as splicing and polyadenylation signals; sequences that stabilize cytoplasmic mRNA; sequences that enhance translation efficiency (e.g., ribosome binding sites); sequences that enhance protein stability; and when desired, sequences that enhance protein secretion. The nature of such control sequences differs depending upon the host organism; in prokaryotes, such control sequences generally include promoter, ribosomal binding site, and transcription termination sequence. The term “control sequences” is intended to include, at a minimum, all components whose presence is essential for expression, and can also include additional components whose presence is advantageous, for example, leader sequences and fusion partner sequences.

[0064] The term “regulatory element” refers to any element which affects transcription or translation of a nucleic acid molecule. These include, by way of example but not limitation: regulatory proteins (e.g., transcription factors), chaperones, signaling proteins, RNAi molecules, antisense RNA molecules, microRNAs and RNA aptamers. Regulatory elements may be endogenous to the host organism. Regulatory elements may also be exogenous to the host organism. Regulatory elements may be synthetically generated regulatory elements.

[0065] The term “promoter,” “promoter element,” or “promoter sequence” as used herein, refers to a DNA sequence which when ligated to a nucleotide sequence of interest is capable of controlling the transcription of the nucleotide sequence of interest into mRNA. A promoter is typically, though not necessarily, located 5' (i.e., upstream) of a nucleotide sequence of interest whose transcription into mRNA it controls, and provides a site for specific binding by RNA polymerase and other transcription factors for initiation of transcription. Promoters may be endogenous to the host organism. Promoters may also be exogenous to the host organism. Promoters may be synthetically generated regulatory elements.

[0066] Promoters useful for expressing the recombinant genes described herein include both constitutive and inducible/repressible promoters. Where multiple recombinant genes are expressed in an engineered organism of the invention, the different genes can be controlled by different promoters or by identical promoters in separate operons, or the expression of two or more genes may be controlled by a single promoter as part of an operon.

[0067] The term “recombinant host cell” (or simply “host cell”), as used herein, is intended to refer to a cell into which a recombinant vector has been introduced. It should be understood that such terms are intended to refer not only to the particular subject cell but to the progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term “host cell” as used herein. A recombinant host cell may be an isolated cell or cell line grown in culture or may be a cell which resides in a living tissue or organism.

[0068] The term “wild-type” (i.e., “WT”) as used herein refers to a comparative strain lacking the modification being discussed. It does not refer to the native, unmodified strain, but rather to a strain lacking a selected modification. For example, when comparing two versions of a recombinant

strain modified to express a recombinant silk polypeptide, where one is a sec72 KO, the KO strain may be referred to as the Δ sec72 strain, while the other version is referred to as “WT.”

[0069] The term “peptide” as used herein refers to a short polypeptide, e.g., one that is typically less than about 50 amino acids long and more typically less than about 30 amino acids long. The term as used herein encompasses analogs and mimetics that mimic structural and thus biological function.

[0070] The term “polypeptide” encompasses both naturally-occurring and non-naturally-occurring proteins, and fragments, mutants, derivatives and analogs thereof. A polypeptide may be monomeric or polymeric. Further, a polypeptide may comprise a number of different domains each of which has one or more distinct activities.

[0071] The term “isolated protein” or “isolated polypeptide” is a protein or polypeptide that by virtue of its origin or source of derivation (1) is not associated with naturally associated components that accompany it in its native state, (2) exists in a purity not found in nature, where purity can be adjudged with respect to the presence of other cellular material (e.g., is free of other proteins from the same species) (3) is expressed by a cell from a different species, or (4) does not occur in nature (e.g., it is a fragment of a polypeptide found in nature or it includes amino acid analogs or derivatives not found in nature or linkages other than standard peptide bonds). Thus, a polypeptide that is chemically synthesized or synthesized in a cellular system different from the cell from which it naturally originates will be “isolated” from its naturally associated components. A polypeptide or protein may also be rendered substantially free of naturally associated components by isolation, using protein purification techniques well known in the art. As thus defined, “isolated” does not necessarily require that the protein, polypeptide, peptide or oligopeptide so described has been physically removed from its native environment.

[0072] The term “polypeptide fragment” refers to a polypeptide that has a deletion, e.g., an amino-terminal and/or carboxy-terminal deletion compared to a full-length polypeptide. In a preferred embodiment, the polypeptide fragment is a contiguous sequence in which the amino acid sequence of the fragment is identical to the corresponding positions in the naturally-occurring sequence. Fragments typically are at least 5, 6, 7, 8, 9 or 10 amino acids long, preferably at least 12, 14, 16 or 18 amino acids long, more preferably at least 20 amino acids long, more preferably at least 25, 30, 35, 40 or 45, amino acids, even more preferably at least 50 or 60 amino acids long, and even more preferably at least 70 amino acids long.

[0073] A protein has “homology” or is “homologous” to a second protein if the nucleic acid sequence that encodes the protein has a similar sequence to the nucleic acid sequence that encodes the second protein. Alternatively, a protein has homology to a second protein if the two proteins have “similar” amino acid sequences. (Thus, the term “homologous proteins” is defined to mean that the two proteins have similar amino acid sequences.) As used herein, homology between two regions of amino acid sequence (especially with respect to predicted structural similarities) is interpreted as implying similarity in function.

[0074] When “homologous” is used in reference to proteins or peptides, it is recognized that residue positions that are not identical often differ by conservative amino acid

substitutions. A “conservative amino acid substitution” is one in which an amino acid residue is substituted by another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, a conservative amino acid substitution will not substantially change the functional properties of a protein. In cases where two or more amino acid sequences differ from each other by conservative substitutions, the percent sequence identity or degree of homology may be adjusted upwards to correct for the conservative nature of the substitution. Means for making this adjustment are well known to those of skill in the art. See, e.g., Pearson, 1994, *Methods Mol. Biol.* 24:307-31 and 25:365-89 (herein incorporated by reference).

[0075] The twenty conventional amino acids and their abbreviations follow conventional usage. See *Immunology-A Synthesis* (Golub and Gren eds., Sinauer Associates, Sunderland, Mass., 2nd ed. 1991), which is incorporated herein by reference. Stereoisomers (e.g., D-amino acids) of the twenty conventional amino acids, unnatural amino acids such as α -, α -disubstituted amino acids, N-alkyl amino acids, and other unconventional amino acids may also be suitable components for polypeptides of the present invention. Examples of unconventional amino acids include: 4-hydroxyproline, γ -carboxyglutamate, ϵ -N,N,N-trimethyllysine, ϵ -N-acetyllysine, O-phosphoserine, N-acetylserine, N-formylmethionine, 3-methylhistidine, 5-hydroxylysine, N-methylarginine, and other similar amino acids and imino acids (e.g., 4-hydroxyproline). In the polypeptide notation used herein, the left-hand end corresponds to the amino terminal end and the right-hand end corresponds to the carboxy-terminal end, in accordance with standard usage and convention.

[0076] The following six groups each contain amino acids that are conservative substitutions for one another: 1) Serine (S), Threonine (T); 2) Aspartic Acid (D), Glutamic Acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Methionine (M), Alanine (A), Valine (V), and 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W).

[0077] Sequence homology for polypeptides, which is sometimes also referred to as percent sequence identity, is typically measured using sequence analysis software. See, e.g., the Sequence Analysis Software Package of the Genetics Computer Group (GCG), University of Wisconsin Biotechnology Center, 910 University Avenue, Madison, Wis. 53705. Protein analysis software matches similar sequences using a measure of homology assigned to various substitutions, deletions and other modifications, including conservative amino acid substitutions. For instance, GCG contains programs such as “Gap” and “Bestfit” which can be used with default parameters to determine sequence homology or sequence identity between closely related polypeptides, such as homologous polypeptides from different species of organisms or between a wild-type protein and a mutein thereof. See, e.g., GCG Version 6.1.

[0078] A useful algorithm when comparing a particular polypeptide sequence to a database containing a large number of sequences from different organisms is the computer program BLAST (Altschul et al., *J. Mol. Biol.* 215:403-410 (1990); Gish and States, *Nature Genet.* 3:266-272 (1993); Madden et al., *Meth. Enzymol.* 266:131-141 (1996); Altschul et al., *Nucleic Acids Res.* 25:3389-3402 (1997); Zhang

and Madden, *Genome Res.* 7:649-656 (1997)), especially blastp or tblastn (Altschul et al., *Nucleic Acids Res.* 25:3389-3402 (1997)).

[0079] Preferred parameters for BLASTp are: Expectation value: 10 (default); Filter: seg (default); Cost to open a gap: 11 (default); Cost to extend a gap: 1 (default); Max. alignments: 100 (default); Word size: 11 (default); No. of descriptions: 100 (default); Penalty Matrix: BLOWSUM62.

[0080] Preferred parameters for BLASTp are: Expectation value: 10 (default); Filter: seg (default); Cost to open a gap: 11 (default); Cost to extend a gap: 1 (default); Max. alignments: 100 (default); Word size: 11 (default); No. of descriptions: 100 (default); Penalty Matrix: BLOWSUM62. The length of polypeptide sequences compared for homology will generally be at least about 16 amino acid residues, usually at least about 20 residues, more usually at least about 24 residues, typically at least about 28 residues, and preferably more than about 35 residues. When searching a database containing sequences from a large number of different organisms, it is preferable to compare amino acid sequences. Database searching using amino acid sequences can be measured by algorithms other than blastp known in the art. For instance, polypeptide sequences can be compared using FASTA, a program in GCG Version 6.1. FASTA provides alignments and percent sequence identity of the regions of the best overlap between the query and search sequences. Pearson, *Methods Enzymol.* 183:63-98 (1990) (incorporated by reference herein). For example, percent sequence identity between amino acid sequences can be determined using FASTA with its default parameters (a word size of 2 and the PAM250 scoring matrix), as provided in GCG Version 6.1, herein incorporated by reference.

[0081] Throughout this specification and claims, the word “comprise” or variations such as “comprises” or “comprising,” will be understood to imply the inclusion of a stated integer or group of integers but not the exclusion of any other integer or group of integers.

[0082] Exemplary methods and materials are described below, although methods and materials similar or equivalent to those described herein can also be used in the practice of the present invention and will be apparent to those of skill

in the art. All publications and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. The materials, methods, and examples are illustrative only and not intended to be limiting.

Overview

[0083] Provided herein are recombinant strains and methods of improving secretion and productivity of recombinantly expressed proteins in yeast strains (e.g., *P. pastoris*).

[0084] Proteins destined for secretion must first cross the membrane of the endoplasmic reticulum (ER translocation). Multiple targeting pathways recruit elongating ribosomes or fully translated proteins to the ER membrane, which polypeptide chains enter via pore-forming protein complexes called translocons. Yeasts including *S. cerevisiae* and *P. pastoris* express two translocons, the SEC61 and SSH1 complexes. Each consists of a core trimeric complex, but the SEC61 translocon also associates with the tetrameric SEC63 complex. FIG. 1 shows the protein subunits contained within each of these translocon complexes.

[0085] SEC72 encodes a nonessential member of the SEC63 complex, but deletion of SEC72 (Δ sec72) results in the accumulation of some secretory precursors. Surprisingly, herein we show that Δ sec72 strains have improved secretion of recombinant silk polypeptides.

[0086] In some embodiments, the strains are modified to delete the sec72 gene (SEQ ID NO: 1). As we describe and shown herein, the deletion of sec72, which expresses SEC72 (SEQ ID NO: 2), an accessory factor for translocating proteins into the ER for secretion, unexpectedly assists silk secretion in bench-scale block model assays (up to +75%). This remained true across expression levels, signal sequences, and different silks.

[0087] In some embodiments, deletion of sec72 is effected using a plasmid having a 5' homology arm to sec72 and a 3' homology arm to sec72 flanking a yeast selection marker, e.g., as shown in FIG. 2. The Δ sec72 deletion elicited transcriptional adaptation that could indirectly assist secretion. Sequences of the sec72 gene and the SEC72 protein are provided in Table 1.

TABLE 1

sec72 open reading frame and SEC72 protein sequences		
Description	SEQ ID NO:	Sequence
sec72 (PAS_chr2-1_0448) ORF sequence (5' to 3')	1	atgctaccatttttcgtacgacgtgagctcaagaaactgaaagtcacaggtgactacgcagacttagaatatgacatacagcagctgaacaccttgagcggagagatcctggccaataaagcagatgttctcttccaccaagt aaggagtcggttgacaagaaattgtccacatggctcagaataatcac gagtcggctgtatccaacataaagacaggcaagtatcctgaggctatc aaattgttgacgacgggtcttgaaatgggttaacagaaggcccaagtac gagagttttcagatgacgttgagtgaatgacgatctttatgtcact agagctgacgccttacatgatgaatggagactttgaaggggcatcatt gatgcagatttactggtaacgctcctgccatccattccagataattac attagaagaggggtagcccttttcaagatggggagatacgtttgatgca aaaaacaattttgagagaggactttcatttgaccagataatgcaaaa ttgaagaaggagtttagattttgtgctggaagaatcgacgaggagaat ggagagttatag
SEC72 (PAS_chr2-1_0448) Protein Sequence	2	MLPFSYDVSSKKLKVTDYADLEYDIQQLNTLSGEILANKADVSPSPS KESFDKKLSHMAQKLHESAVSNIKTGKYPEAIKLLTTGLEMVNRRPKY ESFQMTLSEMTIFIVTRADAYMMNGDFEGAFNDADLLVTLPLSPDNY IRRGVALFKMGRYVDKNNFERGLSFDPDNAKKKELDFVLKKIDEEN GEL

[0088] In some observed cases, the sec72 deletion slowed strain growth, and fermentation production screens of Δsec72 strains struggled with glucose accumulation. Thus, also provided herein, according to some embodiments, are strains recombinantly overexpressing proteins that comprise the SSH1 translocon complex. As shown herein, overexpression of the SSH1 translocon complex in Δsec72 strains improved the strain growth rate and fermentation performance, while maintaining improved secretion. Runs of this combined deletion-overexpression strain with only 4 copies

of of recombinant silk-like block polypeptide expressing genes (i.e., 18B) showed similar titers and improved specific productivity (+18%) over a strain with 6 copies of silk-like block polypeptide expressing genes (i.e., 18B).

[0089] In some embodiments, the SSH1 translocon complex is overexpressed in host cells by inserting a plasmid comprising genes encoding recombinant SBH2, SSS1, and SSH1 (FIG. 3). Sequences for these genes and the expressed protein can be found in Table 2.

TABLE 2

Open reading frame and protein sequences for subunits of the SSH1 translocon complex.		
Description	SEQ ID NO:	Sequence
SSH1 (PAS_chr1-4_0629) ORF sequence	3	atggcggggttgcggttttttagacattgcaagaccatttgctcagctgg atcccggaagttgaacttccttatgaaaactgggggttcgatgaaaag ctgattttactcatttttcaactgctgccatctatttgattctgtccctg cctatatacgggtgtcaaatcctctgaagtcgtggaccagttccccat ttgctgtctgtccttagggagtgagaagggaacattgctggagcttggg ttactgctgtgattacttcggcatttatcttgcaagttgttgctgggt tggaagttttcaaaagtaaaactttgatctgggttagtgacagaatttg ttccaaactttgcaaaagatcacttcagtcgttatcagcatcgatat gctgttcttctcacattttgtgactactttactccaggtgtgtccact gataacgtcttggtggtcccaatttctgatcatcttacagatagtggtg gtcaactcttggttactctactcgttgaagtcattgacaaggattac ggattttcttcaggagctctattgttgcttgcggtttattccgccacc aaactcgtttttggcacgattggtcttagcacctgcaacacctccaga tcgaacgaatctattggtgctctgattcaattattccgcaatttgagc tctaaaccaattgggtgttgccatattgactccttcttcagagtaaac cttctcaacttgactcaatttttatctggggattgccattatttgtgtt tgtctgttcttgaataatgcaagatagcaagtaaccaatgaagccaaac aaggttcgtgccatggcctcagcttaccgaatcaagctacttttcaat ggttcttggcacttctgtacacgtggactgtgctgtacaacttgaac cttattggtttcttctgtcttcaagcttaccgaacttttctcttttaggg aacttcaagtggaaccatttcggcaacaactactacgaattacact ggactgctgtatttattgactcctactttcaacgctgaagctggactt ttaccgaatgttgctaaagcattttgtttcattgctctctatgttggt gttagcactttcttctgtagatcgtgggtccaaactaaacgggtcgta ggcaaggacattgccaagtttttcaaggctcaaggaatctcatgtta ggaaaaagagatgctctgtgtctaaagagtttaacaccctagttcct gttgctctgtcctctggagctttcctattgtctttccagttgccgtc gctgagttattgggtggctctgggtgttccaacctctatcggaatcggt cttttgagtggtttggctattttggaaactgttttgcaagaatggcaa cagctcggaggtgctcagagttctcccaactctccagacttcttag
SSH1 (PAS_chr1-4_0629) Protein Sequence	4	MAGLRFLDIARPFVSWIPEVELPYENWGFDEKLIYSFFTAIYLILSL PIYGVKSSEVVDVPVPHLRSLGSEKGTLELGLLPVITSAPILQLLAG WKVFKVNFDLVSDRILFQTLQKITSVVISIVYAVLLTFCDYFTPGVST DNVLWSQPLILQIVVNFVLVTLVLEVIDKDYGFSSGALLLLAVYSAT NFVFGTIGLSTVNTSRSNESIGALIQLFRLNLSSKPIGVAIYDSFFRVN LPNLTQFYLGAIIICVCLFLNNARYEVPKPNKVRAMASAYPIKLLFN GSLPLLYTWTVLYNLNLIGFFVFKLTNFSLLGNPKVDPPGNNYYEITS GLLYLLTPTFNAEAGLLPNVAKPFVFIAPYVGVSTFFARSWSNINGSS GKDIKFFKAQGISLLGKRDAVSKEFNTLVPVASASGAPLLSPFPAV AELLGGSGVPTSIGIGLLSGLAILETVLQEWQQSGASQFSQYFQTS
SSS1 (PAS_chr1-1_0023) ORF sequence	5	atgtcccaaaaagtcaccgacgtccctctggaatttgtaaggaaggt tccaaattcatctctaaatgtactaaacccctctcagaaggagtaactta aagatagtaagagctgttggaagttgggtttttaatgatggcgctgggt ggttacgttgtaagctcattcatattccaatcagatatttgattggt taa
SSS1 (PAS_chr1-1_0023) Protein Sequence	6	MSQKVTDVPLEFVKEGSKFISKCTKPSQKEYLKIVRAVGVGFLMMGVV GYVVKLIHIPIRYLIV
SBH2 (PAS_chr2-2_0210) ORF sequence	7	atggtaagtggtccagtttgatgagtgcaaatggttccaagttttaga cagttactaatatttaagttctacagcaattccaggaggacagagaa cgttagctaaaagaagagcagcaaaacttggaataagaaacaggatgaac caacctccgccagatctgccggtgctggaggttcttcgtctaccatgc taaagttgtacacagacagggcccaaggtttgaaagttgatccttttaa ttgttctgttcttgctgttggtttcattttcagtgctcattgggttg acgttggtgtaagctgacaggaaagttgatcaactaa

TABLE 2-continued

Open reading frame and protein sequences for subunits of the SSH1 translocon complex.		
Description	SEQ ID NO:	Sequence
SBH2 (PAS_chr2-2_0210) Protein Sequence	8	MSTAIPGGQRTLAKRRAANLDDKKQDEPTSARSAGAGGSSSTMLKLYTD EAQGLKVDPLIVLVLA VGFI FSVIGLHVAKLTGKLIN

Protease Knock-Outs

[0090] In some embodiments, to attenuate a protease activity in *Pichia pastoris*, the genes encoding these enzymes are inactivated or mutated to reduce or eliminate activity. This can be done through mutations or insertions into the gene itself or through modification of a gene regulatory element. This can be achieved through standard yeast genetics techniques. Examples of such techniques include gene replacement through double homologous recombination, in which homologous regions flanking the gene to be inactivated are cloned in a vector flanking a selectable marker gene (such as an antibiotic resistance gene or a gene complementing an auxotrophy of the yeast strain). In some embodiments, the Nourseothricin selection plasmid shown in FIG. 4A can be used as a base plasmid, and homology arms (HA) flanking resistance cassettes (FIG. 4B and FIG. 4C) can be used, where the homology arms specifically target the desired protease to be knocked out. Description of protease knock-outs and methods of modifying host cells to inhibit recombinant protein degradation are provided in U.S. application Ser. No. 15/724,196 and PCT Application No. PCT/US2017/054997, each of which are incorporated herein by reference in its entirety.

[0091] Alternatively, the homologous regions can be PCR-amplified and linked through overlapping PCR to the selectable marker gene. Subsequently, such DNA fragments are transformed into *Pichia pastoris* through methods known in the art, e.g., electroporation. Transformants that then grow under selective conditions are analyzed for the gene disruption event through standard techniques, e.g. PCR on genomic DNA or Southern blot. In an alternative experiment, gene inactivation can be achieved through single homologous recombination, in which case, e.g. the 5' end of the gene's ORF is cloned on a promoterless vector also containing a selectable marker gene. Upon linearization of such vector through digestion with a restriction enzyme only cutting the vector in the target-gene homologous fragment, such vector is transformed into *Pichia pastoris*. Integration

at the target gene site is confirmed through PCR on genomic DNA or Southern blot. In this way, a duplication of the gene fragment cloned on the vector is achieved in the genome, resulting in two copies of the target gene locus: a first copy in which the ORF is incomplete, thus resulting in the expression (if at all) of a shortened, inactive protein, and a second copy which has no promoter to drive transcription.

[0092] Alternatively, transposon mutagenesis is used to inactivate the target gene. A library of such mutants can be screened through PCR for insertion events in the target gene.

[0093] The functional phenotype (i.e., deficiencies) of an engineered/knockout strain can be assessed using techniques known in the art. For example, a deficiency of an engineered strain in protease activity can be ascertained using any of a variety of methods known in the art, such as an assay of hydrolytic activity of chromogenic protease substrates, band shifts of substrate proteins for the selected protease, among others.

[0094] Attenuation of a protease activity described herein can be achieved through mechanisms other than a knockout mutation. For example, a desired protease can be attenuated via amino acid sequence changes by altering the nucleic acid sequence, placing the gene under the control of a less active promoter, down-regulation, expressing interfering RNA, ribozymes or antisense sequences that target the gene of interest, or through any other technique known in the art. In preferred strains, the protease activity of proteases encoded at PAS_chr4_0584 (YPS1-1) and PAS_chr3_1157 (YPS1-2) (e.g., polypeptides comprising SEQ ID NO: 10 and 12) is attenuated by any of the methods described above. In some aspects, the invention is directed to methylotrophic yeast strains, especially *Pichia pastoris* strains, wherein a YPS1-1 and a YPS1-2 gene (e.g., as set forth in SEQ ID NO: 9 and SEQ ID NO: 11) have been inactivated. In some embodiments, additional protease encoding genes may also be knocked-out in accordance with the methods provided herein to further reduce protease activity of a desired protein product expressed by the strain.

TABLE 3

Open reading frame nucleotide sequence and polypeptide sequence for proteases targeted for deletion in <i>P. pastoris</i>		
Description	SEQ ID NO:	Sequence
PAS_chr4_0584 (YPS1-1) ORF sequence (5' to 3')	9	atgttgaaggatcagttcttgttatgggttgctttgatagcgagcgtaccggtttc cggtgtgatggcagctcctagcgagtcgggcataaacacgggttgaaaaacgagatg ccaaaaacgttgttggcggtcaacagttggacttcagcgttctgaggggtgattcc ttcgaaagtgcctcttcagagaacgtgcctcggttgtgaggagagatgacacgct agaagctgagctaatcaaccagcaatcattctacttgcacgactgaaagttggat cacatcaagcggatattggaatcctagtggacacaggaatcctctgatttatgggtg atggactcggtaaacccatactgcagtagccgttccccggtgaagagagatataca cgatgagaagatcgccgaatgggatcccatcaatctcaagaaaaatgaaacttctc agaataaaaaattttgggattgggtcggttggaactagcactagtctctcctccacc gccacggcaactggtagtggtagtggtagtggtagtggtagtggtagtggtagtggtg

TABLE 3-continued

Open reading frame nucleotide sequence and polypeptide sequence for proteases targeted for deletion in <i>P. pastoris</i>		
Description	SEQ ID NO:	Sequence
		tggcacagccgtatcggtaagttctgcacagggcaacattggattgctctacgtatg gaacggtttgatcacgctgattcctcgacgttccatgacaataatacagactttttc atctcatacgctgataccacttttgcttcaggaatctggggttatgacgacgtcat tatcgacggcatagaggtgaaagaactttccttcgccgttgacagacatgaccaatt cctctattgggtgtgtaggtattggactgaaaggcctagaatccacatagctagt gcatcttcgggtcagtgaaatgtatcagtagacaatttggcagccaagatggtcac cgatgggttgatcaacaaaaatgcatactccttgtaactgaaactccaaggacgcct caagtgggtccatcctccttggaggtgtggatcatgaaaaatattcgggacaattg ttgacagttccagtcatacaacacactcgcttcagtggttacagagaggcaattcg tttaaaaaattactttaaatggaatagatgtgaaaaagggtttctgaccagggaactc tttacaaggagagatttgctgcattattggactctggagctacgctaacgtatgct ccttcttctgttttaaatcaattggccggaactgggaggctcctatgattcgtc aagacaagcttataccattcggtgtgtttctgcatcagataccacttctctggtat tcaatttttgggggtgctacagtggaagtttccctgtacgatctacagattgcaaca tattacaccgggggaagtgcacacgaatgtcttattggaatattcagctctggaag tgatgagtttgtgctcgggtgataccttcttgaggtcagcctacgtgggtttacgatc ttgatgggcttgaagtgtcgcttgcccaagccaactcaacgaaacgcatctgat gttgaggtcattacctccagtgtaaccttcgctactcgtgcattccggatcacgttc tacatggtctggttctgccagcgggtacagtttacacttcgggttcagatggaatccg gtgctgcttccagctccaactcctctggatcgaatagggttctcttctctcatcg tctcttcatcgctcctcgacttccagtgagacgaagaaggaggagctccgcca cagggtcccttcagctacctttctctctgtttggtagttattctcggcggtgtgta tagtatag
PAS_chr4_0584 (YPS1-4) Polypeptide Sequence	10	MLKDQFLWV ALIASVPVSG VMAAPSESGH NTVEKRDAKN VVGVLQDLFS VLRGDSFESA SSENVPRLVR RDDTLEAELI NQSFYLSRL KVGSHQADIG ILVDTGSSDL WVMDSVNPYC SRSRVRKRI HDEKIAEWDI INLKKNETSQ NKNFWDWLVG TSTSSPSTAT ATGSGSGSGS GSGSGSAATA VSVSSAQATL DCSTYGTDFH ADSSTFDHNN TDFPISYADT TFASGIWGYD DVIIDGIEVK ELSTFAVADMT NSSIGVLGIG LKGLESTYAS ASSVSEMYQY DNLPAKMVD GLINKNAYSL YLNSKDASSG SILFGVDHE KYSGQLLTVP VINTLASSGY REAIRLQITL NGIDVKKGSD QGTLQGRFA ALLDSGATLT YAPSSVLNSI GRNLGGSYDS SRQAYTIRCV SASDTTSLVF NFGGATVEVS LYDLQIATYY TGGSATQCLI GIFSSGSDEF VLGDFTLRSY YVVDLDGLE VSLAQANFNE TDSDEVAITS SVPSATRASG YSSTWSGSAS GTVYTSVQME SGAASSSSNS GSNMGSSSSS SSSSSSTSSG DEEGSSANR VPFSYLSLCL VVILGVCIV
PAS_chr3_1157 (YPS1-2) ORF sequence (5' to 3')	11	atgatcatcaaccacttggtattgacagccctcagcattgcactagcaagtgcgca actccaatcgcttttcaaggctaaacagttgccattcaaaaagtttatcattccaa cgacccaaaggacggttaattaaagagagatgactacgagtcctcgcattgagac acatcggagtcctgtacactgcagagatccaaattggatctgacgaaactgaaatt gaggtcattgtcgacactggttctgcgcactgtgtgggtcatcgattccgacgctgc cgtctgtgagttatcctacgatgagattgaggccaatagcttttctcggctctcg ccaaaattcatggacaagatagctcctccatcacaaagactcctggatgggctgagt gagtttggatttgctctcgatgggtgaaatttctcaatacctagccgataaatctgg acgtgtttcgaaaagagaggaaaatcaacaagatttcaacattaaccgtgacgagc ctgtgtgtgaacagtttggttctcctgatctcagttcttccgacactttccaaagc acaaatcagcttttggtattgcttaaccttgatggaaccactgctaaccggaacttg ggtcagggcacacagtcgcatcggcgactttgccatcagccaacagagttttgcct tagtcaacatcacagataactacatgggaatcttgggtctcggctcctgtaccctaa caaaccaccaatagtaaccaattgcagcaaacagatttacttatgatgggtgtgtg ggattcattgcggtcccaaggatttatcaattcagcatcggtttctgtttacttgt ctccagatgaagataacgagcacgacgaattcagcgacggagaaattttatttggt gctattgatagggccaaagatagacgggccatttagacttttcccatatgtcaatcc ttacaaaccagtttaccctgatcaataacttctacgttacagtggtccacaattg cgggtgtcttctcgatgaaactctcattattgaaagacgtcctcggttggcatta atcgatacaggtgccaccttctcctatttgccaaacctaccattgattcggttagc gttttccatccatggaggtttgaatagtttctcaattgggactatttgtcatc gtacaagttctctgtctgttgctagaaataagggtgatgagttcaagtttggtgaa gacgttgtgatccaatccccagtttctgatcatctattggacgtctcaggcctttt tactgatggccaacaatactccgcatcactgtacgtgaaagtccttgacggacttt ccattctaggatgatacattcatcaaatccgctacttattctttgacaatgaaac agccagctgggtatttggtcagatcaacgtcactgatgacgaggatattgaggtggt cggtgatttcaactatgaacgagaccagcctactcctcacttggtctagcgatt tacctcatgaaacacccactagggtttgagtactgctcaggggaggccttggt accggaataaaacacggccacaagtctgcaagtctcgttccacatctggctctac ttacgaaacttctctacatctggtctgctctgtgacttcttcagggtgcatctt ctgctactcaaaatgacgaaacatccactgatcttgagctccagctgcatcttta agtgcaacgccatgtctttttgccatcttctgctcatgtttag

TABLE 3-continued

Open reading frame nucleotide sequence and polypeptide sequence for proteases targeted for deletion in <i>P. pastoris</i>			
Description	SEQ ID NO:	Sequence	
PAS_chr3_1157 (YPS1-2) Polypeptide Sequence	12	MIINHLVLTA LSIALANDYE SLDLRHIGVL YTAEIQIGSD ETEIEVIVDT GSADLWVIDS DAAVCELSYD EIEANSFSSA SAKFMDKIAP PSQELLDGLS EFGFALDGEI SQYLADKSGR VSKREENQOD FNINRDEPVC EQFGSFDSSS SDTFQSNNSA FGIAYLDGTT ANGTVWRDVT RIGDFAISQQ SFALVNITDN YMGILGLGPA TQQTNSNPI AANRFTYDGV VDSLRSQGF I NSASFVYLS PDEDNEHDEF SDGEILFGAI DRAKIDGPER LFPYVNPYPK VYPDQYTSYV TVSTIAVSSS DETLIERRP RLALIDTGAT FSYLPTYPLI RLAFSIHGGE EYVSQLGLFV IRTSSLVAR NKVIEFKFGE DVVIQSPVSD HLLDVSGLFT DGQQYSALTV RESLDGLSIL GDTFIKSAYL FFDNENSQLG IGQINVTDE DIEVVGDFTI ERDPAYSSW SSDLPHEPTPT RALSTASGGG LGTGINTATS RASSRSTSGS TSRTSSTSGS ASGTSSGASS ATQNDETSTD LGAPAASLSA TPCLFAILL ML	

Production of Recombinant Strains

[0095] Provided herein are methods of transforming a strain to reduce activity, e.g., using vectors to deliver recombinant genes or to knock-out or otherwise attenuate endogenous genes as desired. These vectors can take the form of a vector backbone containing a replication origin and a selection marker (typically antibiotic resistance, although many other methods are possible), or a linear fragment that enables incorporation into the target cell's chromosome. The vectors should correspond to the organism and insertion method chosen.

[0096] Once the elements of a vector are selected, construction of the vector can be performed in many different ways. In an embodiment, a DNA synthesis service or a method to individually make every vector may be used.

[0097] Once the DNA for each vector (including the additional elements required for insertion and operation) is acquired, it must be assembled. There are many possible assembly methods including (but not limited to) restriction enzyme cloning, blunt-end ligation, and overlap assembly [see, e.g., Gibson, D. G., et al., Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nature methods*, 6 (5), 343-345 (2009), and GeneArt Kit (tools.invitrogen.com/content/sfs/manuals/geneart_seamless_cloning_and_assembly_man.pdf)]. Overlap assembly provides a method to ensure all of the elements get assembled in the correct position and do not introduce any undesired sequences.

[0098] The vectors generated above can be inserted into target cells using standard molecular biology techniques, e.g., molecular cloning. In an embodiment, the target cells are already engineered or selected such that they already contain the genes required to make the desired product, although this may also be done during or after further vector insertion.

[0099] Depending on the organism and library element type (plasmid or genomic insertion), several known methods of inserting the vector comprising DNA to incorporate into the cells may be used. These may include, for example, transformation of microorganisms able to take up and replicate DNA from the local environment, transformation by electroporation or chemical means, transduction with a virus or phage, mating of two or more cells, or conjugation from a different cell.

[0100] Several methods are known in the art to introduce recombinant DNA in bacterial cells that include but are not limited to transformation, transduction, and electroporation,

see Sambrook, et al., *Molecular Cloning: A Laboratory Manual* (1989), Second Edition, Cold Spring Harbor Press, Plainview, N.Y. Non-limiting examples of commercial kits and bacterial host cells for transformation include NovaBlue Singles™ (EMD Chemicals Inc., NJ, USA), Max Efficiency® DH5a™, One Shot® BL21 (DE3) *E. coli* cells, One Shot® BL21 (DE3) pLys *E. coli* cells (Invitrogen Corp., Carlsbad, Calif., USA), XL1-Blue competent cells (Stratagene, CA, USA). Non limiting examples of commercial kits and bacterial host cells for electroporation include Zappert™ electrocompetent cells (EMD Chemicals Inc., NJ, USA), XL1-Blue Electroporation-competent cells (Stratagene, CA, USA), ElectroMAX™ *A. tumefaciens* LBA4404 Cells (Invitrogen Corp., Carlsbad, Calif., USA).

[0101] Several methods are known in the art to introduce recombinant nucleic acid in eukaryotic cells. Exemplary methods include transfection, electroporation, liposome mediated delivery of nucleic acid, microinjection into to the host cell, see Sambrook, et al., *Molecular Cloning: A Laboratory Manual* (1989), Second Edition, Cold Spring Harbor Press, Plainview, N.Y. Non-limiting examples of commercial kits and reagents for transfection of recombinant nucleic acid to eukaryotic cell include Lipofectamine™ 2000, Optifect™ Reagent, Calcium Phosphate Transfection Kit (Invitrogen Corp., Carlsbad, Calif., USA), GeneJammer® Transfection Reagent, LipoTAXI® Transfection Reagent (Stratagene, CA, USA). Alternatively, recombinant nucleic acid may be introduced into insect cells (e.g. sf9, sf21, High Five™) by using baculo viral vectors.

[0102] Transformed cells are isolated so that each clone can be tested separately. In an embodiment, this is done by spreading the culture on one or more plates of culture media containing a selective agent (or lack of one) that will ensure that only transformed cells survive and reproduce. This specific agent may be an antibiotic (if the library contains an antibiotic resistance marker), a missing metabolite (for auxotroph complementation), or other means of selection. The cells are grown into individual colonies, each of which contains a single clone.

[0103] Colonies are screened for desired production of a protein, metabolite, or other product, for reduction in protease activity, for growth, or for increase in secretion activity. In an embodiment, screening identifies recombinant cells having the highest (or high enough) product production titer or efficiency. This includes a decreased proportion of deg-

radiation products or an increased total amount of desired polypeptides secreted from a cell and collected from a cell culture.

[0104] This assay can be performed by growing individual clones, one per well, in multi-well culture plates. Once the cells have reached an appropriate biomass density, they are induced with methanol. After a period of time, typically 24-72 hours of induction, the cultures are harvested by spinning in a centrifuge to pellet the cells and removing the supernatant. The supernatant from each culture can then be tested for to determine whether desired secretion amounts have been achieved.

Silk Sequences

[0105] In some embodiments, the modified strains described herein recombinantly express a silk-like polypeptide sequence. In some embodiments, the silk-like polypeptide sequences are 1) block copolymer polypeptide compositions generated by mixing and matching repeat domains derived from silk polypeptide sequences and/or 2) recombinant expression of block copolymer polypeptides having sufficiently large size (approximately 40 kDa) to form useful fibers by secretion from an industrially scalable microorganism. Large (approximately 40 kDa to approximately 100 kDa) block copolymer polypeptides engineered from silk repeat domain fragments, including sequences from almost all published amino acid sequences of spider silk polypeptides, can be expressed in the modified microorganisms described herein. In some embodiments, silk polypeptide sequences are matched and designed to produce highly expressed and secreted polypeptides capable of fiber formation. In some embodiments, knock-out of protease genes or reduction of protease activity in the host modified strain reduces degradation of the silk like polypeptides. In some embodiments, knock-out of sec72 and overexpression of an SSH1 translocon complex improves secretion while mitigating defects in growth, maintaining, or improving growth of the strain.

[0106] Provided herein, in several embodiments, are compositions for expression and secretion of block copolymers engineered from a combinatorial mix of silk polypeptide domains across the silk polypeptide sequence space, wherein the block copolymers have minimal degradation. In some embodiments provided herein are methods of secreting block copolymers in scalable organisms (e.g., yeast, fungi, and gram positive bacteria) with minimal degradation. In some embodiments, the block copolymer polypeptide comprises 0 or more N-terminal domains (NTD), 1 or more repeat domains (REP), and 0 or more C-terminal domains (CTD). In some aspects of the embodiment, the block copolymer polypeptide is >100 amino acids of a single polypeptide chain. In some embodiments, the block copolymer polypeptide comprises a domain that is at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% identical to a sequence of a block copolymer polypeptide as disclosed in International Publication No. WO/2015/042164, "Methods and Compositions for Synthesizing Improved Silk Fibers," incorporated by reference in its entirety.

[0107] Several types of native spider silks have been identified. The mechanical properties of each natively spun silk type are believed to be closely connected to the molecular composition of that silk. See, e.g., Garb, J. E., et al.,

Untangling spider silk evolution with spidroin terminal domains, *BMC Evol. Biol.*, 10:243 (2010); Bittencourt, D., et al., Protein families, natural history and biotechnological aspects of spider silk, *Genet. Mol. Res.*, 11:3 (2012); Rising, A., et al., Spider silk proteins: recent advances in recombinant production, structure-function relationships and biomedical applications, *Cell. Mol. Life Sci.*, 68:2, pg. 169-184 (2011); and Humenik, M., et al., Spider silk: understanding the structure-function relationship of a natural fiber, *Prog. Mol. Biol. Transl. Sci.*, 103, pg. 131-85 (2011). For example:

[0108] Aciniform (AcSp) silks tend to have high toughness, a result of moderately high strength coupled with moderately high extensibility. AcSp silks are characterized by large block ("ensemble repeat") sizes that often incorporate motifs of poly serine and GPX. Tubuliform (TuSp or Cylindrical) silks tend to have large diameters, with modest strength and high extensibility. TuSp silks are characterized by their poly serine and poly threonine content, and short tracts of poly alanine. Major Ampullate (MaSp) silks tend to have high strength and modest extensibility. MaSp silks can be one of two subtypes: MaSp1 and MaSp2. MaSp1 silks are generally less extensible than MaSp2 silks, and are characterized by poly alanine, GX, and GGX motifs. MaSp2 silks are characterized by poly alanine, GGX, and GPX motifs. Minor Ampullate (MiSp) silks tend to have modest strength and modest extensibility. MiSp silks are characterized by GGX, GA, and poly A motifs, and often contain spacer elements of approximately 100 amino acids. Flagelliform (Flag) silks tend to have very high extensibility and modest strength. Flag silks are usually characterized by GPG, GGX, and short spacer motifs.

[0109] The properties of each silk type can vary from species to species, and spiders leading distinct lifestyles (e.g. sedentary web spinners vs. vagabond hunters) or that are evolutionarily older may produce silks that differ in properties from the above descriptions (for descriptions of spider diversity and classification, see Hormiga, G., and Griswold, C. E., Systematics, phylogeny, and evolution of orb-weaving spiders, *Annu. Rev. Entomol.* 59, pg. 487-512 (2014); and Blackedge, T. A. et al., Reconstructing web evolution and spider diversification in the molecular era, *Proc. Natl. Acad. Sci. U.S.A.*, 106:13, pg. 5229-5234 (2009)). However, synthetic block copolymer polypeptides having sequence similarity and/or amino acid composition similarity to the repeat domains of native silk proteins can be used to manufacture on commercial scales consistent silk-like fibers that recapitulate the properties of corresponding natural silk fibers.

[0110] In some embodiments, a list of putative silk sequences can be compiled by searching GenBank for relevant terms, e.g. "spidroin" "fibroin" "MaSp", and those sequences can be pooled with additional sequences obtained through independent sequencing efforts. Sequences are then translated into amino acids, filtered for duplicate entries, and manually split into domains (NTD, REP, CTD). In some embodiments, candidate amino acid sequences are reverse translated into a DNA sequence optimized for expression in *Pichia* (Komagataella) *pastoris*. The DNA sequences are each cloned into an expression vector and transformed into *Pichia* (Komagataella) *pastoris*. In some embodiments, various silk domains demonstrating successful expression and secretion are subsequently assembled in combinatorial fashion to build silk molecules capable of fiber formation.

[0111] Silk polypeptides are characteristically composed of a repeat domain (REP) flanked by non-repetitive regions

(usually 2-8) in the repeat domain of the silk sequence. Repeated blocks inside a repeat domain or macro-repeat, and repeated macro-repeats within the repeat domain, may be separated by spacing elements. In some embodiments, block sequences comprise a glycine rich region followed by a polyA region. In some embodiments, short (~1-10) amino acid motifs appear multiple times inside of blocks. For the purpose of this invention, blocks from different natural silk polypeptides can be selected without reference to circular permutation (i.e., identified blocks that are otherwise similar between silk polypeptides may not align due to circular permutation). Thus, for example, a “block” of SGAGG (SEQ ID NO: 23) is, for the purposes of the present invention, the same as GSGAG (SEQ ID NO: 24) and the same as GGSGA (SEQ ID NO: 25); they are all just circular permutations of each other. The particular permutation selected for a given silk sequence can be dictated by convenience (usually starting with a G) more than anything else. Silk sequences obtained from the NCBI database can be partitioned into blocks and non-repetitive regions.

Samples of Block Sequences

Samples of Block Sequences		
Species	Silk Type	Representative Block Amino Acid Sequence
<i>Alia typus gulosus</i>	Fibroin 1	GAASSSSSTIIITKSASASAAADASAAATASAASSSSANAASAFAPS FSSILLESYGFCISIFGSSISSSYAAAIASASRAAAESNGYTTHAYA CAKAVASAVERTVSGADAYAYAQAI SDAISHALLYTGR LNTANANSL ASAFAYAFANAAQASASSASAGAASASGAASASGAGSAS (SEQ ID NO: 26)
<i>Plectreurus tristis</i>	Fibroin 1	GAGAGAGAGAGAGAGAGSGASTSVSTSSSSSGAGAGAGSGAGSAG AGSGAGAGAGAGGAGGAGFGSGGLGLGYGVLSSAQQAQAQAAAQAQA QAQAQAYAAAAQAQAQAQAQ (SEQ ID NO: 27)
<i>Plectreurus tristis</i>	Fibroin 4	GAAQKQPSGESSVATASAAATSVTSGGAPVKGPGVPAPIFY PQGLPQ QGPAPGPSNVQPGTQQGPIGGVGGSNAPSSSFASALSLNRGFT EVI SSASATAVASAFQKGLAPYGTAFALSAASAAADAYNSIGSGANAFAY AQAFARVLYPLVQQYGLSSSAKASAFASAIASSFSSGTSGQGPSIG QQPPVTISAASASAGASAAVGGGVGQGPYGGQQQSTASASAAAA TATS (SEQ ID NO: 28)
<i>Araneus gemmoides</i>	TuSp	GNVG YQLGLKVANS LGLGNAQALASSLSQAVSAVGVGASSNAYANAV SNVAGQVLAGQGILNAANAGSLASSFASLSAASVASQSASQSA ASQSQAASAFRQAASQSASQSDSRAGSQSSTKTTSTSTSGSQADSR SASSSASQASAFAPQSSASLSSSSSFSSAPSSATSISAV (SEQ ID NO: 29)
<i>Argiope aurantia</i>	TuSp	GSLASSFASALSASAASVASSAAAQASQSQAASAFSRAASQSASQ SAARSGAQSI STTTTTSTAGSQAASQSASSAASQASASSFARASSAS LAASSSFSAFSSANSLSALGNVG YQLGFNVANNLIGNAAGLGNAL SQAVSSVGVGASSSTYANAVSNVAGQFLAGQGILNAANA (SEQ ID NO: 30)
<i>Deinopis spinosa</i>	TuSp	GASASAYASAINAVGPYLYGLGLFNQANAASFASSFASAVSSAVAS ASASAASSAYAQSAQAQAASSAFSQAAAQSAASASAGASAGAGAS AGAGAVAGAGAVAGAGAVAGASAAAASQAASSSSASAVASAFAPQAS YALASSAFAMAFASATSAGYLGSLAYQLGLTTAYNLGLSNAQAFAS TLSQAVTGVGL (SEQ ID NO: 31)
<i>Nephila clavipes</i>	TuSp	GATAASYGNALSTAAQFFATAGLLNAGNASALASSFARAFSASAES QSFAQSQAFAQQAASFQQAASRSASQSAEAGSTSSSTTTTTSAARSQ AASQSASSSYSAFAQAASSSLATSSALSRAFPSSVSSASAASSLAYS IGLSAARSLGIADAAGLAGVLARAAGALGQ (SEQ ID NO: 32)

TABLE 4-continued

Samples of Block Sequences		
Species	Silk Type	Representative Block Amino Acid Sequence
<i>Argiope trifasciata</i>	Flag	GGAPGGGPGGAGPGGAGFGPGGGAGFGPGGGAGFGPGAAGGPGGPG GPGGPGGAGGYGPGGAGGYGPGGVGPGGAGGYGPGGAGGYGPGGSGP GGAGPGGAGGEGPVTVDVDTVGPEGVGGGPGGAGPGGAGFGPGGGA GFGPGGAPGAPGGPGGPGGPGGPGGVGPGGAGGYGPGGAGGVGP AGTGGFGPGGAGGFGPGGAGGFGPGGAGGFGPAGAGGYGPGGVGPGG AGGFGPGGVGPGGSGPGGAGGEGPVTVDVDSV (SEQ ID NO: 33)
<i>Nephila clavipes</i>	Flag	GVSYGPGGAGGPGYGGPYGPGGEGPGGAGGPGYGGGVGPGGSGPGG YGPGGAGPGGYGPGGSGPGGYGPGGSGPGGYGPGGSGPGGYGPGGSG PGGYGPGGYGPGGSGPGGSGPGGSGPGGYGPGGTGPGGSGPGGYGPG GSGPGGSGPGGYGPGGSGPGGFGPGGSGPGGYGPGGSGPGGAGPGGV GPGGFGPGGAGPGGAAPGAGPGGAGPGGAGPGGAGPGGAGPGGAGP GGAGGAGGAGGSGGAGGSGGTTIIEDLDITIDGADGPITISEELPIS GAGGSGPGGAGPGGVGPGGSGPGGVGPGGSGPGGVGPGGSGPGGVGP GGAGGPGYGGSGPGGAGGAGPGGAYGPGGYGPGGSGPGGAGGP YGPGEPPGGAGGPGYGGAGGPGYGGAGGPGYGGEGGPGYGP (SEQ ID NO: 34)
<i>Latrodectus hesperus</i>	AcSp	GINVDSDIGSVTSLILSGSTLQMTIPAGDDLSGGYPGGFPAGAQPS GGAPVDFGPGSAGGDVAKLARSLASTLASSGVFRAAFNSRVSTPVA VQLTDLVQKIASNLGLDYATASKLKASQAVSKVRMGSDTNAYALA ISSALAEVLSSSGKVADANINQIAPQLASGIVLGVSTTAPQFGVDLS SINVNLDISNVARNMQASIQGGPAPITAECPDFGAGYPPGAPTDLSG LDMGAPSDGSRGGDATAKLLQALVPAKSDVFRAYKRGTRKQVVQ YVTNSALQQAASSLGLDASTISQLQTKATQALSSVSADSDSTAYAKA FGLAIAQVLGTSGQVNDANVNQIGAKLATGILRGSSAVAPRLGIDL (SEQ ID NO: 35)
<i>Argiope trifasciata</i>	AcSp	GAGYTGPSGPSTGPGSGYGPGLGGGAPFGQSGFGGSAGPQGGFGATGG ASAGLISRANALANTSTLRTVLRTVGVSQQIASSVVQRAAQSLASTL GVDGNLRFVAVQAVSRPAGSDTSAYAQAFFSALFNAGVLNASNID TLGSRVLSALLNGVSSAAQGLGINVDSGVSQSDISSSSSFLSTSSSS ASYSQASASSTS (SEQ ID NO: 36)
<i>Uloborus diversus</i>	AcSp	GASAADIATAIAASVATSLQSNGLVTASNVSQLSNQLASYVSSGLSS TASSLGIQLGASLQAGFGASAGLSASTDISSSVEATSASTLSSSSASS TSVVSSINAQLVPALAQTAVLNAAFNTQNAIRIAELLTQQVGRQ YGLSGSDVATASSQIRSAIYSVQQGSASSAYVSAIVGPLITALSSRG VVNASNSSQIASSLATAILOFTANVAPQFGISIPSAVQSDLSSTISQ SLTAISSQTSSSVDSSTSAFGGISGPGSPSYGPQPSGPTFGPGPSL SGLTGFTATFASFSKSTLASSTQFQLIAQSNLDVQTRSSLISKVLIN ALSSLGISASVASSIAASSSQSLLSVSA (SEQ ID NO: 37)
<i>Euprosthenops australis</i>	MaSp1	GGQGGGQGRYQGAGSSAAAAAAAAAAAAA (SEQ ID NO: 38)
<i>Tetragnatha kauaiensis</i>	MaSp1	GGLGGGQAGQGGQAGAGQGGYGSGLGGAGQGASAAAAAAAAA (SEQ ID NO: 39)
<i>Argiope aurantia</i>	MaSp2	GGYGPAGQQGPGSQGPGSGGQQGPGGLGPGPSAAAAAAAAA (SEQ ID NO: 40)
<i>Deinopis spinosa</i>	MaSp2	GPGGYGGPGQQGPGQGYGPGTQQGQGGPSGQQGPAGAAAAAAAAA (SEQ ID NO: 41)
<i>Nephila clavata</i>	MaSp2	GPGGYGLGQQGPGQQGPGQGPAGYGPSGLSGPGGAAAAAAAAA (SEQ ID NO: 42)

[0112] Fiber-forming block copolymer polypeptides from the blocks and/or macro-repeat domains, according to certain embodiments of the invention, is described in International Publication No. WO/2015/042164, incorporated by reference. Natural silk sequences obtained from a protein database such as GenBank or through de novo sequencing are broken up by domain (N-terminal domain, repeat domain, and C-terminal domain). The N-terminal domain and C-terminal domain sequences selected for the purpose

of synthesis and assembly into fibers include natural amino acid sequence information and other modifications described herein. The repeat domain is decomposed into repeat sequences containing representative blocks, usually 1-8 depending upon the type of silk, that capture critical amino acid information while reducing the size of the DNA encoding the amino acids into a readily synthesizable fragment. In some embodiments, a properly formed block copolymer polypeptide comprises at least one repeat domain

comprising at least 1 repeat sequence, and is optionally flanked by an N-terminal domain and/or a C-terminal domain.

[0113] In some embodiments, a repeat domain comprises at least one repeat sequence. In some embodiments, the repeat sequence is 150-300 amino acid residues. In some embodiments, the repeat sequence comprises a plurality of blocks. In some embodiments, the repeat sequence comprises a plurality of macro-repeats. In some embodiments, a block or a macro-repeat is split across multiple repeat sequences.

[0114] In some embodiments, the repeat sequence starts with a Glycine, and cannot end with phenylalanine (F), tyrosine (Y), tryptophan (W), cysteine (C), histidine (H), asparagine (N), methionine (M), or aspartic acid (D) to satisfy DNA assembly requirements. In some embodiments, some of the repeat sequences can be altered as compared to native sequences. In some embodiments, the repeat sequences can be altered such as by addition of a serine to the C terminus of the polypeptide (to avoid terminating in F, Y, W, C, H, N, M, or D). In some embodiments, the repeat sequence can be modified by filling in an incomplete block with homologous sequence from another block. In some embodiments, the repeat sequence can be modified by rearranging the order of blocks or macro-repeats.

[0115] In some embodiments, non-repetitive N- and C-terminal domains can be selected for synthesis. In some embodiments, N-terminal domains can be by removal of the leading signal sequence, e.g., as identified by SignalP (Peterson, T. N., et al., SignalP 4.0: discriminating signal peptides from transmembrane regions, *Nat. Methods*, 8:10, pg. 785-786 (2011)).

[0116] In some embodiments, the N-terminal domain, repeat sequence, or C-terminal domain sequences can be derived from *Agelenopsis aperta*, *Aliatypus gulosus*, *Aphonopelma seemanni*, *Aptostichus* sp. AS217, *Aptostichus* sp. AS220, *Araneus diadematus*, *Araneus gemmoides*, *Araneus ventricosus*, *Argiope amoena*, *Argiope argentata*, *Argiope bruennichi*, *Argiope trifasciata*, *Atypoides riversi*, *Avicularia juruensis*, *Bothriocyrtum californicum*, *Deinopis Spinosa*, *Diguetia canities*, *Dolomedes tenebrosus*, *Euagrus chiseus*, *Euprosthenops australis*, *Gasteracantha mammosa*, *Hypochilus thorelli*, *Kukulcania hibernalis*, *Latrodectus hesperus*, *Megaheura fulva*, *Metepeira grandiosa*, *Nephila antipodiana*, *Nephila clavata*, *Nephila clavipes*, *Nephila madagascariensis*, *Nephila pilipes*, *Nephilengys cruentata*, *Parawixia bistriata*, *Peucetia viridans*, *Plectreurys tristis*, *Poecilotheria regalis*, *Tetragnatha kauaiensis*, or *Uloborus diversus*.

[0117] In some embodiments, the silk polypeptide nucleotide coding sequence can be operatively linked to an alpha mating factor nucleotide coding sequence. In some embodiments, the silk polypeptide nucleotide coding sequence can be operatively linked to another endogenous or heterologous secretion signal coding sequence. In some embodiments, the silk polypeptide nucleotide coding sequence can be operatively linked to a 3× FLAG nucleotide coding sequence. In some embodiments, the silk polypeptide nucleotide coding sequence is operatively linked to other affinity tags such as 6-8 His residues. An example of a vector for delivering a silk polypeptide nucleotide coding sequence to a host cell is shown in FIG. 5, which depicts a plasmid map of a vector comprising an open reading frame (ORF) for expressing a silk-like polypeptide comprising a secretion signal and a

FLAG tag. The ORF is operably linked to a pGCW14 promoter and a tAOX1 terminator. The vector also comprises selection markers to select successfully transformed cells.

[0118] In some embodiments, the silk polypeptide comprises a full length spider silk polypeptide. In some embodiments, the full length spider silk polypeptide is Major ampullate spidroin 1 (MaSp1) or Major ampullate spidroin 2 (MaSp2).

Silk-Like Polypeptides

[0119] In some embodiments, the *P. pastoris* strains disclosed herein have been modified to express a silk-like polypeptide. Methods of manufacturing preferred embodiments of silk-like polypeptides are provided in WO 2015/042164, especially at Paragraphs 114-134, incorporated herein by reference. Disclosed therein are synthetic proteinaceous copolymers based on recombinant spider silk protein fragment sequences derived from MaSp2, such as from the species *Argiope bruennichi*. Silk-like polypeptides are described that include two to twenty repeat units, in which a molecular weight of each repeat unit is greater than about 20 kDa. Within each repeat unit of the copolymer are more than about 60 amino acid residues that are organized into a number of “quasi-repeat units.” In some embodiments, the repeat unit of a polypeptide described in this disclosure has at least 95% sequence identity to a MaSp2 dragline silk protein sequence.

[0120] In some embodiments, each “repeat unit” of a silk-like polypeptide comprises from two to twenty “quasi-repeat” units (i.e., n3 is from 2 to 20). Quasi-repeats do not have to be exact repeats. Each repeat can be made up of concatenated quasi-repeats. Equation 1 shows the composition of a repeat unit according the present disclosure and that incorporated by reference from WO 2015/042164. Each silk-like polypeptide can have one or more repeat units as defined by Equation 1.

$$\text{(Equation 1)} \quad \text{(SEQ ID NO: 13)}$$

$$\{\text{GGY} - [\text{GPG} - \text{X}_1]_{n1} - \text{GPS} - (\text{A})_{n2}\}_{n3}$$

[0121] The variable compositional element X₁ (termed a “motif”) is according to any one of the following amino acid sequences shown in Equation 2 and X₁ varies randomly within each quasi-repeat unit.

$$\text{(Equation 2)} \quad \text{(SEQ ID NO: 14)}$$

$$\text{X}_1 = \text{SGGQQ}$$

or

$$\text{(SEQ ID NO: 15)}$$

$$\text{GAGQQ}$$

or

$$\text{(SEQ ID NO: 16)}$$

$$\text{GQGPY}$$

or

$$\text{(SEQ ID NO: 17)}$$

$$\text{AGQQ}$$

or

$$\text{SQ}$$

[0122] Referring again to Equation 1, the compositional element of a quasi-repeat unit represented by “GGY-[GPG-X₁]n1-GPS” (SEQ ID NO: 18) in Equation 1 is referred to a “first region.” A quasi-repeat unit is formed, in part by repeating from 4 to 8 times the first region within the quasi-repeat unit. That is, the value of n1 indicates the number of first region units that are repeated within a single quasi-repeat unit, the value of n1 being any one of 4, 5, 6, 7 or 8. The compositional element represented by “(A)n2” (SEQ ID NO: 19) (i.e., a polyA sequence) is referred to as a “second region” and is formed by repeating within each quasi-repeat unit the amino acid sequence “A” n2 times (SEQ ID NO: 19). That is, the value of n2 indicates the number of second region units that are repeated within a single quasi-repeat unit, the value of n2 being any one of 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20. In some embodiments, the repeat unit of a polypeptide of this disclosure has at least 95% sequence identity to a sequence containing quasi-repeats described by Equations 1 and 2. In some embodiments, the repeat unit of a polypeptide of this disclosure has at least 80%, or at least 90%, or at least 95%, or at least 99% sequence identity to a sequence containing quasi-repeats described by Equations 1 and 2.

[0123] In additional embodiments, 3 “long” quasi repeats are followed by 3 “short” quasi-repeat units. Short quasi-

repeat units are those in which n1=4 or 5. Long quasi-repeat units are defined as those in which n1=6, 7 or 8. In some embodiments, all of the short quasi-repeats have the same X₁ motifs in the same positions within each quasi-repeat unit of a repeat unit. In some embodiments, no more than 3 quasi-repeat units out of 6 share the same X₁ motifs.

[0124] In additional embodiments, a repeat unit is composed of quasi-repeat units that do not use the same X₁ more than two occurrences in a row within a repeat unit. In additional embodiments, a repeat unit is composed of quasi-repeat units where at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 of the quasi-repeats do not use the same X₁ more than 2 times in a single quasi-repeat unit of the repeat unit.

[0125] Thus, in some embodiments, provided herein are strains of yeast that recombinantly express silk-like polypeptides with a reduced degradation to increase the amount of full-length polypeptides present in the isolated product from a cell culture. In some embodiments, the strain expressing a silk-like polypeptide is a *P. pastoris* strain comprises a PAS_chr4_0584 knock-out and a PAS_chr3_1157 knock-out. In some embodiments, the strain expressing a silk-like polypeptide is a *P. pastoris* strain comprising an sec72 knock-out and/or an overexpressed SSH1 translocon complex.

TABLE 5

		18B vector									
Description	SEQ ID NO:	5' to 3' Sequence									
18B silk-like polypeptide encoding sequence	20	ggtgggttacg	gtccaggcgc	tgggtcaacaa	ggtccaggaa	gtgggtgggtca	acaaggacct	60			
		ggcgggtcaag	gaccttacgg	tagtggccaa	caagggtccag	gtggagcagg	acagcagggt	120			
		ccggaggagcc	aaggacctta	cggaccagggt	gctgctgctg	ccgcccgctgc	cgctgccgga	180			
		ggttacgggtc	caggagccgg	acaacagggt	ccagggtggag	ctggacaaca	aggtccaggga	240			
		tcacaagggtc	ctgggtggaca	aggtccatac	ggctcctgggtg	ctgggtcaaca	gggaccagggt	300			
		agtcaggagac	ctgggttcagg	tgggtcagcag	ggtccaggag	gacagggtcc	ttacggccct	360			
		tctgccgctg	cagcagcagc	cgctgccgca	ggaggatacg	gacctgggtgc	tggacaacga	420			
		tctcaaggac	caggaggaca	aggtccttat	ggacctggcg	ctggccaaca	aggacctggt	480			
		tctcagggtc	cagggttcagg	aggccaacaa	ggcccaggag	gtcaaggacc	atacggacca	540			
		tccgctgcgg	cagctgcagc	tgetgcagg	ggatatggcc	caggagccgg	acaacagggt	600			
		cctgggtcac	aagggtccagg	atctgggtggt	caacaggggac	caggcgccca	gggaccttat	660			
		ggtccaggag	ccgctgcagc	agcagcagct	gttgagggtt	acggccctgg	tggcgggtcaa	720			
		caaggcccgag	gatctcagg	tcctggatct	ggaggacaac	aaggtcctgg	aggtcagggt	780			
		ccatacggac	cttcagcagc	agctgctgct	gcagccgctg	gtgggttatgg	acctgggtgct	840			
		ggtcaacaag	gaccgggttc	tcagggtccg	ggttcaggag	gtcagcagg	ccctgggtgga	900			
		caaggacctt	atggacctag	tgcggctgca	gcagctgccg	ccgcagggtg	ttacgggtcca	960			
		ggcgctgggt	aacaagggtc	aggaagtgg	ggtcaacaag	gacctggcg	tcaaggacct	1020			
		tacggtagtg	gccaacaagg	tccagggtgga	gcaggacagc	agggtccggg	aggccaaggga	1080			
		ccttacggac	cagggtgctgc	tgetgccgcc	gctgccgctg	ccggagggtta	cggtccaggga	1140			
		gccggacaac	agggtccagg	tggagctgga	caacaagggtc	caggatcaca	aggtcctggt	1200			
		ggacaaggtc	catacgggtc	tgggtgctggt	caacaggggac	caggtagtca	aggacctggt	1260			
		tcagggtggtc	agcagggttc	aggaggacag	ggtccttacg	gcccttctgc	cgctgcagca	1320			
		gcagccgctg	ccgcaggagc	atacggacct	ggtgctggac	aacgattcca	aggaccaggga	1380			
		ggacaaggtc	cttatggacc	tggcgtctggc	caacaaggac	ctgggtctca	gggtccagggt	1440			
		tcaggaggcc	aacaaggccc	aggaggtcaa	ggaccatacg	gaccatccgc	tgcggcagct	1500			
		gcagctgctg	cagggtggata	tggcccaggga	ccgggacaac	agggtcctgg	ttcacaagggt	1560			
		ccaggatctg	gtgggtcaaca	gggaccaggc	ggccaggggac	cttatgggtcc	aggagccgct	1620			
		gcagcagcag	cagctgttgg	aggttacggc	cctgggtgcc	gtcaacaagg	cccaggatct	1680			
		cagggtcctg	gatctggagg	acaacaagg	cctggagggtc	agggtccata	cggaacctca	1740			
		gcagcagcgtg	ctgctgcagc	cgctgggtggt	tatggacctg	gtgctgggtca	acaaggaccg	1800			
		ggttctcagg	gtccgggttc	aggaggtcag	cagggccctg	gtggacaagg	accttatgga	1860			
		cctaggtcgg	ctgcagcagc	tgccgccgca	ggtgggttacg	gtccaggcgc	tgggtcaacaa	1920			
		ggtccaggaa	gtgggtggtca	acaaggacct	ggcgggtcaag	gacctcacgg	tagtggccaa	1980			
		caagggtccag	gtggagcagg	acagcagggt	ccgggaggcc	aaggacctta	cggaccagggt	2040			
		cctgctgctg	ccgccgctgc	cgctgccgga	gggtacgggtc	caggagccgg	acaacagggt	2100			
		ccagggtggag	ctggacaaca	aggtccaggga	tcacaagggtc	ctgggtggaca	aggtccatac	2160			
		ggtcctgggtg	ctgggtcaaca	gggaccagggt	agtcaggagc	ctgggttcagg	tgggtcagcag	2220			
		ggtccaggag	gcagggttc	ttacggccct	tctgccgctg	cagcagcagc	cgctgccgca	2280			
		ggaggatagc	gacctgggtgc	tggacaacga	tctcaaggac	caggaggaca	aggtccttat	2340			

TABLE 5-continued

		18B vector	
Description	SEQ ID NO:	5' to 3' Sequence	
		ggacctggcg ctggccaaca aggacctggt tctcagggtc caggttcagg aggccacaac	2400
		ggcccaggag gtcaaggacc atacggacca tccgctgcgg cagctgcagc tgcctgcaggt	2460
		ggatatggcc caggagcccg acaacagggt cctggtttcac aaggttcagg atctggtggt	2520
		caacaggggac caggccggca gggaccttat ggtccaggag ccgctgcagc agcagcagct	2580
		gttgaggagtt acggcccttg tgcgggtcaa caaggcccag gatctcaggg tccctggatct	2640
		ggaggacaac aaggtccttg aggtcagggt ccatacggac cttcagcagc agctgctgct	2700
		gcagccgctg gtggttatgg acctggtgct ggtcaacaag gaccgggttc tcagggtccg	2760
		ggttcaggag gtcagcaggg ccctggttga caaggacctt atggacctag tgcggctgca	2820
		gcagctgcgc cgcga	2835
18B polypeptide sequence	21	GGYGPAGAGQQPGSGGQQPGGQGPYSGSQQPGGAGQQPGPGQGPYPGPAAAAAAAAAAGGYGPGAGQQG PGGAGQQPGSGQPGGQGPYPGAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQR SQGPGGQGPYPGAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQQGPGSGQPGSGG QQPGGQGPYPGSAAAAAAAAAVGGYGPAGAGQQPGSGQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGA GQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQQGPGSGGQQPGGQGPYPGSA AGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQQGPGGAGQQPGSGSQPGGQGPYPGAGQQQPGSGSQPG SGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQRSQPGGQGPYPGAGAGQQPGSGSQPGSGGQQPGGQ GPYPGSAAAAAAAAAAGGQGPYGPAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAVGGYGPAGAGQ QPGSGGQGPYPGSAAAAAAAAAAGGYGPGAGQQGPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAA GGYGPAGAGQQPGSGGQQPGGQGPYSGSQQPGGAGQQPGPGQGPYPGPAAAAAAAAAAGGYGPGAGQQG PGGAGQQPGSGQPGGQGPYPGPAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQR SQGPGGQGPYPGAGAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQQGPGSGQPGSGG QQGPGGQGPYPGSAAAAAAAAAVGGYGPAGAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGA GQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAA	
Repeat sequence of a silk-like polypeptide	22	GGYGPAGAGQQPGSGGQQPGGQGPYSGSQQPGGAGQQPGPGQGPYPGPAAAAAAAAAAGGYGPGAGQQG PGGAGQQPGSGSQPGGQGPYPGAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQR SQGPGGQGPYPGAGQQPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGAGQQGPGSGQPGSGG QQGPGGQGPYPGSAAAAAAAAAVGGYGPAGQQGPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAAAGGYGPGA GQQGPGSGSQPGSGGQQPGGQGPYPGSAAAAAAAAA	

Equivalents and Scope

[0126] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments in accordance with the invention described herein. The scope of the present invention is not intended to be limited to the above Description, but rather is as set forth in the appended claims.

[0127] In the claims, articles such as “a,” “an,” and “the” may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include “or” between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The invention includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The invention includes embodiments in which more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process.

[0128] It is also noted that the term “comprising” is intended to be open and permits but does not require the inclusion of additional elements or steps. When the term “comprising” is used herein, the term “consisting of” is thus also encompassed and disclosed.

[0129] Where ranges are given, endpoints are included. Furthermore, it is to be understood that unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or

subrange within the stated ranges in different embodiments of the invention, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

[0130] All cited sources, for example, references, publications, databases, database entries, and art cited herein, are incorporated into this application by reference, even if not expressly stated in the citation. In case of conflicting statements of a cited source and the instant application, the statement in the instant application shall control.

[0131] Section and table headings are not intended to be limiting.

EXAMPLES

[0132] Below are examples of specific embodiments for carrying out the present invention. The examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way. Efforts have been made to ensure accuracy with respect to numbers used (e.g., amounts, temperatures, etc.), but some experimental error and deviation should, of course, be allowed for.

[0133] The practice of the present invention will employ, unless otherwise indicated, conventional methods of protein chemistry, biochemistry, recombinant DNA techniques and pharmacology, within the skill of the art. Such techniques are explained fully in the literature. See, e.g., T. E. Creighton, *Proteins: Structures and Molecular Properties* (W.H. Freeman and Company, 1993); A. L. Lehninger, *Biochemistry* (Worth Publishers, Inc., current addition); Sambrook, et al., *Molecular Cloning: A Laboratory Manual* (2nd Edition, 1989); *Methods In Enzymology* (S. Colowick and N. Kaplan eds., Academic Press, Inc.); Remington's *Pharmaceutical*

Sciences, 18th Edition (Easton, Pennsylvania: Mack Publishing Company, 1990); Carey and Sundberg Advanced Organic Chemistry 3rd Ed. (Plenum Press) Vols A and B (1992).

Example 1: Production of Recombinant Yeast Expressing 18B

[0134] Strains of *P. pastoris* were modified to recombinantly express 18B silk-like polypeptides as follows:
[0135] First, we transformed a strain of *P. pastoris* to abrogate KU70 function to facilitate further editing and engineering. A HIS+ derivative of *Pichia pastoris* (Komagataella phaffii) strain GS115 (NRRL Y15851) was electroporated with a DNA cassette consisting of homology arms flanking a zeocin resistance marker and targeting the KU70 locus. Sequences are provided in Table 12. Transformants were plated on YPD agar plates supplemented with zeocin. This resulted in abrogation of KU70 function.
[0136] Then, we modified this strain to express a recombinant gene encoding a silk-like polypeptide. A HIS+ derivative of *Pichia pastoris* (Komagataella phaffii) strain GS115 (NRRL Y15851) was transformed with a recombinant vector (SEQ ID NO: 20) to cause expression and secretion of a silk-like polypeptide (“18B”) (SEQ ID NO: 21). Transformation was accomplished by electroporation as described in PMID 15679083, incorporated by reference herein.
[0137] Each vector includes an 18B expression cassette with the polynucleotide sequence encoding the silk-like protein in the recombinant vectors flanked by a promoter (pGCW14) and a terminator (taOX1 pA signal). The recombinant vectors further comprised dominant resistance markers for selection of bacterial and yeast transformants, and a bacterial origin of replication. The first recombinant vector included targeting regions that directed integration of the 18B polynucleotide sequences immediately 3' of the AOX2 loci in the *Pichia pastoris* genome. The resistance marker in the first vector conferred resistance to G418 (aka geneticin). The second recombinant vector included targeting regions that directed integration of the 18B polynucleotide sequences immediately 3' of the TEF1 loci in the *Pichia pastoris* genome. The resistance marker in the second vector conferred resistance to Hygromycin B.

Example 2: Production of Recombinant ΔSec72 Strain

[0138] Cells modified to express 18B were transformed with a vector comprising a DNA cassette with 5' and 3' homology arms targeting sec72 (SEQ ID NO: 1). The homology arms flank a yeast selection marker, as shown in the plasmid map in FIG. 2, which knocks out the sec72 gene when inserted into the sec72 gene. Transformants were plated on YPD agar plates supplemented with yeast selection medium, and incubated for 48 hours at 30° C.

Example 3: Production of Recombinant ΔSec72Strain Overexpressing an SSH1 Translocon Complex

[0139] *P. pastoris* (Δsec72) cells modified to express 18B (SEQ ID NO: 21) were transformed with a vector for overexpression of the SSH1 translocon complex. A plasmid map of the vector is shown in FIG. 3. The vector comprises open reading frames for SSH1 (SEQ ID NO: 3), SSS1 (SEQ ID NO: 5), and SBH2 (SEQ ID NO: 7). Each open reading

frame is operatively linked to a promoter and a terminator. Transformants were plated on YPD agar plates supplemented with yeast selection medium, and incubated at 48 hours at 30° C.

Example 4: Production of Protease Double Knock-Out Strain

[0140] To generate AAprtease strains (i.e., double protease knock-out), a selected yeast strain was transformed with vector comprising a DNA cassette with ~1150 bp homology arms flanking a nourseothricin resistance marker. A plasmid map comprising the nourseothricin resistance marker is shown in FIG. 4A, and sequences provided in Table 13. Homology arms used for each target were amplified by the primers provided in Table 8, and inserted into the nourseothricin resistance plasmid. Homology arms were inserted into the nourseothricin plasmid to generate cassettes comprising a nourseothricin resistance marker flanked by 3' and 5' homology arms to the target protease as shown in FIG. 4B and FIG. 4C. In FIG. 4B, the resistance cassette (Nour Resistance Cassette) is shown flanked by homology arms (HA1 and HA2). In FIG. 4C, details of the nourseothricin marker are shown, including the promoter from ILV5 gene from *Saccharomyces cerevisiae* (pILV5), the Nourseothricin acetyltransferase gene from *Streptomyces noursei* (nat), and the polyA signal from CYC1 gene from *Saccharomyces cerevisiae*.
[0141] A vector with homology arms targeting YPS1-1 (SEQ ID NO: 77) was used to transform modified yeast strains. Transformants were plated on YPD agar plates supplemented with nourseothricin, and incubated for 48 hours at 30° C.

Proteases targeted for deletion in <i>P. pastoris</i> strain.		
Protease Gene Symbol	Protease ORF Sequence (SEQ ID NO:)	Protease polypeptide sequence (SEQ ID NO:)
PAS_chr4_0584 (YPS1-1)	9	10
PAS_chr3_1157 (YPS1-2)	11	12

[0142] To generate double knockouts, nourseothricin resistance was eliminated from the single protease knock-out strains produced above. A vector with homology arms targeting YPS1-2 (SEQ ID NO: 80) was used to transform the single protease knock-out strains. Transformants were plated on YPD agar plates supplemented with nourseothricin, and incubated for 48 hours at 30° C.

Example 5: ΔSec72 Improves MFa-18B Secretion Across Integration Copy Numbers

[0143] *Pichia pastoris* strains modified to express two, four, or six copies of 18B (Ab MaSp2 79 kDa) comprising the MFa1 (sc) pre-pro leader sequence were prepared using techniques described in Example 1.
[0144] Δsec72 strains of each of the above were prepared using techniques described in Example 2. Δsec72 and WT strains for each of the 2x, 4x, and 6x 18B expressing strains were prepared. Secretion of MFa-18B from WT and Δsec72

strains (i.e., . . . SEC72 KO strains) from each of the 2×, 4× and 6× 18B expressing strains was measured by ELISA, with the results shown in FIG. 6.

[0145] Referring to FIG. 6, “silk copies” are the number of expression cassettes for 18B (Ab MaSp2 79 kDa) using the MFa1 (sc) pre-pro leader sequence in each strain. The 18B is C-terminally tagged with a 3×FLAG epitope for ELISA detection. Error bars show standard error of the mean among n≥4 biological replicates. The results show that deletion of the sec72 gene improves MFa-18B secretion across integration copy numbers (i.e., 2×, 4×, and 6× silk copies).

Example 6: ΔSec72Improves Secretion from Non-MFa Signals

[0146] *P. pastoris* strains were modified according to Example 1 to express 4 different silk polypeptides each comprising a different secretion signal: PEP4 (sc), CPY+4 (sc), DAP2 (sc), and MFa1 (sc). Each complete leader sequence is a hybrid composed of the indicated signal peptide with the MFa1 (sc) propeptide (*S. cerevisiae* ortholog systematic name: YPL187W). (sc) indicates the signal peptide derives from *S. cerevisiae*. Strains recombinantly expressing polypeptides with PEP4 (sc), CPY+4 (sc), and DAP2 (sc) each comprise a single expression cassette (i.e., 1 silk copy). Strains recombinantly expressing MFa1 (sc) comprise 2 expression cassettes (i.e., 2 silk copies). Δsec72 strains of each of the above were prepared using techniques described in Example 2 to compare secretion from WT and Δsec72 strains. For the Δsec72strain comprising DAP2 (sc), the strain was additionally modified to knockout YPS1-1 and YPS1-2 proteases, as described in Example 4. Secretion of polypeptides with each secretion signal from WT and Δsec72 strains was measured by ELISA, with the results shown in FIG. 7.

[0147] The results indicate that the deletion of the sec72 gene improves secretion for polypeptides comprising non-MFa1 (sc) signal peptides. Error bars show standard error of the mean among n≥4 biological replicates.

TABLE 7A		
MFa1(sc) propeptide sequence		
	Sequence	SEQ ID NO:
MFa1 (sc) propeptide	gctccagtcacactacaacagaagatga	105
Nucleotide	aacggcacaaaattccggctgaagctgtca	
Encoding Sequence (5' to 3')	tcggttactcagatttagaaggggatttc	
	gatgttgctgttttgccattttccaacag	
	cacaaataacgggttattgtttataaata	
	ctactattgccagcattgctgctaaagaa	
	gaaggggtatctctcgagaaaagagaggc	
	tgaa	
MFa1 (sc) propeptide	APVNTTTEDETAQIPAEAVIGYSDLEGDF	106
Polypeptide Sequence	DVAVLPFSNSTNNGLLFINTTIIASIAAKE	
	EGVSLEKREAE	

TABLE 7B		
Silk Secretion Signals		
Secretion Signal	Sequence	SEQ ID NO:
PEP4 Nucleotide Encoding Sequence (5' to 3')	atgttcagcttgaagcattattgccatt ggccttgttgttggtcagcgccaaccaag ttgctgca	83
PEP4 Polypeptide Sequence	MFSLKALLPLALLLVSANQVAA	84
PRC1, mutant (i.e., CPY + 4) Nucleotide Encoding Sequence (5' to 3')	atgaaagcattcctgtgttactactttt actaggcctgtccactacactcgtaag ca	85
PRC1, mutant (i.e., CPY + 4) Polypeptide Sequence	MKAFLLLLLLLGLSTLAKA	86
DAP2 Nucleotide Encoding Sequence (5' to 3')	atggaaggtggcgaagaagaagttgagcg cattcctgatgaacttttcgatacaaaaa agaagcatttgtagataagctcataagg gtcggaataatccttgtagctcctgatatg gggcactgtttgttgctaaaaagtatt	87
DAP2 Polypeptide Sequence	MEGGEEEVERIPDELFDTKKHLLDKILR VGIILVLLIWGTVLLKSI	88
MF(alpha) 1, variant Nucleotide Encoding Sequence (5' to 3')	atgagatttccttcaatttttactgctgt tttattcgcagcatcctccgcattagct	89
MF(alpha) 1, variant Polypeptide Sequence	MRFPSIFTAVLFAASSALA	90

Example 7: ΔSec72 Improves Secretion of Longer Silks and Distinct Silk Sequences

[0148] *P. pastoris* strains were modified according to Example 1 to express long silk polypeptides Argioppe bruennichi (Ab) MaSp2 (106 kDa) (SEQ ID NO: 43), *Latrodectus hesperus* (Lh) MaSp1 (55 kDa) (SEQ ID NO: 44). Ab MaSp2 106 kDa (aka 24B) is a longer concatenmer of Ab MaSp2 79 kDa (18B). Lh MaSp1 55 kDa is distinct sequence from another class of spidroins.

[0149] Δsec72 strains of each were prepared using techniques described in Example 2 to compare secretion from WT and Δsec72 strains. The strains comprising Lh MaSp1 (55 kDa) were additionally modified to knockout YPS1-1 and YPS1-2 proteases, as described in Example 4. Secretion of the long silk polypeptides from WT and Δsec72 strains was measured by ELISA, with the results shown in FIG. 8. Error bars show standard error of the mean among n≥4 biological replicates.

Argioppe bruennichi MaSp2 protein amino acid
sequence (SEQ ID NO: 43):
MNWSIRLALLGFVVLSTQTVFAVGQAATPWENSQLAEDFINSFLRFIAQS

GAFSPNQLDDMSSIGDTLKTAEKMAQSRKSSKSLQALNMAFASSMAEI
AVAEQGLSLEAKTNAIANALASAFLETTGFVNQQFVSEIKSLIYMIAQA
SSNEISGSAAGGGSGGGSGGGYGGQASASASAAAAYGSAPQGAGG
PAPQGPSQQGPVSQGPYFGAAAAAAAGGYGPGAGQQGPGSQGPGSGGQ
QGPSSQGPSSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQG
PGSGGQQGPGGQGPYFGAAAAAAAVGGYGPAGQQGPGSQGPGSGGQQG
PGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSGGQQGPGGQGPY
PSAAAAAAAGGYGPGAGQQGPGSGGQQGPGGQGPYSGGQQGPGGAGQQG
PGGQGPYFGAAAAAAAGGYGPGAGQQGPGAGQQGPGSQGPGGQGPY
GPGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQG
PGSQGPGSGGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPG
AGQQGPGSQGPGSGGQQGPGGQGPYFGAAAAAAAVGGYGPAGQQGPGS
QGPGSGGQQGPGGQGPYFGSAAAAAAGGYGPGAGQQGPGSQGPGSGGQ
QGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSGGQQGPGGQGPYSGGQ
QGPAGAGQQGPGGQGPYFGAAAAAAAGGYGPGAGQQGPGAGQQGPGG
SQGPGGQGPYFGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAA
GGYGPAGQRSQGPGGQGPYFGAGQQGPGSQGPGSGGQQGPGGQGPYGP
SAAAAAAGPGAGRQPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGP
GARRQPGSQGPGSGGQQGPGGQGPYSGGQQGPGAGQQGPGGQGPYFG
AAAAAAAGGYGPGAGQQGPGAGQQGPGSQGPGGQGPYFGAGQQGPG
SQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSGG
QQGPGGQGPYFGAAAAAAAVGGYGPAGQQGPGSQGPGSGGQQGPGGQ
PYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAA
AAAAAGGYGPGAGQQGPGSGGQQGPGGQGPYSGGQQGPGAGQQGPGGQ
PYFGAAAAAAAGGYGPGAGQQGPGAGQQGPEGPGSQGPGSGGQQGP
GGQGPYFGAAAAAAAVGGYGPAGQQGPGSQGPGSGGQQGPGGQGPYGP
SAAAAAAGGYGPGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAA
AGGYGPGAGQQGPGSGGQQGPGGQGPYSGGQQGPGAGQQGPGGQGPYGP
GAAAAAAAGGYGPGAGQQGPGAGQQGPGSQGPGGQGPYFGAGQQGP
GSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSG
GQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGS
QGPSSGGQQGPGGQGPYFGAAAAAAAVGGYGPAGQQGPGSQGPGSGGQ
QGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSGGQQGPGGQGP
YGPSAAAAAAAGGYGPGAGQQGPGSGGQQGPGGQGPYSGGQQGPGAGQ
QGPGGQGPYFGAAAAAAAGGYGPGAGQQGPGAGQQGPGSQGPGGQGP
PYFGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQ
QGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSGGQ

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QGPGGQGPYSGGQQGPGAGQQGPGGQGPYGGYGPAGQQGPGSQGPGS
GGQQGPGGQGPYGPSAAAAAAAGPGARRQPGSQGPGSGGQQGPGGQGP
YSGGQQGPGAGQQGPGGQGPYFGAAAAAAAGGYGPGAGQQGPGGAG
QQGPGSQGPGGQGPYFGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAA
AAAAAGGYGPGAGQQGPGSQGPGSGGQQGPGGQGPYFGAAAAAAAVGGY
GPGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQG
PGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSGGQQG
PGGQGPYSGGQQGPGAGQQGPGGQGPYFGAAAAAAAGGYGPGAGQQG
GPGAGQQGPGSQGPGGQGPYFGAGQQGPGSQGPGSGGQQGPGGQGPY
PSAAAAAAAGPGAGRQPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAG
PGARRQPGSQGPGSGGQQGPGGQGPYSGGQQGPGAGQQGPGGQGPYGP
GAAAAAAAGGYGPGAGQQGPGAGQQGPGSQGPGGQGPYFGAGQQGP
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GQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSGGQQGPGGQGPYGS
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SQGPGSGGQQGPGGQGPYFGAAAAAAAVGGYGPAGQQGPGSQGPGSGG
QQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSGGQQGPGGQ
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QQGPGGQGPYFGAAAAAAAGGYGPGAGQQGPGAGQQGPGSQGPGGQ
GYPYFGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAG
QQGPGSQGPGSGGQQGPGGQGPYFGAAAAAAAVGGYGPAGQQGPGSQG
PGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSQGPGSGGQQG
PGGQGPYGPSAAAAAAAGGYGPGAGQQGPGSGGQQGPGGQGPYSGGQQG
PGAGQQGPGGQGPYFGAAAAAAAGGYGPGAGQQGPGAGQQGPGSQG
GPGGQGPYFGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGY
GPGAGQQGPGSQGPGSGGQQGPGGQGPYGPSAAAAAAAGGYGPGAGQQG
PSSQAPVASAAASRLSSPQASARVSSAVSTLVSSGPTSPAALSNAISSVV
SQVSASNPLSGCDVLVQALLEIVSALVHILGSSSIGQINYAASSQYAQM
VGNSSVAQALG

Latrodectus hesperus MaSp1 protein amino acid
sequence (SEQ ID NO: 44):
MTWSTRLALSFLFVLCTQSLYALAQANTPWSSKANADAFINSFISAASNT
GSFSDQMEDMSLIGNTLMAAMDNMGGRTIPSKLQALDMAFASSVAEIAA
SEGGDLGVTTNAIADALTSFYQTGTGVNRSFISEIRSLIGMFAQASAND
VYASAGSSGGGYGASSASASASAAAPSGVAYQAPAQISFTLRGQQP
VSYGQGGAGPGGAGAAAAAAAGGAGQGGQGGYGGYGGYGGAGQGGSG
AAAAAAAGGTGQGGAGQGGAGAAAAAAAGGAGQGGQGGYGGYGGYGGY
GGTGQGGAGAAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGGYGGYGGY
AAAGGAGQGGQGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGY

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GAGQGGAAAAAAGAGQGGYGGQGGAGQGGSGAAAAAAGGAGQGGQGG
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 AAAAAAGGAGQGGQGGYGGQGGYGGQGGAGQGGAGAAAAAAGGAGQGGQGG
 GYGQGGYGGQGGAGQGGAGAAAAAAGGAGQGGQGGYGGQGGYGGQGGAGQGG
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 AAGAGQGGYGGQGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGQGGAG
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 QGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGQGGAGQGGAGAAAAAA
 GGAGQGGQGGYGGYGGYGGQGGAGQGGAGAAAAAAGGAGQGGYGGYGGAGQGG
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 QGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGQGGAGQGGAGAAAAAA
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 GQGGAAAAAGAGQGGYGGQGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGG
 YGGQGGAGQGGAGTAAAAAAGGAGQGGQGGYGGYGGYGGQGGAGQGGAGAAA
 AAAAGGAGQGGYGGYGGAGQGGAGAAAAAAGAGQGGYGGQGGAGQGGAGAA
 AAAAAAGGAGQGGQGGYGGYGGYGGQGGAGQGGAGAAAAAAGGASQGGQGG
 GYGQGGYGGQGGAGQGGAGAAAAAAGGAGQGGYGGYGGAGQGGAGAAAAAGAGQ
 GGYGGQGGAGQGGAGAAAAAAGGAGAGQGGQGGYGGYGGYGGAGQGGAGAGA
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 RGAGQGGAGAAAAAAGAGQGGYGGQGGAGQGGAGAAAAAAGGAGQGG
 QGGYGGYGGYGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGG
 QGGAGQGGAGAAAAAAGGAGQGGYGGYGGAGQGGAGAAAAAGAGQGGYGGQGA
 GQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGGAGQGGAGAAAAAAG
 GAGQGGQGGYGGYGGYGGAGQGGAGAAAAAAGGAGQGGYGGYGGAGQGG
 AAAAAAAGSGQGGYGGYGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGG
 YGGQGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGGYGGAGQGG
 AAAAAAAGGAGQGGYGGYGGAGQGGAGAAAAAGAGQGGYGGYGGYGGYGGAG
 AAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGG
 QGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGG
 AAGAGQGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGGYGG

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QGGAGAAAAAAGGASQGGQGGYGGQGGYGGQGGAGQGGAGAAAAAAGGAG
 QGGYGGYGGAGQGGAGAAAAAGAGQGGYGGQGGAGQGGAGAAAAAAGGAGRG
 QGGYGGYGGYGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGG
 QGGAGAAAAAAGGAGQGGYGGYGGAGQGGAGAAAAAGAGQGGYGGYGGAGQGG
 GAGAAAAAAGGAGRGQGGYGGYGGYGGAGQGGAGAAAAAAGGAG
 QGGQGGYGGYGGYGGAGQGGAGAAAAAAGVGGAGQGGYGGYGGYGGYGGYGG
 AAAAAAGSGQGGYGGYGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGG
 QGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGGYGGYGGYGG
 AAGGAGQGGYGGYGGAGQGGAGAAAAATGAGQGGYGGYGGYGGYGGYGGYGG
 GGAGQGGQGGYGGYGGYGGYGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGG
 YGGQGGAGQGGAGAAAAAAGGAGQGGYGGYGGYGGYGGYGGYGGYGGYGGYGG
 YGGQGGAGQGGAGAAAAAAGGAGQGGQGGYGGYGGYGGYGGYGGYGGYGGYGG
 AGGAGQGGQGGYGGYGGYGGYGGAGQGGAGAAAAAAGGAGQGGYGGYGGYGG
 GGAGAAAAASGPGQIYYGPQSVAPAAAAASALAPATSARISSHASAL
 LSNPTNPASISNVISNAVSISSNPGASACDVLVQALLELVLTALLTII
 GSSNIGSVNYDSSGQYQVVTQSVQNAFA

Example 8: Secretion Improves with SEC72
 Deletion, but Neither Knockdown Nor
 Overexpression

[0150] *P. pastoris* strains were modified according to Example 1 to comprise 4 copies of DNA cassettes comprising recombinant genes expressing 18B silk-like polypeptide. Δsec72 strains were prepared using techniques described in Example 2. The strains were additionally modified to knockout YPS1-1 and YPS1-2 proteases, as described in Example 4.

[0151] From this strain, a strain overexpressing sec72 was prepared by transforming with a vector comprising a recombinant sec72 gene operably linked to a THI11 promoter (pTHI11). pTHI11 is de-repressed in minimal media that lacks the vitamin thiamine. From prior RNAseq and promoter fusion studies, pTHI11 is among the strongest promoters in block and tank minimal media, which lack thiamine.

[0152] Additionally, from the *P. pastoris* strain modified to express 4 copies of recombinant genes expressing 18B silk-like polypeptide, instead of knocking out sec72, expression of SEC72 was knocked down using DAmP (decreased abundance by mRNA perturbation), a transcriptional knockdown strategy that disrupts the 3' UTR of a gene with a marker cassette.

[0153] Secretion of the 18B silk-like polypeptides from WT, Δsec72, Δsec72 with recombinantly overexpressed sec72 (operably linked to pTHI11), and sec72 knockdown strains was measured by ELISA, with the results shown in FIG. 9A.

[0154] The Δsec72 phenotype reverts upon complementation with an inducible allele of sec72. Similarly, the knockdown of SEC72 did not measurably affect silk secretion as compared to WT. Thus, secretion improves with sec72 deletion, but not with knockdown nor overexpression of sec72.

Example 9: Other Disruptions to SEC63 Complex
Stoichiometry do not Affect Silk Secretion

[0155] *P. pastoris* strains were modified according to Example 1 to comprise 4 copies of DNA cassettes comprising recombinant genes expressing 18B silk-like polypeptide. Δ sec72 strains were prepared using techniques described in Example 2.

[0156] To test whether the Δ sec72 phenotype is a direct effect of removal of SEC72 or an indirect effect of disrupting the SEC63 complex stoichiometry, pTHI11 was used to overexpress the 3 non-SEC72 complex members (SEC62, SEC63, and SEC66) simultaneously (by transforming the strain with a vector comprising 3 concatenated expression cassettes with sec62, sec63, and sec66). In addition, pTHI11 was used to overexpress SEC72. Secretion of 18B was measured from each strain and compared with Δ sec72 strains using ELISA, with the results shown in FIG. 9B.

[0157] Neither stoichiometric change (overexpression of SEC62, SEC63, and SEC66, or overexpression of SEC72) measurably affected silk secretion. Error bars show standard error of the mean among $n \geq 3$ biological replicates.

Example 10: RTqPCR Shows Transcriptional
Adaptation to Δ Sec72, Including Translocon Levels

[0158] Δ sec72 strains were prepared using techniques described in Example 2. mRNA was isolated after growth and RTqPCR was performed on selected markers to analyze transcriptional adaptation to Δ sec72.

[0159] Reference normalized expression (ALG9 or ACT1) as a fold change from the “WT” strain RMs71 in minimal media is shown in FIG. 10A. Shading indicates range and center bar represents median among $n=3$ biological replicates (except $n=2$ for Δ sec72). SSH1 transcript levels increase 4-8 fold across all 3 perturbations. Δ sec72 leads to a 2-4 fold reduction in SEC61 translocon component expression.

[0160] FIG. 10B shows a projection of centered, log-scaled, reference normalized expression data onto principal component space, left=PC1 vs. 3, right=PC2 vs. 3. Replicates are colored by strain genotype, showing low within-genotype variance compared variance across genotypes. Examining the loading coefficients revealed that most variation along PC1 is driven by strong SSA3 repression and along PC3 by SEC61 translocon complex repression. These 3 PCs reconstruct 99% of the variance from the original 20 dimensions.

Example 11: Overexpression of SSH1 Improves
Growth Rate of the Δ Sec72 Silk Secretion Strain

[0161] Though SSH1 also functions in ER translocation, it is distinct from SEC61 in its dispensability for *S. cerevisiae* growth and lack of interaction with the SEC63 complex. SSH1 may show translocation substrate preferences, but they broadly overlap with those of SEC61.

[0162] To test whether overexpression of SSH1 could aid growth of Δ sec72 strains, *P. pastoris* Δ sec72 strains were prepared using techniques described in Example 2. The regulatory and coding sequences for the SEC61 and SSH1 complexes (SSS1, SBH2, and one of SEC61 or SSH1) were assembled by PCR onto an integrating plasmid. The resulting integration duplicates the gene copy number of one of the SEC61 or SSH1 complexes. *P. pastoris* (Δ sec72) cells were transformed with a vector for overexpression of the

SSH1 translocon complex as described in Example 3. Similarly, *P. pastoris* (Δ sec72) cells were transformed with a vector for overexpression of SEC61. Selected strains (as indicated by “AA” in FIG. 11) were additionally modified to knockout YPS1-1 and YPS1-2 proteases, as described in Example 4.

[0163] Pre-cultures in YPD were diluted 1:1600 into minimal media (RMm17). OD600 was recorded over 6 time-points spanning 8 hours of exponential growth beginning 19 hours after dilution. A linear model fit log (OD600) vs. time to estimate doubling rate and the standard error of the slope among $n=6$ biological replicates. The measured growth for each strain is shown in FIG. 11. Doubling times are shown as the reciprocal of doubling rate. Error bars are asymmetrical due to the reciprocal transformation of the estimates.

[0164] SEC61 and SSH1 duplication promote faster doubling of the SEC72 silk production strain (FIG. 11, left). This effect may be due to decreasing the load of intracellularly accumulated silk. However, Δ sec72 leads to a significant growth defect, which is partially restored by duplication of SSH1 but not SEC61 (FIG. 11, right).

Example 12: SSH1 Duplication Improves
Fermentation Performance of Δ Sec72 Strain

[0165] *P. pastoris* strains were modified according to Example 1 to comprise 4 copies or 6 copies of DNA cassettes comprising recombinant genes expressing 18B silk-like polypeptide. Δ sec72 strains were prepared from recombinant cells expressing 4 copies of the DNA cassette using techniques described in Example 2. Some of the resulting *P. pastoris* (Δ sec72) cells were also transformed with a vector for overexpression of the SSH1 translocon complex as described in Example 3. Secretion of 18B from each strain was measured using ELISA, with the results shown in FIG. 12A and FIG. 12B.

[0166] In FIG. 12A, each trajectory represents an independent batch-fed fermentation. Marked points are samples taken for analytes including the concentration of secreted silk protein in the broth (“18B titer”, y-axis). The x-axis is cumulative glucose fed, since the feed program is time-varying. Shading depicts the range of all runs of the same strain. The Δ sec72 deletion strain shows highly variable performance. When SSH1 is duplicated in this strain (“2N SSH1”), performance improves and variability decreases. The Δ sec72 2N SSH1 strain with only 4 silk expression cassettes secretes approaches the range of a reference strain with 6 silk expression cassettes.

[0167] Specific productivity (the ratio of product to biomass) was measured for each strain culture and shown in FIG. 12B. Δ sec72 backgrounds grow more slowly and generate less biomass, elevating their metrics above that of the higher-expressing, highest-titer strain. Due to high variability in the Δ sec72 strain, summarized data show only the top $n=2$ runs. Error bars are standard error of the mean.

Other Embodiments

[0168] It is to be understood that the words which have been used are words of description rather than limitation, and that changes may be made within the purview of the appended claims without departing from the true scope and spirit of the invention in its broader aspects.

[0169] While the present invention has been described at some length and with some particularity with respect to the

several described embodiments, it is not intended that it should be limited to any such particulars or embodiments or any particular embodiment, but it is to be construed with references to the appended claims so as to provide the broadest possible interpretation of such claims in view of the prior art and, therefore, to effectively encompass the intended scope of the invention.

[0170] All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including definitions, will control. In addition, section headings, the materials, methods, and examples are illustrative only and not intended to be limiting.

Sequences

TABLE 8

Forward (F) and Reverse (R) Primers for 5' and 3' homology arms (HA) targeting protease ORF		
Description	SEQ ID NO:	5' to 3' Sequence
KO_PAS_chr4_0584 5' HA F	45	TACTACAGGCTGGCTGTTCC
KO_PAS_chr4_0584 5' HA R	46	CTCACTTAATCTTCTGTACTCTGAAGAA GTCCAACCTGTTGAACGCC
KO_PAS_chr4_0584 3' HA F	47	AGAAGTTGATTGAGACTTTCAACGAGGG TCCCTTCAGCTACCTTT
KO_PAS_chr4_0584 3' HA R	48	TCCCTGCTAAGCCCTAATCG
KO_PAS_chr3_1157 5' HA F	49	CTCTGATTGCACGAGAAGGC

TABLE 8-continued

Forward (F) and Reverse (R) Primers for 5' and 3' homology arms (HA) targeting protease ORF		
Description	SEQ ID NO:	5' to 3' Sequence
KO_PAS_chr3_1157 5' HA R	50	CTCACTTAATCTTCTGTACTCTGAAGTG AAAGGCGATTGGAGTTGC
KO_PAS_chr3_1157 3' HA F	51	AGAAGTTGATTGAGACTTTCAACGAGCT GGCTCTGCTTCTGGTACT
KO_PAS_chr3_1157 3' HA R	52	GATGTTAGAGCGGGCATAAG

TABLE 9

Forward and reverse primers for amplifying modified sequences		
Description	SEQ ID NO:	Sequence (5' to 3')
KO_PAS_chr4_0584 Verification F	53	ACTTGTCAGGACGATACGGA
KO_PAS_chr4_0584 Verification R	54	CCGGTCTCCCTGGAAATAGA
KO_PAS_chr3_1157 Verification F	55	AGTTGTCCGTCATTAGCCCT
KO_PAS_chr3_1157 Verification R	56	TGTTCCCTTTTCGGCTAGACA

TABLE 10

sec72 KO vector		
Description	SEQ ID NO:	Sequence (5' to 3')
SEC72 KO full plasmid	91	ggcgccgctgtttaaacccctccaccagccatataccactacaacaccacagaagagaaagagctcatatcatccgctcatgagagagtac cagcacagaaatcacgtcaagtaaaactagtagcgaagcattacgtaataatagcaactttatgacaaatcattccattttttccac tggagcggtgcaactgcgtaaatcattctctttggaaggcaagggaagaacaacaaattttccctccggtatcaaaacattgaatcat gtctactgaaccacttttaaatgtgctcctgtcggtgatgggtaccggtaaagtaagtgcaaatattttgatgagtcggataatg ttttcgcgccccttagttccctcctcatgattactaacaattcatagaccaccttcgttaagagacaccttactggagaggttcgtaagaa gtacattgctactttgggagtcgaagttcatcccttgctcattccacactaactgtggtccctatcacattcaacggtttgggacactgct ggacagagaaggtttgggtggactgagagatgggttattacattaacgggtgactgtggtatcatcatggtcgacgttacatcgagaatta ctacaagaacggttccaaactggcaccgtgacttggtcagagtggtgagaaacattccaattgtgctttgtggttacaaggttgatgt caaggaaagaaaggtcaaggctaaagaccatcactttccacagaaagaagaacttgcaattcttgacattttctgccaaggtccaactac aactttgagaagccattcttggtggttagctagaaaagttgctggtgagcccccaattagagttcggtgctgctccgacttgcaagccc cagaggttcaaattgatgccgatttaataaagaagtagagcaagagaacgcccagggtgcccgtatgccattgacctgatgaagatga tgcgacttgtaagcttttacttacagtacattgagaaccatcatagggcagctatcgtaagtttagttgttctgtagtgaagctga gttgtttctgtagtgttgcaggtcgagagggatctctcagccttagacaaaaaaaggttgacacgttgatcacactctctg tttcatccgatctttcacctacgaggtcccactcctctcagagtagacagaagattaaagtgagagaattctaccggttcgtatagcataca ttatacgaagttatttcagtaaatgtctgtttctttgttgcaagtggtgagccattttgacttcgtgaaagttttctttagaagttgt tttccagagggccaaacattccaccgtagtaaaagtgcaagcgttaggaaggtccaagactggcataaatcaggtatgaagtcgagcact ggcaggtgatcttctgaaagtttctactagcagataagatccagtagtcatgcatatggcaacaatgtaccgtgtggaatcgaagacg cgtcctactaaccttcgcatctgttggtccagtttggttggtatcgatcaacgtgacaaggttgctgattcccggtgaagcatgcatacc caaggacgctgttgcaattccaagtgagccagttcccaaatctttgttaattatagagcacttcattgtgtgctgtgaaagttaa atgcgaaacaaattagagataatctcgaaacccgagacttcaaacgccaatgatggtgcccgcacacaataagcgttcataatccgctgg gtgactttctcgctttaaaaaattatccgaaaaaatttttgacggctagctcagtcctaggtacgctagcataaagaggagaaaaatg gctaaactgacctctgctgttcgggtctgacccgctcgtgacgttgctggtgctgttgagttctgggacgacccgtctgggtttctctc gtgacttcggttgaggacgacttcgctggtgtgtgctgacgacggttaccctgttcattctctgctgttcaggacacaggtttgctccgga caacaccctggctgggtttgggttcggtgtcggaagaactgtacgctgaatggtctgaagttgtttctaccaacttcggtgacgct ctggttcggtatgacggaatcggtgaacagccggtgggtcgtagttcgctctcggtgacccgggtgaactcggtcacttcg ttgctgaagaacaggactaacacgtccgacggcgccacgggtcccaggcctcgagatccgtcccccctttctctgtcgatatca

TABLE 10-continued

		sec72 KO vector
Descrip- tion	SEQ ID NO:	Sequence (5' to 3')
		tgttaattagttatgtcacgcttacattcacgccccccccacatccgctctaaccgaaaaggaggttagacaaacctgaagtcta gggtccctattttttttttatagttatgttagtattaagaacgttatttatatttcaaatttttctttttttctgtacagacgcgtg tacgcatgtaacattatactgaaaacctgtgtgagaaggttttgggacgctcgaaggctttaatttgcagctataaacttcgtatag catacattataacctgttatcgccgcccagatctaacatccaaagacgaaaggttgaaatgaaacctttttgacctccgacatccacag gagagttatagagctgctatgcgtgttaaaaatagtttaaatcttcgtaaagtagtttagtccatgtaatttgcgtatgaatcgatagc taatctggatgctgaacggatgcttactggcatgcatatttcattaccatctaaagctgcgcacaaacctgaatagtcagtgaggg aagcttccctgtaaacgtcctgtccctttagggaccatcgatccccacgatcaaatcgcgatcacatctatcaactgtcccttccat ctatctatgcaaggtaatgacagactctgttaactctgatgattctgatctggaatcatagaggtgactgagcctactccaaaagt gaccttttggcccccaatccagcatttaattttactgccccataagcaacagtaacggcacaactccaataaggagaaaacttgatg accaatccaactccaattcttttggcagactggaatcggtacgggaatcatcagtgaaaccacaagctagtagtcaatagtagtag gttcatcccccaagccgaccaattttccaataatcagaataatgaacttgataacaacaatggattcgccgactggatttctaaagtc caacctgaatttccctttccacttaatgatggacaaaaaagtcagcaatcaacctacaaaactcaaattttgaagagatcatcgatt taactgaagatcgagataaatacatctgtccccgcattcatcatcttaccctagttccctccagcacacagaatcagagcca tcataatagccaaacaacacagcacagatgcgcattcttccaagggaacgacctctccaatcatattcagatgatgaagacgaa gatttgaanaattttaggatccaattttgttcagcagcctctaggaattatgcccaggaaacttccaacgccccctgcaaacatactccatt ttgacggttcaaacagaaatgaacaagccagatggctggacttgcggataaaaagatttgttagataatcttcacaatcttcgagttca tgctcagtcgaatattatggagatcaataggttctattccactttggggcatttaaacagagaagtttaaacctgcaggcgctcgg tcgttcggctgcggcgagcggtatcagctcactcaaaaggcggttaacggttatccacagaatcaggggataacgcaggaagaaacat gtgagcaaaaagccagcaaaaagccaggaacgttaaaaagcccgcttgctggcgcttttccataggctccgccccctgacgagcat cacaaaaatcgacgctcaagtgcagaggtggcgaacccgacaggaactataagataccaggcgcttccccctggaagctccctcgtgc gctctcgtgtccgacctgcgccttaccggataacctgtccgccttctcccttcgggaagcgtggcgcttctcctatagctcacgctg taggtatctcagttcgggtgtaggtcgttcgtccaaagctgggctgtgtgcaacgaacccccgttcagcccgacgctgcgccttatcc ggtaactatcgtcttgagtcacacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggt atgtaggcggtgctacagagttcttgaagtgggtggcctaactacggctacactagaagaacagatttgggtatctgcgcctctgctgaa gccagttaccttcggaaaaagagtttgtagctcttgatccggcaaaaacacaccccgctggtagcggtgggtttttggtttgcaagcag cagattacgcgcagaaaaaaggatctcaagaagatcctttgatctttctacggggtctgacgctcagtggaacgaaaaactcacgtt aagggttttgggtcatgagattatcaaaaaggatcttcaactagatccttttaaatataaaatgaagttttaaatcaatctaaggtat atatgagtaaaacttggtctgacagttaccaatgcttaactcagtgaggcaccatctcagcgatctgctctattcogttcaacgttctg gcctgactccccctcgtgtagataaactacgatacgggaggggttaccatctggccccagtgctgcaatgatccgcgcgacccacgct cacggctccagatttatcagcaataaacacagccagccggaagggccgagcagagaagtggtcctgcaactttatccgcctccatcca gtctatttaattgttgcgggaagctagagtaagtagtttgcgcagtttaaatagtttgcgcaacggttggtaggaatagtttcccgccct gtgtcacgctcgtcgtttggtatggcttcattcagctccgggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaa aagcggttagctccttcggtcctccgatcgttgtcagaagtaagtggccgcagtggttatcactcatggttatggcagcactgcataa ttccttcaacagcggtttctgggtgagcaaaaacaggaagcaaaatgcccgaaaaagggaataaggggcgacagaaatggtgaata ctcatactcttcttttcaatattatgaagcatttatcagggttatttgtctcatgagcggatcacatatttgaatgtatttagaaaa ataaaacaatagggttcgcgcacatttccccgaaaagtgcacct
SEC72 5' homology arm	92	aaacctccaccagccatataccactacaacaccacagaagaaagagctcatatcatccgtcatgagagagtagccagcacagaat acagtcagtagtaaaactagtagcaagcattacgttaataatagcaactttatgacaaatcattccattttttccactggagcgtgcac tgcgtaaatcattctctttggaaggcaggggaagaacaacaaaattttctcctcgttatacaacattgaatcatgcttactgaacc caacttttaaatggctccttgctcggtgatgggtgacggtaagtgcaaatatttgatgagtcggatgaatgtttcccgccct tagttccctcatgattactaacaattcatagaccacctcgttaagagacacctactggagagttccgtgaagaagtagatgtctac tttgggagtcgaagttcatcccttgtcattccacactaactgtggtcctatcacattcaacggttgggacactgctggacaagagaag tttgggtgagtagagatggttattacattaacggtgactgtggtatcatcatggtcgacgttacatcgagaattacttacaagaacg ttccaaactggcaccgtgacttggtcagagtggtgagaacattccaatttgtcttttgtggttaacaaggttgatgtcaaggaaagaaa gggtcaaggctaaagaccatcatttccacagaagaagaacttgcaatcatttgacatttctgccaagtccaactacaactttgagaag ccattcttgtggttagctagaaagtgtctggtgagcccccaattagagttcgttgctgctccgacttgcaagccccagaggttcaaa ttgatgccgatttaataaagaagtacgagcaagagaacccgaggtgcgcgtatgccattgacctgatgaagtagatgccgacttgta agcttttacttacagtagacttgaagaacatcatagggcagctatcgtaagtttagttgtttgctgatgaagtagttgtttctgt agtggttcgaggtgcagagggatctctcagcttagacaaaaaaagggttgacagcttgatcacctctctgtttcatccgac tttcaactacagag
SEC72 5' homology arm	93	ggagagttatagagctgctatgcgtgtaaaaatagtttaaatcttcgtaaaagtatgttagtccatgtaatttgcgtatgaatcgatacg ctaactctggatgctgaacggatgcttactggcatgattattcattaccatctaagctgcgcccacacccagtaaatgcaagtggg gaagcttccctgtaacgctcctgtccctttagggacacatcgatccccacgatcaaatcgcgatcacatcaactgtccctttcca tctatctatgcaaggtaaatgacagactctgttaactctgatgattctgatctggaaatcatagaggtgactgagcctactccaaaagt ggaccttttggcccccaatccagcatttaattttactgcccccataaagcaacagtaacggcacaactccaataaggagaaaaacttgat gaccaatccaaactccaattcttttgcagactggaatcgttacgggaatcatcagtgaaaccacaagctagtagcttaaatagtagta ggttcatcccccaagccgaccaattttccaataatcagaataatgaacttgataacaacaatggattcgccgactggatttctaagtc ccaactcgaatttccctttccacttaatgatggacaaaaaagttccagcaatcaacctacaaactcaaattttgaagagatcatcgat ttaactgaagatcgcagataaatacatctgtccccgcattcatcatcttctaccocagttccctccagcacacagatcagagcc atcatatagccaacaacaacacagcacagatgcgcattcttccaagggaagcagccttccaatcatatcagatgatgaagacga agatttgcgaattttaggatccaatatttgttcagcagcctctaggaattatgccaggaaacttccaacgccccgcaaacatactccat tttgacgggttcaaacagaaatgaacaagccagatggctggacttgcggataaaaagatttgtagataatcttccaactcttcaggttc atgctcagtcgaatattatggagatcaataggttcatttccactttggggcatttaaacagagaagttt

TABLE 10-continued

sec72 KO vector		
Description	SEQ ID NO:	Sequence (5' to 3')
Yeast selection marker (Sh ble) for SEC72 KO (nucleotide sequence)	94	ttcagtaatgtcttgtttcttttgggtgcagtggtgagccatttgcattcgtgaaagtttctttagaatagttgttccagaggccaa acattccaccctgtagtaagtgcagcgttaggaagtccaagactggcataaatcaggtataagtgtcgagcactggcaggtgatcttc tgaaggtttctactagcagataagatccagtagtcatgcataatggcaacaatgtaccgttggtatctaagaacgcgtcctactaacct tcgcatcgttgggtccagttgttgggtatcgatcaacgtgacaaggttgatcgattccgcgtgaagcatgcatacccaaggacgctgtt gcaattccaagtgcagcagttccaacaatcttggtaataatagagcacttcattgtgtgctgtgaaagtaaaatgcgaacaataa agagataatctcgaacccgcagcttcaaacgccaatatgatgtgcccgcacacaataagcgttcataatccgctgggtgactttctcgct ttaaaaaattatccgaaaaaatttttgacggctagctcagtcctaggtacgctagcattaaagaggagaaaaatggctaaactgacctc tgctgttccggttctgaccgctcgtgacgttgcgtggtgctgttgagtcttggaaccgacgctcgggtttctctcgtgacttcgttgag gacgacttcgctggtgttgcgtgacgacgttacccgttcctctcgtgttcaggaccaggttgcgtccggacaacacccctggctt ggggttgggttcggtggtcgtggacgaactgtacgctgaatgggtcgtgaagtgttctaccacactccgtgacgcttctggtccggtat gaccgaaatcgtgaacagcgtgggtcgtgagttcgctcgtgacccggtcgtgaactcgttcacttcgtgtcgaagaacag gactaacacgtccgacggcgccacgggtcccaggcctcggagatccgtcccccctttctcttgcgtatcatgttaattagttatg tcacgcttacattcacgccctccccacatccgctcaaccgaaaaggaggttagacaacctgaagtctaggtccctatttatt tttttatagttatgttagtattaagaacgttatttatatttcaaatttttctttttttctgtacagacgcgtgtacgcgtgtaacat tatactgaaaaccttgcttgagaaggtttgggacgctcgaaggcttaatttgcaagct
Yeast selection marker (Sh ble) for SEC72 KO (protein sequence)	95	MAKLTSAVPLTARDVAGAVEFWTDLRGLFSRDFVEDDFAGVVRDDVTLFISAVQDQVVPDNTLAWVVRGLDELYAEWSEVVSTNFRD ASGPAMTEIGEQPWGREFALRDPAGNCVHFVAEEQD

TABLE 11

SSH1 Complex Overexpression vector		
Description	SEQ ID NO:	Sequence (5' to 3')
Full Plasmid	96	ccagccaggacagaaatgcctcgacttcgctgctgcccaaggttgccgggtgacgcacaccgtggaacggatgaaggcacgaacc cagtgagacataagcctgttcggttcgttaagctgtaatgcaagtagcgtatgcgctcacgcaactggccagaaccttgaccgaacg cagcgggtggtaacggcgagtgccggttttcatggctgttatgactgttttttgggggtacagttctatgcctcgggcacccaagc agcaagcgcgttacgcccgtgggtcgatgtttgatgttatggagcagcaacgatgttacgcagcagggcagtcgcccataaaacaaag taaacatcatgagggaagcgggtgatcgccgaagtatcgactcaactatcagaggtagttggcgtcatcgagcgcctatctcgaacc gacgttgcctggcgttacatttgcggctccgcagtggtggcgccgtgaagccacacagtgatattgatattgctgggttacgggtga ccgtaaggcttgatgaaacaacgcggcgagctttgatcaacgaccttttggaaccttcggctcccccggagagagcgagattctc cgcgctgtagaagtaccattgttgtgcacgacgacatcattccgtggcgttatccagctaacgcgcaactgcaatttgagagaatg gcagcgaatgacattcttcaggtatcttcgagccagccagatcgacattgatctggctatcttgcgtgacaaaagcaagagaaac atagcgttgccctggtaggtccagcggcgaggaactctttgatccgggttcctgaacaggatctatttgaggcgctcaaatgaaacc ttaacgctatggaactcgcccccagctgggctgggtgatgagcgaatgtagtgttacctgttccgcatttggtacagcgcagtc aaccggcaaaatcgccgcgaaggatgtcgctgccgactgggcaatggagcgcctgcggccagctatcagcccgctcatactgaaag ctgacagcgcttatcttggaacaagaagaatcgcttggcctcgccgcagatcagttggaagaatttgcactacgtgaaaggc gagatcaccaaggtagtcggcaaatataactactcgcattccattctcaggctgtctcgtctcgtctccaaactttctgcgcccaatc tcccttccattccaaatctctacaacaaggcaggttcgtccgttgatgtgaagtgtgcaaaagccaaacacgaaaaacacatt gtcaaaatggtagttgaaggcataaaggcaaggcaagctaaactttgtcataacgctcaaaagtgtctcttttttatggcaaaag ggatgttctttatcacatagcagtggaatgacatcttcattcatcattcgtatcatcaataccatcgagttctgaaatccaaaggt ttctgtgttgggataaggcatcagattctttgagcaggtcgctccttggcgtcctcctctagatgaaatttgccatcgttaggagg tgagttatgggaaacgttgacttgttgcctcaatccaaggatcaacgttgaccatcggattgcggttctgagagacagacaggcat tataccgtgggtgacccggtttacgcgcagaggatgttgcgtggtgttgttgacagaagacgcataccagacgctcgggatgac atcgacgaatgaaagggcgagcttgcataccatttgcgtgatccagaagacttgggttccagtcattgtctcaggtccat attgatcaatgtgttcaactgtttcaatctgggtccggagggtggctattacacaagctgtggctactatctaaagtgggagaagt aacggaacacatgatgagacacaggaattacagggcgtgcattccccaataacaattagtcagatttcaaccaattacgtaagc gctcaaccccttttgcgaacgctatcgagcaaaagtcagggtgaaaccttcattccattatccaaagtcgacgcgaagctttaaca acatccataatggtaagtgtccagtttgatgagtgagcaatgggtccaaagttttagaccagttactaatatttaaagttcacagca attccaggaggacagagaacgttagctaaaagaagacgacgaacttggaataagaacaggatgaaccaacctccgcccagatctgc cgggtgctggagggtctctcgtctacatgctaaagtgttacacagacgagggcccaaggtttgaaagtgtatcctttaaattgtctctg tcttctgtgtgggtttcatcttcaagtgtcatttgggttgcacgttgggtgcaagctgacaggaaggttgatcaactaaagacctat taaacaggtttcatcatatctgtactatatttacaagtcactgcgtttagggtatataactaaagacattcaagaagcacatccac aacttgcgaagtcctgtcaaatgtactagatgcttttcagaacatcctgcggtttgaggagatctcgaatttccagtcaccaag tcttctcttctgtagaggtctttagttcttggtaagctgaattgggggttcttacctcaatttctatttagtgggaaatgcttccc acaattatttgcaatgggatcccggaactttacttgccttcaacttatgtccatactgaactttccgtcacgggtgtgcacttg aacgtcgaaatgagctcagttattcgggtgacagtgctagattcttttacttgatgttttattcgaaagtgcggtgatttgatgat gtgcgataatttggccagggtgggggttcgtgagcggaaacacaaacgcagaatccaagcagacaaagtgaggagtggaacaattc ttgatgttaattcgttaagatccttaacaaatcttagcaaaactgaaagtgaagactacgatagggttacttgaaaaatatat

TABLE 11-continued

SSH1 Complex Overexpression vector	
Descrip- tion	SEQ ID NO: Sequence (5' to 3')
	tcaagttttgtcagaatacccaactccatattaaacgatattattgtccctagaactattacgtaccacttgtgtgataattatactg atgagttggaagggttatgttttagtgggcttctggatttgaagaagaagaagattacgaggaagaagatgaagtcctcggtcct atcgaaaaaaaccagagatactgataattcagacgatgaggcctctgataccgatccagaacagagtgacgaagaagacgaagg tgagaataactgaaacagagtcagaagaggaaccgatcaaaactctctgcggaagaagaacggctctgtggaaacaagaatggaga ttatccatgaaactcctatcgacaaattactctcacttttgtctcattgaagaagatacctcaaataaaccattaaaatgaaa atccaagaagcagttcttctgctgacaaagactcggcaaatggaaagttaaatcgtaccgaaagcctaaacccaaacccaaacccctc tcagggtgctacagaacagttatcgaaacagatcaaatataaaagggttaagcagtcacaggtgaagacgcgatgaactcaag acgaagacgaagatgacgatgacgaggtcatctcggagagtgaaagctgataaactctgatgaagaagaagatgaagaatggaatggc tttggaaactatctagaatcatatgtaaaaccatcatcaggcaataaacaatttctcgtctatttgcaccccaacacctcgaccgacg tggtcagttcgaccctttacctacagtgcttttaactcccatttggatctcctaaataacccacacaaatgtcccaaaagtcacc gacgtccctctggaatttgttaaggaaggttccaaattcatctctaaatgtactaaacccctctcagaaggagtagcttaaatgag aagagctgttggagttgggttttaaatgatggcggtgggtgggtacgttgcaggtcatttcatattccaatcagatatttgattg tttaaaagttttagtttgaatacaatgtgtatgcttaattatattcacttcgtttcattgatttttctgtatccctgttgtgctga atcatctctatcgtgatcctctcaagtgcacctcaatagaagacaacttatggaggtgtagtaccatcatcagttctgacgtt tctagtttctgtgtatgctgcataccaatttcagttcttctcgtagtggtttgagcataggtctgtttaccgctctctttacactc tgatagagcaccttatcaaacctcttatcacagctgaaaggagaaggcggtgggtgggttcttctgtcagaatcgccccagaaactg tggatcctgctgttgcagaatgaaagcagaaggaagatggaaagagagaagcaagctcttaagagggtcagcaagccaggaa gctcgggaaaggcgaagaattagtcagtcacccagataggtttgagaccaacagctaaagctcccagcactgaaggtcctcaatc gcactggaaacatcaaggtcgatgaccgcactaacgaaggagctaaactattcgaaaaggagattcctacattacagaattaga gggtgatgtcgaaggaatgaaattcatatcaaggcgagggtactggtagcgtactaccggtacgattaaagcaaaagtacatct gtacaacaggtgaccttctgttccgtgggctactctggtagcactttgtcttatggagttcaatgtttgtctaaatacccttcg cacattaaagacttttcaaaagtgcaatgctgagggtctatctcaggagagaaacatattcttcgaaggagatgggtgtgataa gactagggtatgggtcacgtatgaaagaggatccatctacaatagagtaactttaactggtgaaacttcaaaaggacgggtcaca tcccttagaagaatgttgcctttcaatgccaccatccatctgtacattttgccagacacagttacaatgggtatcagagttgag tttaaccaagcttatgacatagagggtgtcaccgaaagtgggttacaataatgttcacagatgaatcgctccctggcaggatcagc tgccgtccatcccacgttaccatcatatcacttatcatccaagctgtccaagatcgtgatgagagaagggtatcacatgtgtt tgggtgaagtggttaaggcgtggatttggatacttaccatgactgacctctgcagcaaatagtaagcaaacacgcaaggtctc tgaacgggtctttgagctgtctgtgtcatcaaacatcttcatctgtatgttattgtttcttttcatcactaggatccatcct tctggatgggtgtcttgcacttatttagccactccaaaatgccgggtctgggagatgaccgcagcaccattgacaccatatacaga gagtagtcagagactcctctggtttccagctcttgcattgggaatttcataaaatgtctcctatcttacccttggtccatcagttgtt ttccaccctcatctgggtcgacaaatgggttaactgtttccaaataatcaaccttatcacaaactttatcgagaaatctctgtca cgccggcgaataccggcaccaaagtctataccatatttgattttgtcaggatcgttagagaaacgactctttaactttcttaaaag gtcatcgatcccaacaacatctgcttttggaaaccgaagtgtctcaccgtaccaccacaaactcctgcagacagatgctgggacgt gttccagatttcaggggcatcgtcatagttgtatgacaaagtgtccactgtttagggtcatcccatatttgctcactgaggtggtta aaatgggatacagtgatataatgaagggaatgggttaagagcctacctgagaaataaaattattatgcgccatccgacatccag aaaaattgatgaaagattggctattgttgacggttcttgatccccaaaaaaacaaagaaatgctgccgtcctagttttgtc ttcaaaagattgggttcgtgctatgccattcccaaccacaaagagctgtcccatccattaaagttgtgctgaggtatattgttgca atccagtgctgtgattacctccaacatcgacgcggaatttcgcatgggtgggaaacccaaattcttcggtcgtccatcaaacctc gaagtcatttcaacaccaaactcaacagctatcatagaaaaatattggcggggttgcggttttttagacattgcaagaccatttgtca gctggatcccggaagttgaacttcttatgaaaactgggggttcgatgaaaagctgatttactcattttcactgtgccatctcat ttgatctctgctccctgcctatatacgggtgtcaaatcctctgaagtctggaccacagttcccatattgctgtctgctttagggagtga gaagggaaacattgctggagcttgggttactgcctgtgattactctggcatttatcttgcagttgttggctgggtggaaagttttca aagtaaaactttgatctgggttagtgacagaatattgttccaaactttgcaaaagatcacttcagtcgttcttgatgatttaattg gttctctcacattttgtgactactttactccaggtgtgtccactgataacgtcttgggttcccaatttctgacatcttcaagat agtggtgtgtcaacttcttgggttactctactcgttgaagtcattgacaaggattacggattttcttcaggagctctattgttggctg cggtttatctcgccaccaaactcgtttttggcagcagttgggtcttagcaccgtcaacacccctccagatcgaaagaaatcttctg ctgatccaattattccgcaatttgagctctaaaccaaatttggtgttgccatatactgactcctctctcagagtaaaccttctcaact gactcaattttatctggggatggccatttttgggtttgtctgttcttgaataatgcaagatcggaagtaaccaatgaagccaaaca aggttcgtgccatggcctcagctacccaatcaagctactttccaatgggtcttctggccacttctgtcagctcgtctgtgtac aacttgaaccttatttggtttcttctgtcttcaagcttaccacacttttctcttttagggaacttcaagtggaaccattcgccaacaa ctactacgaaatcacatctggactgctgtattttatgactcctactttcaacgctgaagctggacttttaccacatgttgcctaagc cattgttttctattgccttctatgttgggttagcactttcttctgctagatcgtgggtccaaacttaacgggtgctgaggaaggac attgccaagtttttcaaggctcaaggaatctcatgttaggaaaaagagatgcctctgtgtctcaagagtttaacaccctagttcc tgttgcctctgcctctggagctttccatattgtcttttccagttgcgctgctgagttattgggtggctctgggtgtccaaacctca tcggaactcgtcttttgagttgttggctattttggaaactgttttgcaagaatggcaacagctcgggtgcgcatgaaagcgcattcc caatacttccagacttcttaggtttagaatccttgaagactatccagacattcaccgcagcctcaatttaccttctacatacatca acatactctatagaggagagttccattgctcgtactgaacccacacacactctctttataccctacaaactcttctgtccaaactca atggcgtcatctcgtgtcggtatagacaaatgggttaacccagtcctatttcaagttgtcttttctgatatccatgatttaattggg cattatttgaagttcgaaactgttttctggcactttagctttgatgatcgtttgatataatttcatccttggagttatcacagtagtg gctttctctccaggttggtatcgtaaaacctgggaaggattgtgtcgaagacccaaactccctttacaacctcactcaagtcggttag agggcgcgccgcacatgaagctgtacatggaaagcaggtgaataaacacacacttcaaatgcaccagcgcaggggttaagccg tatgaagccacccaaacgatgcgtatcaaaagttgttgagggtggccggttgccgtttgctgtcgacatttttagcgacgagcttat gtatggctctcgtacgtttatcaagtaaccgaagggtattccggactttttcaaaacattctttccagaggggtttcacctggggagc gcgtgactcgtcgaagatggcggtcgtgacgcgaacgcaggatacctcctggaaagatggctgttcggtctaccaggtctcag gtccgtggtgtcaatttcccgagcaatggtccggttatgcagaagaaaacccctgggttgggaacccgaacacccagagattgtgatcc tgcagatgggtggcctggaaggtcgacgcagatggcatgaaactgggtgggtggcgccatctgagctgtgagctcgtgaccacgt atcgttcgaagaacacgggtcgtgaacatacaaaatgccgggtattcacgggttgaccacgctcgtgtgcgcatgaaagacgcga aaagagacttacgtggagcaacatgaagtgcggttgcgaatttgctgggtttggcggtgggtatggagcaactgtacagttcctt atcatctggcgcaatcggaaccacaaagagcactgggttccgttttcatctccaggaagagtttcagtaagtgtcttcttcttgggt gcaggttgagccattttgacttcgtgaaagttgttttagaataagttgtttccagagggccaaacattccacccgtgtaaaagtga agcgtaggaagacaaagactggcataaatcaggtataagtgtagcactggcaggtgatctctgaaagtttctactagcagata

TABLE 11-continued

SSH1 Complex Overexpression vector	
Descrip- tion	SEQ ID NO: Sequence (5' to 3')
	agatccagtagtcatgcatatggcaacaatgtaccgtgtggatctaagaacgcgtcctactaaccttcgcattcgttgggtccaggt tgttgttatcgatcaacgtgacaaggttgcgattccgcgttaagcatgcataccgaaggacgcgtgtgcaattccaagtgcgca gttccaacaatcttgttaattatagagcacttcatttgttgcgcttgaaagtaaaatgcgaacaaatgaagataatctcgaaa ccgcgacttcaaacgccaatatgatgtgcggcacacaataagcgttcatatccgctgggtgactttctcgtttaaaaaatatcc gaaaaaatttttgacggctagctcagtcctaggtacgctagcataaaggaggagaaaaatgactactcttgatgacacgcctacag atataggacatcagttccgggtgacgcagaggtatcgaaagccttgagcgttcatcactactgatacgggtgttagagtcacccg ctacaggtgatggcttcaccttgagagaggttctgttagaccaccccttaacgaaagtttccctgatgacgaatcggatgacgag tctgatgctggtgaggacggtgacctgattccgaacatttgcgcatacggagatgatggtagacctggctggcttgttgggt gtctacagcggatggaatcgtagactcacagttgaggacatcgaaagttgcacctgaacatcgtgggtcacgggtgtgggtcgtgcac tgatgggactggcaacagagtttgcctagagaaaggagcgagacatttgggttagaagtgaccaatgtcaacgctccgctgatt cacgcataataggcaatgggttccacttggcggcttgatactgcttggatgacggaactgctctgatgggtgaacaagctct ttacatgagtagtccatgtccatagcagctccgacggcgccacgggtccaggcctcgagatccgctccccctttcccttggc gcatcatgttaattagttatgtcacgcttacattcacgccccccccacatccgctcctaacgaaaagggaaggagtagacaacc tgaagttcaggtccctattttatgtttagtttagtatttaagaacgttatttatatttcaaattttctttttttctgt acagacgcgtgtacgcatgtaacattatactgaaaccccttgccttgagaaggttttgggacgctcgaaggctttaatttgaacgtc cgaataaactctcgtagcatacattataccttgttattacagcggcgcaaatattttatctgattaaagatgactctcttgag atcgttttggctcgcgctaattctctgtctgaaacgaaaaaacgccttgacggcggttttctgaaggttctctgagctacc aactcttgaaccgaggttaactggcttggaggagcgcagtcacccaaacttgccttccagttagccttaacggcgcatgactt caagactaacctcctaaatcaattaccagtggtcgtgcagtggtgcttttgcctgcttccgggtggactcaagacgtag ttaccggataaaggcgcagcgtcgagtgaaacgggggttctgtcgcacagctccagcttgagcgaactgctaccggaaactgag tgtcaggcgtggaatgagacaaaacgcggccataaacgcgggaatgacacgggttaaacgaaaggcaggaacaggagagcgcacgagg gagccgccagggggaaacgcctgggtattttatagtcctgtcgggttgcgccaccactgatttgagcgtcagatttctgtgatgctt gtcagggggggcggagcctatggaaaaacggcttggcggcgccctctcacttccctgttaagtatctctggcatctccaggaa atctccgccccgttcgtaagccatttccgctcgcgcagtcgaacgacggagcgtagcaggtcagtgagcgggaagcgggaatata tccgtgtatcacatattctgtgacgcacgggtgcagccttttctcctgccacatgaagcacttactgacacccctcatcagtg caacatagtaagccagtagtactccgctagcgtgatgtccggcggtgcgacgtc
SBH2 Promoter	97 caactttctgcgccaatctcctttccattcaaatctctacaacaaggcaggtctcgtccgttgatgtgaagtttgcaaaagcca aaccacgaaaaacacccattgtcaaatggtagttgaaggcataaggcaaggcagctaaactttgtcataacgtctaaaagttgc tctttttttatggcaaaagggatgtctttatcacatagcagtggaatgacatcttcatcatcattcgtatcatcaataccatc gagttctgaatccaaaggtttctgttgggtgaagggcatcagattctttagcaggtcgtccttggtcctcctctagtgaa atttgccatcgttaggaggtgagttatgggaaacgttgacttgttgctcaatccaaggatcaacgttgaccatcggattgcgcttc tgaggagacagccaggcattataccgtggttgacggcggtttacgcgcagaggatgttgcgtgggtgttggtagacaagacgata ccagagagctcggatgacatcgacgaaattgaaggcgacgttgcataccatttgggtgctgatccaagaaacttctcgtgttcag tcaattgtctcaggtccatattgatcaatgtgttcaactgtttcaatctgggtccggagggtggctattacgacaagctgtggcta ctatctaagtgggagaagtaacggaacacattgatgagacacaggaattacagggcggtgcatccaccaataacaattagtcgagat ttcaaccaattacgtaagcgtcacaaccccttttctgaacacgtatcgagcaaaagtcagggtgaaaccttcatccattatataccaa gtcgaccgaagctttaacaacatccata
SSH1 Promoter	98 agacaagtggagagtggaacaaattcttgatgttaattcgttaagatccttaacaaactcttagcaaaactgaaaagtgaagactacg atagggtattacttgaaaaatataattcaagtttgtgcagaatacccaactccatataacgatattattgtccctagaactattacg taccacttgtgtgatatttatactgatgagttggaaaaggttatgttttagtgggttccctggatttgaagaagaagaagattacga ggaagaagatgaagtcctcggctcctatcgaaaaaaaacacagagatactgataattcagacgatgagggcctctgatccgatccag aaacgagtgacgaagaagacgaaggtgagaatactgaaaacgagtcagaagaggaaacggatcaaaactctctgcggaagaagaacg gctctgtggaaaaacaaagatggagattatccatgaaactcctatcgacaaattactctcaccttttgcctcattgaagaagaagata ctcaaataaaacattaaaattgaaaattcaagaagcagttcttgctgatccaagactcggcaaatggaagaaactcaatcgtacccgaa agcctaaaacaaaacccaaacctcttcaggtgctacagaaacagttatacgaacagatcaaatataaaaagggtaaaagcagtcaca ggtgaagacgacgatgaactcaagacgaagacgaagatgacgatgacgaggtcatctcggagagtgaaagctgataaactcgtatga agaagaagatgaagaatggaatggcttggaaactatctagaatacatatgtaaaacacatcagggcaataacaatttctcgtcatt ttgcatcccaacacctcgaccgacgtgttcagttgcagccctttacctacagtgctctttaactcccatttggatctcctaataaac tacaaa
SSH1 Promoter	99 tctgaacgggtctttgagctgtctgtgtcatcaaacatatcttcatctgtagttgtattgtttcttttatcactaggagtcacatc ctctcggatggctgtcttgcacttatttagccactccaaaaatgccgggtctgggagatgaccgcagccatttgacaccatatca gagagtagcgactcctctgggttccagttctgcagtggaatttcataaatgtctcctatcttcaacttccatcagtggtttgt agttccaccctcatctggtcgacaaatgggttaactgggtccaataaatcaaccttatcacaacatttatgcagaaaaatctctgt cagcgggccaataccggcaccaaaagctataccatatttgattttgtcaggatcggttagagaacgactctttaactttcttaaaa aggtctcgatccacaacacatctgcttttggaaacggaattgtctcacgctacccacccaaaactccgctgcacagatcgtggagc ctggtccagttattcagggtcctgcatagttgatcaaacgtgtccacctgtttagggctcatcccatattgttctgtcattgtggt aaaaatgggatacagtgatatttgaaggggaatgggtataagagctacactgagaataaaaattattatgcccattccgacatcc agaaaaattgatgaagattggctattgtgacggttcttgatcccaaaaaaaacaaagaaatgctgcgctcctagttttt gcttcaagaatgggttctgtgctatgccattcccaacccaaagagctgtcccatccattaaagtgtgtgctgactgatattgttgca caatccagtgctgattacctccaacatcgcacgcgaatttcgcctgggtgggaaacccaaattctcgtgctgctccatcaaac ctgaagtcatttcaacacaaactcaacagctatcatagaaaaat
SBH2 Terminator	100 gacctatatttaaacaggtttcatcatatctgtactatatttacaagtcactcgttttaggtatataactaaagacattcaagaag cacatccacaacttgtgcaagtcctgtcaaatgtactagatgcttttcagaacatcctcgggtttagaggatctcctgaatttccc agtcccaagcttttctctgtagaggtctttaggttctctgtgaatgctgaattgggggttcttacctcaatttctattagtgggaaa tgctttcccacaattatttgaatggggtcccggaactttactttgcttcaacttatgtcccatcactgaactttccgtcacgggt gtcaactgaaagtcgaatgagtcagtttccggtagacgtgtctagattcttttacttgatgttttatttcgaagtagcgtga ttgatattggtgcgataatttgtggccagg

TABLE 11-continued

SSH1 Complex Overexpression vector			
Description	SEQ ID NO:	Sequence (5' to 3')	
SSS1 Terminator	101	aagtttaggtttgaatacaatgtgtatgcttaattatattcacttcggtttcattgatttttgctatccctgtgtgctggttaaatcatctctatcgtgatcctctcaagttgcacctcaaatagaagacaacttatggaggtgtactaccaatatacagtccttgacgtttctagtttcgtgtgatgtgcataccaatttcagttcttcctgtagtgttttgagcataggcttgtttaccctgtctctttacactctgatatgagcaccttatcaaacctcttatcacagctgaaaggagaaggcggtggtggttctctggtcagaatcgccccagaaactgtggatcctgctgttgcaagattgaaagcagaaaggaagatggaaagagagaagcaagctcttaagaggctcagcaagcccaggaagctcggaaaggcgaagaattagtcacgtaccagataggtttgagaccaaccagtaaaagctccca	
SSH1 Terminator	102	gtttagaatccttgaagactatccagacattcccccgcacctcaatttaccttctacatacatcacatattctatagaggagagtccattgtcgtactgaacccacacccactcttctttataccctacaaactcttcgtccaactcaatggcgctcatctggtgcggtatagacaataatggtagccagtcacatttcaaaagttgtctttttcgatatccatgattaatttgggcatatttgaagttcgaaactgtttctctggcacttagctttgatgatcggttgatatatttcatccttggaggtatacagtagtggctttcctcccaggtggtatcgtaaaacctgggaaggattgtgctcaa	
Yeast Selection Marker (Nat) for SSH1 overexpression genes (Nucleotide)	103	ttcagtaatgtctgtttctttttgttcagtggtgagccattttgacttcgtgaaagtttctttagaatagtgtttccagaggccaaacattccaccgtagttaaagtgcaagcgttaggaagaccaagactggcataaatcaggtataagtgctcgagcactggcaggtgatcttctgaaagtttctactagcagataaagatccagtagtcatgcatatggcaacaatgtaccgtgtggatctaaagaacgcgtcctactaaccttcgcattcgttggtccagtttggtgttatcgatcaacgtgacaagggtgtcgattccgcgttaagcatgcataccaaggaagcctgttgcaattccaagtgagccagttccaacaactcttgtaataattagagcaactcattgtgttgcgcttgaaagtaaaatgcgaacaataaagagataatctcgaacccgcgacttcaaacgcccaatgatgtgcggcacacaataagcgttcatatccgctgggtgactttctcgctttaaaaaattatccgaaaaaatttttgacggctagctcagtcctaggtacgctagcataaagaggagaaaaatgactactcttgatgacacagcctacagatataggacatcagttccgggtgacgcagaggtatcgaagccttgagcggttcattcacactgatacgggtgttttagagtcacgcctacaggtgatggcttcacctgagagaggttccgtgtagaccacccctaacgaaagttttccctgatgacgaatcggatgacgagtcgatgctggtgaggacgggtgacctgattccagaacatttgtcgcataccggagatgatggtagacctggctggctttgtgtggtgtctacagcggtggaatcgtagactcacagttgaggacatcgaagttgcacctgaacatcggtgtcacgggtgttggtcgtgcatgagggactggcaacagagtttgctagagaaagaggagccggacatttgtggttagaagtgaccaatgtcaacgctcctgctattcacgcataataggcgaatgggtttcactttgtgcggctcttgatctgctttgtatgacggaactgcttctgatggtgaacaagctctttacatgagtagtgcacatgcatagcagctccgacggcgcccccaggggtcccaggcctcgagatccgtcccccttttctttgtcgatcatcgtaattagttatgtcacgcttacattcacgcctccccccacatccgctctaaccgaaaaggaaggagttagacaacctgaagtcctaggtccctatttttttttatagttatgttagtatttaagaacgttatttatatttcaaatttttctttttttctgtacagacgcgtgtacgcatgtaacattatactgaaaaccttgcttgagaaggttttgggagctcgaaggctttaatttgcaagct	
Y Yeast Selection Marker (Nat) for SSH1 overexpression genes (Amino Acid)	104	MTTLLDDTAYRYRTSVPGDAEAEIALDGSFTTDTVFRVTATGDGFTLREVPVDPPLTKVFPDDDESDDSDAGEDGDPDSRTFVAYGDDGLAGFVVVSYSGWNRRLTVEDIEVAPEHRHGTVGRALMGLATEFARERAGAGHLWLEVTNNVNAPIAHAYRRMGFTLCGLDATALYDGTASDGEQALYMSMPCP	

TABLE 12

Zeocin Cassette with HA arms for KU70 deletion in <i>P. pastoris</i>			
Description	SEQ ID NO:	5' to 3' Sequence	
Plasmid sequence	57	ggagttgaatcacatcttactggatagcgagctttttgacgaagtgaataatttctaattttaaacaaaggaaggggtcaaaaacggagatattctatacttggaaaaaagagatgacaatcagtgatttcatcaattttgtatctagtggccttctgtgttttcgtggaagcagcaacgaggaaggggtatcctagatgatttttacaacgaactgaacgactgctttgaggggggtaaacatgaaagtaatatggaactccgtcctagttatttgccaggaggaagcaaggggtgtataggcttagtacttatagaggaaacggggttacgtgcaagcgcgcagtcgctgagctttgaggggggggactttcacatctctctctcacacttagcctaacacagagaataataaaaagcattgcaagatgagtggtgtgagcaagcaatcagacatccacgaagcatttatctttgtaattgaattgaccccgagcttcacgcgcgcggttcagaagggaatctcagctccagatcatcttagagaatgtcagtgaggttatttctgagctaatcattaccttgcgccgtacaggaatagggtgtaccttattaattacgacgggtggtcaaaacgcgaatttaccctattttgagttacaagacctgaatttggaatgatgaaacaattgtaccaagtcctggaggaacatgtaagtgggttaattcctctcgagaagcaattcccaattgaacacagtaaacctgtatcagccactctgttcttccacttaaggctctctttttacatggcgaagactcataagcgtactggaagacattacaactgaaaaagattttctgttcactaataacgataaaccttacaatggaaactctcagctgagagttcccttgagaaaaacccctggctgattacaatgacgtagacattactttgattccgtttcttctgaacaagccttcaggtgtcaagtttgacaagacggaataactcagaaattttgttctatgataaagatgctgttcgatgtcaattgaggagatccgccaacgaatttctagacataaggagatcaagcgggttacttcacctgtcctttgaaatcgcaaaataactgtgtgcatttctgtgaaagggttattctatgtttatcatgaaactccaggaagatcaaatttgtcgtcaatgaggggttcaactttcaagatgtggagacaaaactcagtttgtcgatccaacatccggaaaagagttttccagtgaaacagctgatcaagcatatcctctaggtgccgatgcttacattcctttaactcagag	

TABLE 12-continued

Zeocin Cassette with HA arms for KU70 deletion in <i>P. pastoris</i>			
Descrip- tion	SEQ ID NO:	5' to 3' Sequence	
		caagtcaaaacaataaatcgatttaatgatatcatcaatatccctcttttgaaattctaggtttcagggatatactaa ttggttgccacagatcagtttggaagcagtcgttttatccctcaataactatggatgtttacacattcgcagagaa catttagttgtcttcagtaaatgtcttgtttcttttgttgacgtggtgagccattttgacttcgtgaaagtttctttagaa tagttgtttccagaggccaacattccaccgtagtaaaagtgcagcgtaggaagaccaagactggcataaatcagggtat aagtgctcagcactggcagggtgatcttctgaaagtttctactagcagataagatccagtgtcatatggcaacaat gtaccgtgtggatctaagaacgcgtcctactaaccttcgcattcgttgggtccagtttgttattcgatcaacgtgacaa ggttgctcattccgcgtgaagcatgcatacccaaggacgcctgttgcaattccaagtgcagcagttccaacaattctttgta atattagagcacttcattgtgttgccgttgaaagtaaaatgcgaacaaattaaagataaatctcgaaacccgcagcttcaa acgccaatatgatgtgcccacacaaataagcgttcataatccgcctgggtgactttctcgtttaaaaaattatccgaaaaa atttttgacggctagctcagtcctaggtagcgtatgaatgaaggagaaaaatggctaaactgcactctcgtgttccgtg tctgacgcgtcgtgacgttgctgggtgctgttgagttctggacgcagcgtctgggtttctcgtgacttcgttgaagacg acttcgctggtgtgttcgtgacgcgttacctgttcatctcgtgttcaggaccaggttgttccggacaacaccctg gcttgggtttgggttcgtgggtcgtgacgaactgtacgctgaatgggtctgaagtgtttctaccaactccgtgacgttc tgggtccggtatgaccgaaatcgggtgaacagcgtgggtcgtgagttcgtctcgtgacccggtcggtaactgcgttc acttcgttgctgaagaacaggactaacacgtccgacggcggccacgggtccaggcctcggagatccgtccccccttttc ctttgctgatatacgtataatagttatgtcacgcttacattcacgcctccccccacatccgtctaacccgaaaaggaa gagttagacaacctgaagtcctaggtccctatttttttatagttatgttagtattaaagacgttatttatatttcaa atttttctttttttctgtacagacgcgtgtacgcatgtaacattatactgaaaaccttgcttgagaaggttttgggacg ctcgaaaggctttaatttgcaagctgtattagtttcaacttttcagcaacctggtcggaaagatccacatcaagaattgga caaccccaagagtgataaaatccttccctacaatggcacttcaaaatgttacgtgacgattaccttcaattggaacacg atatcgacatcagtgaccccttgagaacaaaagtacataaacagcctcgatgagacaaaaaccaagatcatgaaacta cgggactatgtcaaggaaactgccgatgatgacgaccttcacggcttgccaaactctcaagagctcaaccaaagct gaacaaaatttccaactttgatatactgcacaaataagaagccaaagacccacgacagtagacccctgttctactgatg atgacatcatcaacgcttggaaggcaggaactctgaacgggttcaaggtggatcaattacgaaaaatcgtgaaggtcacga aacaactttctggagacggcctccaaaaggcagatctcatcgcaacattgacaagtaactttcagcagaagttcaaaga gactaaggcctgattcgtgttcttacttttctcgcgaacgtgttttttccaccacattgcctatgttgtaatgcaa tgcagatgctggccagtttttgacgattctcgaaaaatggcattttcgtcgatgccattggccaaactgaaaaattcaag caaaaaatagattggatttttatctgcaacgtcttccacctacacaacctctacaaacttcagacaaactcgtttataaa agcagctactagatccaaaatgacaagttcgttattctctactacgtttgttggtggcatttggttggtggttagcaaca acctcttgccatgtcctgttgaccactctatgaataacgagactccgcaagaattgaaaccttgacggctgaattctct actgaagaagtggactcttccgtttaagtcaataaaaactactgacacagatgatgcacagaaacacacgggtatctct tgactgattagtcctcgtcatttttggttctcattttctcacagtcacctatcaatgtatgatcacctggaaggatttccc tacgatacttcaaatcttttacttgataatattactcattatgggtcaggaatgcagactgcctgattcaagacgctgct ctctctatttaacacttgtacactaaaccccggaagccagggaagggaataaacctctctctggtataaagctcgttct tattttatgcatagaaaaggaatctattatatttcgttctatttggcactctgctaactgtagattaacgggtctcgtaaat tcaaaatctcttccgatcaaacccgggtgaaatattactctcgtgcatagctaatttcaaaataacgcgtcctaaaaatg aacggctatttacctggactctcttgcacaaatgggcaacaaaaacataaagctgatcagaacgtaactagctctcgtgaat ccat	
HA F	58	ggagtgtaatcacatcttactg	
KU70 HA 1	59	gacaactaaatgttctctcggaatgtgtaaaatcaccatagttattaggggataaaaacgatgctttgccaactgatac tgtggcaaccaattagatataccctgaaacctagaatttccaaagaggggatattgatgatattcaaatcgatttat tgttttgacttgctctgagtttaaaaggaatgtaagcatcggcacctagaggatagctttgatcagctgttcaactggaaa actcttttccggatgttggtatgcacaaactgagattttgtctccacatcttgaaagttgaacctcattgacgacaaat ttgatcttcttggagtttctatgataaaaacataaactaacctttcacagaaatgcacaggtattttgagattttcaaagg acaggtgaagtgaacccgcttgatctccttatgtctagaaatcgttggcggatctcctcaattgacatcgaacaaagcat ctttatcatagaacaaaatttctgagatttccgtctgtcaaaacttgacacctgaaggcttggtcagaagaacagggaatc aaagtaaatgtctcagtcattgtaatcagccagggttttcttcaagggaactctcagctgagagatttccattgtaaggttt atcgttattagtgaaacaagaaaatcttttcaagttgtaatgtcttccagtcacgttatgagttctcgccatgtaaaaaa gagaccttaagtgaagaacagagtggtgataacgggttactgtgttcaattgggaattgcttctcgagaggattaaagc ccacttacatgggtcctccaagacttggtacaattgtttcatcatttccaaattcaggtcttgtaactcaaaaatggggtga aatttctcgtctttgaccacgctcgtaatataaaggtaaacacctatttctgtaccgggcaaggtaatgatagctcag aaataacctcactgacattctctaagatgatctggagctgagatttccctctgaagccggcgcgtgaagctccgggggtc aattcaattacaaagataatgcctcgtggatgtcgtattgtctgtgacaacactcat	
KU70 HA 2	60	tcaggccttagtctctttgaacttctcgtgaaagtacttgtcaatgttgccgatgagatctgctttttggaggccgctct ccgaaaagtgtttcgtgaccttacgtattttcgtaatgtatccaccttgaaacccgttcagagttctcgtctccaggcg ttgatgatgtcatcatcagtaggaacagggtctactgtcgtgggggtctttggcttcttattggcgatgatataaagtt ggaaaattttgttcagctcttgggtgagctctttgagagttgtggcaagccgtgaagggtcgtcatcatcggcagtttct tgacatagtcocgtagtttctatgataaaaactaacctttcacagaaatgcacaggtattttgagattttcaaagg acaggtgaagtgaacccgcttgatctccttatgtctagaaatcgttggcggatctcctcaattgacatcgaacaaagcat ctttatcatagaacaaaatttctgagatttccgtctgtcaaaacttgacacctgaaggcttggtcagaagaacagggaatc aaagtaaatgtctcagtcattgtaatcagccagggttttcttcaagggaactctcagctgagagatttccattgtaaggttt atcgttattagtgaaacaagaaaatcttttcaagttgtaatgtcttccagtcacgttatgagttctcgccatgtaaaaaa gagaccttaagtgaagaacagagtggtgataacgggttactgtgttcaattgggaattgcttctcgagaggattaaagc ccacttacatgggtcctccaagacttggtacaattgtttcatcatttccaaattcaggtcttgtaactcaaaaatggggtga aatttctcgtctttgaccacgctcgtaatataaaggtaaacacctatttctgtaccgggcaaggtaatgatagctcag aaataacctcactgacattctctaagatgatctggagctgagatttccctctgaagccggcgcgtgaagctccgggggtc aattcaattacaaagataatgcctcgtggatgtcgtattgtctgtgacaacactcat	
pILV5	61	ttacagtaatgtcttgtttcttttgttgacgtgggtgagccattttgacttcgtgaaagtttctttagaatagttgtttcca gaggccaaacattccaccgtagtaagtgcaagcgttaggaagaccaagactggcataaatcaggtataagtgctcgagca ctggcagggtgatcttctgaagtttctactagcagataagatccagtagtcatgcatatggcaacaatgtaccgtgtgga tctaagaacgcgtcctactaaccttcgcattcgttgggtccagttttgttggattatcgatcaacgtgacaaggttgcgattc ccgctaagcatgcatacccaaggacgcctgttgcaattccaagtgcagccagttccaacaattcttgtaatatatagagcac ttcatttggttgccgttgaaagtaaaaatgcgaacaaataaagagataatctcgaacccgcgacttcaaacgccaatatga tgtgcccacacacataagcgttcatatccgctgggtgactttctcgtcttaaaaaattatccgaaaaaattt	

TABLE 12-continued

Zeocin Cassette with HA arms for KU70 deletion in <i>P. pastoris</i>		
Description	SEQ ID NO:	5' to 3' Sequence
RM2734; testR	62	cagaggccaacattccacc
pporRBS	63	ttaaagaggagaaa
Sh ble (codon optimized)	64	atggctaaactgacctctgctgttccgggtctgacccgctcgtagcgttgcgtggtgctgtgagttctggaccgacgcgtctgggtttctctcgtagcttcggtgaagacgacttcgctgggttggttcgtagcagcgttacccgtgtcatctctgctgttcaggaccaggttggtccggacaaacccctggcttggttggttcgtagcgttggaacgaactgtacgctgaatggtctgaa gttgtttctaccaacttcggtgacgcttctggtccggctatgaccgaaatcggtgaacagccgtgggtcgtaggttcgc tctgctgacccggctggttaactgcgttcacttcggtgctgaagaacaggactaa
CYC1 terminator	65	cacgtccgacggcgccacgggtcccaggcctcggagatccgtcccccttttctttgtcgatatcatgtaattagttatgtcacgcttacattcacgccctccccccacatccgctctaacgaaaaggaggttagacaacctgaagctcagggtccctatttattttttatagttatggttagtattaagaacggttatttatatttcaatttttcttttttctgtacagacg cgtgtacgcatgtaacattatactgaaaaccttgcttgagaagggtttgggacgctcgaaggctttaatttgcaagct
Rm3386; F test oligo	66	aggagttagacaacctgaag
HA R	67	gtaactagtcctctcggaatccat

TABLE 13

Template Nourseothricin Cassette for protease deletion in <i>P. pastoris</i>		
Description	SEQ ID NO:	5' to 3' Sequence
Plasmid sequence	68	cttcagagtacagaagattaagtgaagaaattctacggttcgtatagcatatcattatacgaagttatttcagtaaatgtct tgtttcttttgttgcagtggtgagccattttgacttcgtgaaagtttctttagaatagttgtttccagaggccaacatt ccaccgtagtaaaagtgcagcgtaggaagaccaagactggcataaaatcaggtataagtgtagcactggcaggtgatc ttctgaaagtttctactagcagataaagatccagtagtcatgcatatggcaacaatgtaccgtgtggatctaagaacgcgt cctactaaccttcgcattcggttggtccagtttggtgttatcgatcaacgtgacaaggttgtagattccgcgtgaagcatgc ataccgaaggacgcctgttgcaattccaagtgagccagttccaacaatctttgtaatatagagcacttcatttgtgttc gctgaaagtaaaatgcgaacaaattaagagataatctcgaacccgcgacttcaaacgccaatatgatgtgcggcacaca ataaagcgttcataatccgctgggtgactttctcgcttataaaaaattatccgaaaaaatttttgacggctagctcagtccta ggtacgctagcattaaagaggagaaaaatgactactcttgatgacacagccctacagatataggacatcagttccgggtgac gcagaggctatcgaagccttgagcgttcattcactactgatacgggtgttttagagtcacgcgtacaggtgatggcttcac cttgagagaggttcctgtagaccacccttaacgaaagtttccctgatgacgaatcggatgacgagttgatgctgggtg aggacggtgaccctgattccagaacatttgctgcatacggagatgatggtagcctggctggctttgttggtgtcctac agcggatggaatcgtagactcacagttgaggacatcgaagttgcacctgaacatcgtggtcacgggtgttggtcgtgcact gatgggactggcaacagagtttgctagagaaagaggagccggacatttggtgttagaagtgaccaatgtcaacgctcctg ctattcacgcatataggcgaatgggtttcaactttgtgcggtcttgatactgctttgtatagcgaactgctctgatggt gaacaagctctttacatgagtatgcatgtccatagcacgtccgacggcgccacgggtcccaggcctcggagatccgt cccccctttctttgtcgatatcatgtaattagttatgtcacgcttacattcacgccctccccccacatccgctctaac gaaaaggaaggagtttagacaacctgaagtcaggtccctatttattttttatagttatggttagtattaagaacgctatt tatatttcaaattttctttttttctgtacagacgcgtgtacgcatgtaacattatactgaaaaccttgcttgagaagg ttttgggacgctcgaaggctttaatttgcaagctataaacttcgtatagcatatattataccttgttatgcgccgcaaga agttgattgagactttcaacgag
AOX1 pA terminator	69	cttcagagtacagaagattaagtgaaga
Lox71 F	70	taccgttcgtatagcatatcattatacgaagttat
pILV5	71	ttcagtaatgtcttgtttcttttgttgcagtggtgagccattttgacttcgtgaaagtttctttagaatagttgtttcca gaggccaacattccaccgtagtaaaagtgcagcgtaggaagaccaagactggcataaaatcaggtataaagtgtagcga cttggcaggtgatcttctgaaagtttctactagcagataaagatccagtagtcatgcatatggcaacaatgtaccgttgga tctaagaacgcgtcctactaaccttcgcattcggttggtccagtttggtgttatcgatcaacgtgacaaggttgtagattc ccgctgaagcatgcataccaaggacgcctgttgcaattccaagtgagccagttccaacaatctttgtaatatagagcac ttcatgtgttgcgcttgaaagtaaaatgcgaacaaattaagagataatctcgaacccgcgacttcaaacgccaatatga tgtgcgccacacaataagcgttcataatccgctgggtgactttctcgcttataaaaaattatccgaaaaaattt

TABLE 13-continued

Template Nourseothricin Cassette for protease deletion in <i>P. pastoris</i>		
Description	SEQ ID NO:	5' to 3' Sequence
pproRBS	72	ttaaagaggagaaa
nat (Nourseothricin resistance)	73	atgactactcttgatgacacagcctacagatataggacatcagttccgggtgacgcagaggctatcgaagccttgacggttccattcactactgatacgggtgttagagtcacgcgtacaggtgatggcttcaccttgagagagggttcctgtagaccaccccttaacgaaagtttccctgatgacgaatcggatgacgagtcgatgctggtaggacgggtgacccctgattccagaaca tttgtgcatacggagatgatgggtgacctggctggctttgtgtgggtgctcctacagcggatggaatcgtagactcacagt tgaggacatcgaagtgcacctgaacatcgtgggtcacgggtgttggtcgtgcactgatgggactggcaacagagtttgcta gagaaaggaggagccggacatttgtgggttagaagtgaacaaatgtcaacgctcctgctattcacgcataataggcgaatgggt ttcactttgtgcgggtcttgatactgctttgtatgacggaactgcttctgatgggtgaacaagctctttacatgagtatgcc atgtccatag
CYC1 terminator	74	cacgtccgacggcggccacgggtcccaggcctcggagatccgtcccccttttctttgtcgatatcatgtaattagttat tgtcacgcttacattcacgccctccccccacatccgctctaaccgaaaaggaggttagacaacctgaagtctagggtc cctattttttttatagttatgttagtattaagaacgttatttataatttcaaattttctttttttctgtacagacg cgtgtacgcattgaacattatactgaaaacctgttgtagaagggttttgggacgctcgaaggctttaatttgcaagct
LoxKR3 F	75	ataacttcgtatagcatatacattataccttggtaa
HSP82	76	gcggccgcaagaagttgattgagactttcaacgag

TABLE 14

Exemplary nourseothricin cassettes with HA arms for YPS1-1 and YPS1-2 protease deletion in <i>P. pastoris</i>		
Description	SEQ ID NO:	5' to 3' Sequence
Nourseothricin cassette with homology arms targeting PAS_chr4_0584 (YPS1-1)	77	tactacaggtggtgttctctcgcatgggttttaagtctcgtactgggttttctgttatcggtattacggagccaccttg actgtaagggaacgatactggactaagagagtaaatcgaaaggcaacagcgtttctggcgaaacctaatcaatgacgggttac gagtttactactcctaagaccagtcctatttttgcctagagcaggtcaacgccttacttaaggggccaggacctaattatgac atcgattttgacgagcaggaggcgttcattaaagaaatggaggagttgaggacctctggtggatagagaacagatactca tattcaggaaccgatgaaacacccagagatccgggttgctgtttcttccattgctttaaatgaatggcactttgatgtg ctagactgctgaggatatacggtaactcaggaagatctggaatctaaattattaagtgttcagcaattgggtgtacaatgt tgcataagcagcagtggtgactccagacatgggtcttggcaacggaagtgcctcagaagccgacctcgaagacgacata gtttgtgatgatattgacgcttatgcccagggggtgtgtgtctagattattgttacgcgccaagcaatctactccagaact ttagaaattcatggcaagattgctaccttacaacgagagctggggctatgctataatattctcggaattttggaccggtttt tcgatttaaggttttttagctccattgcgcacaccccgctctccagactccttcgttatccagcattcagcatggacaggt tcaaaaaataaaaatttcttgataggggtccacttcaaacatgcgcctacctgtaggaaaaaaaagagaaacataaattgc cgcaacagaaaaacgtaatgtactgttctatatataaaactgttcagatcaatcataaattctcagtttcaaaccttccgct cagccagattttattcgtaaagaacgcattcattggctctatgttgaaaggatcagttcttggttatgggttgccttgatagcg agcgtaccggtttccggcgtgatggcagctcctagcagatccgggcataaacgcgttgaaaaacgagatgccaaaaacgtt gttggcgttcaacagttggacttctcagagttacagaagattaaagtgcagataatctacggttcgtatagcatatacattata cgaagttatttcagtaattgtcttggttcttttgggtgcagttgggtgcagtttgggtgttatcgatcaacgtgacaagggtgtgc gatccgcgtaagcatgcatacccaaggacgctgttgcaattccaagtgcagcagttccaacaactcttggtaattataga gacttcatgtgttgcgcttgaaagtaaaatgcgaacaaatgaagagataatctcgaacccgcgacttcaaacgccaaata tgatgtgcggcacacaataagcgttcataccgctgggtgactttctcgttttaaaaaattatccgaaaaaatttttgacg gctagctcagtcctaggtacgctagcattaaagaggagaaaaatgactactcttgatgacacagcctacagataggacat cagttccgggtgacgcagaggctatcgaagcctggacgggtcattcactactgatacgggtgttagagtcacccgtacag gtgatgggttcaccttgagagaggttccctgtagaccaccccttaacgaaagtttccctgatgacgaatcggatgacgagt ctagctgctggtaggagcgggtgacccctgattccagaacatttgctgcatacggagatgatggtagcctgggtttgttg tggtgtcctacagcggatggaatcgtagactcacagttgaggacatcgaagtgcacctgaacatcgtgggtcacgggtgttg ctcgtgcactgatgggactggcaacagagtttgcctagagaaaggaggacggacatttgggttagaagtgaccatgtca acgctcctgctattcacgcataataggcgaatgggtttcactttgtgcgggtcttgatactgctttgtatgacggaaactgctt ctgatgggtgaacaagctctttacatgagtagtgcctatgctcactagcacgtccgacggcggccacgggtcccaggcctcgga gatccgtcccccttttctttgtcgatatcatgtaattagttatgtcacgcttacattcacgccctccccccacatccgct ctaaccgaaaaggagaggttagacaacctgaagtctagggtccctatttattttttatagttatggttagtataagaacg ttatttatatttcaaattttctttttttctgtacagacgcgtgtacgcattataactgaaaaccttgcctgag aagggttttgggacgctcgaaggctttaattttgcaagctataacttcgtatagcatatataacctgttatgcggccgca agaagttgattgagactttcaacgagggtcccccttcagctaccttctctcgtgttggtagttatctcggcggtgtgata gtatagatataaaaggcctacattggatagggttcaacattcctcaataaacaacatccaacatcgcgcatccgcattt

TABLE 14-continued

Exemplary nourseothricin cassettes with HA arms for YPS1-1 and YPS1-2 protease deletion in <i>P. pastoris</i>	
Description	SEQ ID NO: 5' to 3' Sequence
	cgcatttcacatttcgcgcctgccttcctttaggttccttgaatcatcatcaatcgtcgcgctctacatcagagcaggact tatctttgccttccccaaaaattgccactccgtcaaatagattcttttgaatccttgactattttgcctaaataggtttt tgttagtttttcttcaaaagcccaaaagaactctatttagattcatccagaacaatctttttccacccatttcgaagt gccgtggagcacagacataaaaaagatgactaccgttcaacctacagggccagacaggctcaccctgccgatattctactg gaattcaacgatggctcctcgagcatgcagtgatcgagctaagcatgaacgaggggattaatataccacccatgagtg aatccatccactaatgagcaatcgccacgggaagagagagcaccaccccaacaatccaatccatcgcatccagaatca tcgaacatagctactcaaaagtcgcctcaggaacccgagactcagcccgccattccaggactagataggctgcctttgat acctcggaacgggtcgtcagaacaggttgacccagtagaggaaggatcctggatgatattataggccaatcattaagg acttccgaagaagacgataccgaatcccgccagagaccagagaccagaagaacattatgatcccgatgaattacttgtac gcagacgacacaaattccagaagtgcatacaaaacacagacgcccataaacacttctagaacttccgacagtgaaactg gtgggctccttatcggtgcacgttccggatctaccagataatgccgacgattactatcgatgtactcattaaactaacc caagcattgcctcagcgtcatcagctccatgatcaagaaacgattagggttagcaggga
PAS_chr4_0584 Homology Arm 1	78 tactacaggctggctgttccctcgcattgggtttaatgtcctgactgggttttgcgtttatcggtattaccggagccaccttg actgtaagggaacgatactggactaagagagtaaatgcgaagggaacacgcgtttctggcgaacctaatcaatgacggttac gaggttactactcctaaagccagtcctattttgctagagcagtgcaacgcttacttaaaggccaggacctaattatgac atcgattttgacgagcaggaggcgttcattaaagaaatggaggagttgaggacctcgtggtgatagagaacagatactca tattcaggaaccgatgaaacacccagagatccgggttgctgttttctccattgctttaaataaatggcactttgatgtg ctgactgcctgaggatatacggtagctcaggaagatctggaatctaaattattaagtgttcagcaattgggtgtacaatgt tgcataagcacagtggtgatctccagacatgggtcttgcgaacggaagtgcctcagaagccgacctcgaagacgacata gtttgtgatgatattgacgcttatgccaggggggtgatgtctagattattgttacacgccaagcaattactccagaact ttagaattcatggcaagattgctacctacaacgagagctggggctatgctataatattctcggaattttggaccgtttt tccgattaaaggtttttagctccattgcgcacaccccgctctccagactccttcgttatccagcattcagcatggacaggt tcaaaaaataaaattttctgatattgggtccacttcaaacatgcgcctacctgtaggaaaaaaagagaaacataaatatgc ccgaacagaaaaacgtaattgactgttctataataaactgttcagatcaatcataaattctcagtttcaaaacttccgct cagccagattttattcgtaaagaaacgcatcattggctctatgttgaaggatcagttcttgttatgggttgctttgatagcg agcgtaccggtttccggcgtgatggcagctcctagcgagtcgggcataacacggttgaaaaacgagatgccaaaaacggt gttggcgttcaacagttggaactt
PAS_chr4_0584 Homology Arm 2	79 ggtcccttcagctacctttctctctgttttggtagttattctcggcgtgtgtatagatagatataaaaggccctacattgg ataggcttcaacattcctcaataaaacaaacatccaacatcgcgcattccgcatttcgcatttcacatttcgcgcctgcctt cctttagggttccttgaatcatcatcaatcgtgcgcgtctacatcagagcaggacttatctttgccttccccaaaaattgcc actccgtcaaatagattcttttgaatccttgactatttttgcctaaataggtttttgttagtttttcttcaaaagcccaaaa gaaactctatttagattcatccagaacaatctttttccacccatttcgaagtgcgctggagcacagacataaaaagat gactaccgttcaacctacagggccagacaggctcaccctgccgcatttctactggaattcaacgatggctcctcgacga tgcagtgatcgagctaaagcatgaacgaggggatataatataccacccatgagtggaatccatccactaatgagcaatcgcc acgggaagagagagcaccaccccaacaatccaatccatcgcatcatccagaatcatcgaacatagctactcaaaagtcgcc tcaggaacccgagactcagcccgccattccaggactagataggctgcctttgatacctcggaacgggggtcgtcagaaca ggttgacccagtacaggggaaggatcctggatgatattataggccaatcattaaaggacttccgaagaagcagcatccgaatc ccgacagagaccacgagaccagaagaacattatgatcccgatgaattacttgtacgcagacgacacaaattccagaagtgc taatacaaaacacagacgcccataaacacttctagaacttccgacagtgaaactggtgggctccttatcggtgcacgttcc ggaatcaccagataatgccgacgattactatcgatgtactcattaaactaaccacaagcatgcctcagcgtcatcac gtccatgatcaagaaacgattagggttagcaggga
Nourseothricin cassette with homology arms targeting PAS_chr3_1157 (YPS1-2)	80 gccttctcgtgcaatcagagctgttgaagagagaagagggcacacggaagctgctgttcaattgtgtgaattgacccgat tacaacctgctggagtgataggagagctgggtcgtgacgaggacggctctatgatgcatagacgactgtgttcagtttg gtctccgccacaacgtaaaaattatcaaccttgaccagatcattgaatacatggattccaagaacagctagatcacgattgga taggaatcacagatatacatgattgaggaacgtgaagacttttgcgaagtgtgagttgtgggtgagggccaggcggtggg gagggtgggggagcctccttggtcgaatgtagatataagtaagcaagacacaagagcgcggaagtcttcaacgagggcggc gttgggtcttgtagcgaacgtaatgactacacagttgagcttgctgcgaacccggtcgacatttgatcatgcatactatgt tgagacacatctcgtactattgcggcaacagctgtaaatgtactaataaagctgatgaaggatgcagggcgctcgtca atttttgattgattgcatttaattgtttgagccattcaaggctgaatgcccgccacctcttgtagtactata taaacccgcaggcagggtacccttggccttctgcgagactaccagtcataacgtatatacccaatgtactagtaaatagccc cggaaaaactataatcccacagaacgtctaacgcctcctatgtcatcgatacccatcgcactactgccatggccccctta cgtgatecatttcaacttactccgcctaaagcttcgcccacatgcctgcgttttgccaagatttactgacgagtttgggttac tcatcctctattataactactagactttcaacatttctcaccacccctcgtgcgaatgatcatcaaccttggattgac agccctcagcattgcactagcaagtgcgaacctccaatcgctttcacttcagagtacagaagattaaagtgagagaattct accgttcgtatagcatacatatacgaagtatttcagtaaatgtctgtttcttttggtgacgtgggtgagccatttgact tcgtgaaagtcttcttagaattgtgtttccagaggccaacattccaccgtagtaaaagtgcaagcgttaggaagaccaag actggcataaatcaggtataagtgctgagcagctggcaggtgatctctgaaagtctcactacagatagaatcagtagt catgcatatggcaacaatgtaccgtgtggatctaaagacgctcctactaaccttcgactcgttgggtccagtttgggtgtt atcgatcaacgtgacaaggtgtcgattccgcgtgaagcatgcatcccaaggacgctgttgcaattccaagtgaagcaggt tccaacaactcttgtaatatatagagcacttctatgtgttgccgttgaaagttaaagtgcaacaaatgaagagataatctcg aaacccgcacttcaaacgcaatatgattgtgagggcacacaataagcgttcataccgctgggtgacttctcgcttataaa aattatccgaaaaaatttttgacggctagctcagtcctaggtacgctagcattaaaggagagaaatgactactcttgatg

TABLE 14-continued

Exemplary nourseothricin cassettes with HA arms for YPS1-1 and YPS1-2 protease deletion in <i>P. pastoris</i>		
Description	SEQ ID NO: 5' to 3' Sequence	
	acacagcctacagatataggacatcagttccgggtgacgcagaggctatcgaagccttggacgggttcattcactactgatacgggtgttagagtcaccgctacaggtgatggcttcaccttgagagagggtcctgtagaccacccttaacgaaagttttccctgatgacgaatcggatgacgagctctgatgctgggtgaggacgggtgacctgatccagaacatttgcgcacacggagatg atggtgacctggctggctttgttgggtgtctacagcggatggaatcgtagactcacagttagggacatcgaagtgcac ctgaacatcgtgggtcacgggtgttggctcgtgcaactgatgggactggcaacagagtttgctagagaaaggaggccggacatt tgtgggtagaagtgaccaatgtcaacgctcctgctattcagcctataggcgaatgggtttcactttgtgcggtcttgata ctgctttgtatgacggaactgcttctgaggtggaacaagctctttacatgagtatgccatgcccagctgcagcggcggc ggcccacgggtcccaggcctcggagatccgtcccccttttctttgtcgatatcatgtaattagttatgtcacgcttacat tcacgcccctccccccacatccgctctaaccgaaaagggaaggagttagacaacctgaagctcaggtccctatttatttttt atagttatgttagtattagaacgcttatttatatttcaaatttttcttttttctgtacagacgctgtacgcatgtaac attatactgaaaaccttgcttgagaagggttttgggacgctcgaaggctttaatttgcaagctataacttcgtatagcat acattataccttggttatgcggccgcaagaagttgattgagactttcaacgagctggctctgctctcgtgactcttcaggtgc atcttctgctactcaaaatgacgaaacatccactgatcttggagctccagctgcactcttaagtgcacgcgcatgtctttt tgccatcttgcgtcctcatgtttagtagactttttttcactgagtttttatgtactactgattacattgttaggtgta atgatgtgcactataataactaatatagttcaaaatgctacagaggaaagtgacgggttgctctggtgggtttttcttattagc acctctgaacactctttaactctaactcctcagccatgctaatcgcgcataaaaataaatcttgaactttttccattt tatgctcataaagcttcttactgtcaccttatcaaaagagcttttgccactaaagtgtcacacccagaattgtctccga atatcgtccaacaatgctaggatctgtggaagtttgacaaataattgaacaccttgagcttgaagcttctgaaagttaa tatccaggctcctttccagaagtaacccagtggaaccttttgagaaactacatcactcaagaacttagtaaaattctgg agttgacaaagaattgattttccagccttgggaatgggtaccacactggaaaaagggtgatctttgatcccagttcctcg tctgagaataaaagggtgctaatcctaaagatttagccgaacaatgggtgctgcatcccaagggtggatattctaaaga cgttattgcgcaaggaccttcttgcagttctttttaacacatcggttctgtacaagtgggtgatctctgatgctctgga gagagcgatgacttttgggtgaccttctctaggaagggaacaaaagttatagtgaggttttctctccaaatattgccc aaacctttccagcgtggccatcttagaagtacaatcatcggtgggttttatttccaatctgtatgaaaagctgggtcatgaagt tatgaggtgaaattatttgggagactggggaacaaatttggtgttcttgacgtaggatttgagcgttacggtgatgaggc aaaaataagactgatccaatcaaccatttggttgaggtctatgtttaaatacaaccaagatatgaaggtcaatcagagtc taatgaggagattgcagaagggaacatttagtagaccaggcaagagcttttttcaagaaaatggaaaatggcgacgaatc ggctgtaagcttgtggaagaagattccgtgagttatccattgagaagtacattgatacttatgccgcctcaacatc	
PAS_chr3_1157 Homology Arm 1	81	gccttctcgtgcaatcagagctgttgaaagagagaagagggcacacgggaagctgctgttcaattgtgtgaattgacccgat tacaacctgctggagtgataggagagctgggtcgtgacgaggacggctctatgatgcatagacgactgtgttcagtttg gtctccgccacaacgtaaaaattatcaaccttgaccagatcattgaatacatggattccaagaacagctagatcagatgga taggaatcacagagatatcatgattgaggaacgtgaagagcttttgcgaaggtgtagtttgtgggtgagggccaggcgggtggg gaggtgggtggggagcctccttgggtcgaatgtagatataagtaagcaagacacaaagagcgcgcaaggtcttcaacgaggcggc gttgggtctgtacgcaacgtaatgactacacagttgagcttgcgcgaacccggtcgacattttgatcatgcatattgt tgagacacatctcgtactattgcggcaaccagctgtaaattgactaataaagctgatgaaggatgcagggcgtcgtca atttttgattgattgcatttaattgtttgagccattcaaggctgaatgcccgccacctagacccttctgtgagtaacta aaacccgcagggcagggtaaccttggccttctgcgagactaccagtcataacgtatataccacatgtaataagccc cggaaaaactctaatccacagaaactctaacgctcctatgtcatcgataccatttcgcactactgcatggaagccctta cgtgacatttcaacttactccgcctaaagcttcgcgccacatgcctgcttttgccaagatttactgacgagtttgggttac tcatcctctatttataactactagactttcaccattcttcaccacccctcgtgccaatgatcatcaaccacttgggtattgac agccctcagcattgcactagcaagtgcgcaactccaatcgcccttca
PAS_chr3_1157 Homology Arm 2	82	ctggctctgcttcttggtacttcttcagggtgcatcttctgctactcaaaatgacgaaacatccactgatcttggagctccag ctgcatctttaagtgcaccccatgtcttttggccattctgtgctcattgtttagtagaacttttttttcaactgagtttt tatgtactactgattacattgtgtagggtgtaattgatgtgcaactataataactaatatagttcaaaatgctacagaggaaagtg caggttgctggtgggtttttcttattagcaccctctgaacactctttacctctaactcctcagccatgctaatcgcgc ataaaaataaatcttgaacttttttccattttatgctcataaagcttccctactgtcaccttatcaaaagagcttttgcca ctaaagtgtcacaccagaattgtccccaatatacgtccaacaatgctaggatctgtggaagtttgacaaataatttga acaccttgagcttgaagcttctgaaagttaatatccaaggctcctttccagaagtaacccagtggaaccttttgagaact acatcactcaagaacttagtaaaatttctggagttgacaaagaattgattttccagccttgggaatgggttaccacactgg aaaaagggtgatcttttgatcccagttcctcgtctgagaataaagggtgctaatcctaaagatttagccgaacaatgggctg ctgcatcccaagggtggatattctaaagagcttattgcgcaaggaccttcttgcagttcttttttaacacatcggttc tgtacaagttgggtgatctctgatgctctggagagaggcgatgactttgggtgcaactcctctaggaagggaacaaaagttt atgtggagttttcttccaaatatttgccaaacctttccagcgtggccatcttagaagtaacaaatcatcggtgttttattt ccaatctgtatgaaaagctgggtcatgaagttatgaggatgaattatttgggagactggggaaaacaatttggtgttcttg cagtaggatttgagcgttacggtgatgaggcaaaattaaagactgatccaatcaaccatttgggtgaggtctatgtaaaaa tcaaccaagatataaggtcctcaatcagagttctactgaggagattgcagaagggaacatttagatgaccaggcaagagctt ttttcaagaaaatggaaaatggcgacgaatcgggtgaagcttgtggaagaagattccgtgagttatccattgagaagtaca ttgatacttatgccgcctcaacatc

TABLE 15

Information on genes and sequences disclosed herein					
<i>S. cerevisiae</i> ortholog standard name	<i>S. cerevisiae</i> ortholog systematic name	<i>P. pastoris</i> (<i>K. phaffii</i>) gene ID	<i>P. pastoris</i> (<i>K. phaffii</i>) ORF coordinates (strand)	Open Reading Frame (ORF) SEQ ID NO:	Peptide SEQ ID NO:
SEC72	YLR292C	PAS_chr2-1_0448	chr2: 827202-827789(-)	1	2
SBH2	YER019C-A	PAS_chr2-2_0210	chr2: 1994927-1995252(+)	7	8
SSS1	YDR086C	PAS_chr1-1_0023	chr1: 580942-581136(+)	5	6
SSH1	YBR283C	PAS_chr1-4_0629	chr1: 2584924-2586363(-)	3	4
PEP4	YPL154C	n/a	n/a	83	84
PRC1, mutant	YMR297W	n/a	n/a	85	86
DAP2	YHR028C	n/a	n/a	87	88
MF(alpha)1, variant	YPL187W	n/a	n/a	89	90

TABLE 16

Information on promoters and terminators for the SSH1 complex genes				
<i>S. cerevisiae</i> ortholog standard name	<i>S. cerevisiae</i> ortholog systematic name	Promoter/ Terminator	<i>P. pastoris</i> (<i>K. phaffii</i>) ORF coordinates (strand)	Nucleotide Sequence SEQ ID NO:
SBH2	YER019C-A	Promoter	chr2: 1994039-1994926(+)	97
SBH2	YER019C-A	Terminator	chr2: 1995253-1995712(+)	100
SSS1	YDR086C	Promoter	chr1: 579989-580941(+)	98

TABLE 16-continued

Information on promoters and terminators for the SSH1 complex genes				
<i>S. cerevisiae</i> ortholog standard name	<i>S. cerevisiae</i> ortholog systematic name	Promoter/ Terminator	<i>P. pastoris</i> (<i>K. phaffii</i>) ORF coordinates (strand)	Nucleotide Sequence SEQ ID NO:
SSS1	YDR086C	Terminator	chr1: 581137-581628(+)	101
SSH1	YBR283C	Promoter	chr1: 2586364-2587353(-)	99
SSH1	YBR283C	Terminator	chr1: 2584553-2584923(-)	102

SEQUENCE LISTING

Sequence total quantity: 107

SEQ ID NO: 1 moltype = DNA length = 588
 FEATURE Location/Qualifiers
 source 1..588
 mol_type = other DNA
 organism = Pichia pastoris

SEQUENCE: 1
 atgctaccat tttcgtacga cgtgagctca aagaaactga aagtcacagg tgactacgca 60
 gacttagaat atgacataca gcagctgaac accttgagcg gagagatcct ggccaataaa 120
 gcagatgttc cttctccacc aagtaaggag tcgtttgaca agaaattgtc ccacatggct 180
 cagaaattac acgagtcggc tgtatccaac ataaagacag gcaagtatcc tgaggctatc 240
 aaattgttga cgacgggtct tgaatgggtt aacagaaggg ccaagtacga gagttttcag 300
 atgacgttga gtgaaatgac gatctttatt gtcactagag ctgacgctta catgatgaat 360
 ggagactttg aaggggcatt caatgatgca gatttactgg taacgctcct gccatccatt 420
 ccagataatt acattagaag aggggtagcc cttttcaaga tggggagata cgttgatgca 480
 aaaaacaatt ttgagagagg actttcattt gaccagata atgcaaaatt gaagaaggag 540
 ttgatttttg tgctgaagaa gatcgacgag gagaatggag agttatag 588

SEQ ID NO: 2 moltype = AA length = 195
 FEATURE Location/Qualifiers
 source 1..195
 mol_type = protein
 organism = Pichia pastoris

SEQUENCE: 2
 MLPFSYDVSS KKLKVTGDYA DLEYDIQQLN TLSGEILANK ADVSPSPSKE SFDKKLSHMA 60
 QKLHESAVSN IKTGYPEAI KLLTTGLEMV NRRPKYESFQ MTLSEMTIFI VTRADAYMMN 120
 GDFEGAFNDA DLLVTLPLSI PDNYIRRGVA LFKMGRYVDA KNNFERGLSF DPDNAKLKKE 180
 LDFVLKKIDE ENGEL 195

SEQ ID NO: 3 moltype = DNA length = 1440
 FEATURE Location/Qualifiers

-continued

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source                1..1440
                      mol_type = other DNA
                      organism = Pichia pastoris

SEQUENCE: 3
atggcggggtg tgcgtttttt agacattgca agaccatttg tcagctggat cecggaagtt 60
gaacttcctt atgaaaaactg ggggttcgat gaaaagctga tttactcatt tttcactgct 120
gccatctatt tgattctgtc cctgcctata tacggtgtca aatcctctga agtcgtggac 180
ccagttcccc atttgcgttc tgccttaggg agtgagaagg gaacattgct ggagcttggg 240
ttactgcctg tgattacttc ggcatattatc ttgcagttgt tggctggttg gaaagttttc 300
aaagtaaaact ttgatctggg tagtgacaga atattgttcc aaactttgca aaagatcact 360
tcagtcgtta tcagcatcgt atatgctgtt cttctcacat tttgtgacta ctttactcca 420
ggtgtgtcca ctgataacgt cttgtggtcc caatttctga tcatcttaca gatagtgggtg 480
gtcaacttct tggttactct actcgttgaa gtcattgaca aggattacgg attttcttca 540
ggagctctat tgttgcttgc ggtttattcc gccaccaact tcgttttttg cacgattggt 600
cttagcaccg tcaacacctc cagatcgaac gaatctattg gtgctctgat tcaattattc 660
cgcaatttga gctctaacc aattggtgtt gccatatatg actccttctt cagagtaaac 720
cttcttaact tgactcaatt ttatctgggg attgccatta tttgtgtttg tctgttcttg 780
aataatgcaa gatacgaagt accaattaag ccaaacaagg ttctgtgccat ggctcagct 840
taccacaatc agctactttt caatggttct ttgccacttc tgtacacgtg gactgtgctg 900
tacaaactga accttattgg tttctttgtc ttcaagctta ccaacttttc tcttttaggg 960
aacttcaaa gggacccatt cggcaacaac tactacgaaa ttacatctgg actgctgtat 1020
ttattgactc ctacttttca cgctgaagct ggaacttttac ccaatgttgc taagccattt 1080
gttttcattg ccttctatgt tgggttttag actttctttg ctagatcgtg gtccaacatt 1140
aacgggtcgt caggcaagga cattgccaag tttttcaagg ctcaaggaa ctcattgtta 1200
ggaaaaagag atgcctctgt gtctaaagag tttaacaccc tagttcctgt tgcttctgcc 1260
tctggagctt tcttattgtc ttttccagtg gccgtcgtg agttattggg tggctctggt 1320
gttccaacct ctatcggaat cggctctttg agtggttttg ctattttgga aactgttttg 1380
caagaatggc aacagctctg aggtgcctca cagttctccc aatacttcca gacttcttag 1440

SEQ ID NO: 4          moltype = AA length = 479
FEATURE               Location/Qualifiers
source                1..479
                      mol_type = protein
                      organism = Pichia pastoris

SEQUENCE: 4
MAGLRFLDIA RPFVSWIPEV ELPYENWGFD EKLIYSFFTA AIYLILSLPI YGVKSSEVVD 60
PVPHLRSALG SEKGTLLLELG LLPVITSAFI LQLLAGWKVF KVNFDLVS DR ILFQTLQKIT 120
SVVISIVYAV LLTFCDYFTP GVSTDNLWS QFLIILQIVV VNFLVTLLE VIDKDYGFSS 180
GALLLLAVYS ATNFVFGTIG LSTVNTSRN ESIGALIQLF RNLSSKPIGV AIYDSFFRVN 240
LPNLTQFYL G IAIICVCLFL NNARYEVPK PNKVRAMASA YPIKLLFN GS LPLLYTWTVL 300
YNLNLIGFFV FKLTNFSLLG NFKVDPPGNN YYEITSGLLY LLTPTFNAEA GLLPNVAKPF 360
VPIAFYVGVS TFFARSWNI NGSSGKDIK FPKAQGISLL GKRDAVSKE FNTLVPVASA 420
SGAFLLSFPV AVAELLGGSG VPTSIGIGLL SGLAILETVL QEWQQSGGAS QFSQYFQTS 479

SEQ ID NO: 5          moltype = DNA length = 195
FEATURE               Location/Qualifiers
source                1..195
                      mol_type = other DNA
                      organism = Pichia pastoris

SEQUENCE: 5
atgtcccaaa aagtcaccga cgtccctctg gaatttgta aggaagggtc caaatcattc 60
tctaaatgta ctaaaccttc tcagaaggag tactaaaga tagtaagagc tgttgagatt 120
gggtttttta tgatgggcgt ggttggttac gttgtcaagc tcattcatat tccaatcaga 180
tatttgattg ttttaa 195

SEQ ID NO: 6          moltype = AA length = 64
FEATURE               Location/Qualifiers
source                1..64
                      mol_type = protein
                      organism = Pichia pastoris

SEQUENCE: 6
MSQKVTDVPL EFKVEGSKFI SKCTKPSQKE YLKIVRAVG VFLMMGVVGY VVKLIHIPIR 60
YLIV 64

SEQ ID NO: 7          moltype = DNA length = 326
FEATURE               Location/Qualifiers
source                1..326
                      mol_type = other DNA
                      organism = Pichia pastoris

SEQUENCE: 7
atggtaagtg tccagtttga tgagtgcaga atgggtccaa gttttagacc agttactaat 60
atttaaagtc tacagcaatt ccaggaggac agagaacgtt agctaaaaga agagcagcaa 120
acttgatata gaaacaggat gaaccaacct ccgccagatc tgcgggtgct ggaggttctt 180
cgtctaccat gctaaagtgt tacacagacg aggcccaagg tttgaaagtt gatcctttaa 240
ttgttcttgt tcttgctggt ggtttcattt tcagtgatcat tggtttgcac gttgttgcta 300
agctgacagg aaagttagtc aactaa 326

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-continued

SEQ ID NO: 8 moltype = AA length = 86
 FEATURE Location/Qualifiers
 source 1..86
 mol_type = protein
 organism = Pichia pastoris

SEQUENCE: 8
 MSTAIPIGGQR TLAKRRAANL DKKQDEPTSA RSAGAGGSSS TMLKLYTDEA QGLKVDPLIV 60
 LVLAVGFIFS VIGLHVVAKL TGKLIN 86

SEQ ID NO: 9 moltype = DNA length = 1800
 FEATURE Location/Qualifiers
 source 1..1800
 mol_type = other DNA
 organism = Pichia pastoris

SEQUENCE: 9
 atgttgaagg atcagttctt gttatgggtt gctttgatag cgagcgtacc ggtttccggc 60
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 gttgttgccg ttcaacagtt ggaacttcagc gttctgaggg gtgattcctt cgaaagtgcc 180
 tcttcagaga acgtgcctcg gcttgtgagg agagatgaca cgctagaagc tgagctaatc 240
 aaccagcaat cattctactt gtcacgactg aaagttggat cacatcaagc ggatattgga 300
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 agtagccgtt cccgcgtgaa gagagatata cacgatgaga agatcgccga atgggatccc 420
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 actagcacta gttctccttc caccgccacg gcaactggta gtggtagtgg tagtggtagt 540
 ggtagtggtg gtggtagtgc tgccacagcc gtatcggtaa gttctgcaca ggcaacattg 600
 gattgctcta cgtatggaac gtttgatcac gctgattcct cgacggtcca tgacaataat 660
 acagactttt tcatctcata cgctgatacc acttttgctt caggaatctg gggttatgac 720
 gacgtcatta tcgacggcat agaggtgaaa gaactttcct tcgccgttgc agacatgacc 780
 aattcctcta ttggtgtggt aggtattgga ctgaaaggcc tagaatccac atatgctagt 840
 gcatcttcgg tcagtgaatg gtatcagtat gacaatttgc cagccaagat ggtcacggat 900
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 aatggaatag atgtgaaaaa ggggttctgc cagggaaactc ttttacaagg gagatttgct 1140
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 tctcttcat cgtctcgac ttccagtga gacgaagaag gagggagctc cgccaacagg 1740
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SEQ ID NO: 10 moltype = AA length = 599
 FEATURE Location/Qualifiers
 source 1..599
 mol_type = protein
 organism = Pichia pastoris

SEQUENCE: 10
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 SSENVPRLVR RDTLEAELI NQSFYLSRL KVGSHQADIG ILVDTGSSDL WVMDSVNPYC 120
 SSRSRVKRDI HDEKIAEWDI INLKKNETSQ NKNFWDWLVG TSTSSPSTAT ATGSGSGSGS 180
 GSGSGSAATA VSVSSAATL DCSTYGTFDH ADSSTFHDNN TDFPISYADT TFASGIWGYD 240
 DVIIDGIEVK ELSFAVADMT NSSIGVLGIG LKGLESTYAS ASSVSEMYQY DNLPAKMVTD 300
 GLINKNAYSL YLNSKDASSG SILFGVDHE KYSGQLLTVP VINTLASSGY REAIRLQITL 360
 NGIDVKKGSQ QGTLQGRFA ALLDSGATLT YAPSSVLNSI GRNLGGSYDS SRQAYTIRCV 420
 SASDTTSLVF NFGGATVEVS LYDLQIATYY TGGSATQCLI GIFSSGSEDF VLGDFTLRS 480
 YVVYDLGLE VSLAQANFNE TDSVVEAITS SVPSATRASG YSSTWSGSAS GTVYTSVQME 540
 SGAASSNSSS GSNMGSSSSS SSSSSSTSSG DEEGSSANR VPFSYLSLCL VVILGVCIV 599

SEQ ID NO: 11 moltype = DNA length = 1781
 FEATURE Location/Qualifiers
 source 1..1781
 mol_type = other DNA
 organism = Pichia pastoris

SEQUENCE: 11
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 aggaccgtt aattaagaga gatgactacg agtccctcga cttgagacac atcgaggtct 180
 tgtacactgc agagatccaa attgagctg acgaaactga aattgaggtc attgtcgaca 240
 ctggttctgc cgaattgtgg gtcatcgatt ccgacgctgc cgtctgtgag ttatcctacg 300
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ctccatcaca agagctcctg gatgggctga gtgagtttg atttgcctc gatggtgaaa 420
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atttcaacat taaccgtgac gagcctgtgt gtgaacagtt tggttccttc gattctagtt 540
cttccgacac ttcccaaagc aacaattcag cttttggtat tgcttacctt gatggaacca 600
ctgctaaccg aacttgggtc agggacacag tccgcatcgg cgactttgcc atcagccaac 660
agagttttgc cttagtcaac atcacagata actacatggg aatcttgggt ctgggtcctg 720
ctacccaaca aaccaccaat agtaacccaa ttgcagcaaa cagatttact tatgatggtg 780
ttgtgggattc attgcggtcc caaggattta tcaattcagc atcgttttct gtttacttgt 840
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ttgaatatgt ttctcaattg ggactatttg tcattcgtag aagttctctg tctgttgcta 1200
gaaataaggt gattgagttc aagtttggtg aagacgttgt gatccaatcc ccagtttctg 1260
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SEQ ID NO: 12      moltype = AA  length = 562
FEATURE            Location/Qualifiers
source             1..562
                   mol_type = protein
                   organism = Pichia pastoris

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SEQUENCE: 12
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VSKREENQDD FNINRDEPVC EQFGSFDSSS SDTFQSNNSA FGIAYLDGTT ANGTWVRDTV 180
RIGDFAISQQ SFALVNITDN YMGILGLGPA TQQTNSNPI AANRFTYDGV VDSLRSQGF 240
NSASFSVYLS PDEDNEHDEF SDGEILFGAI DRAKIDGPRF LFPYVNPYPK VYPDQYTSYV 300
TVSTIAVSST DETLIERRP RLALIDTGAT FSYLPTYPLI RLAFSIHGGF EYVSQLGLFV 360
IRTSSLVAR NKVIEFKPGE DVVIQSPVSD HLLDVSGFLT DGQQYSALTV RESLDGLSIL 420
GDTFIKSAYL FFDNENSQLG IGQINVTDEE DIEVVGDFTI ERDPAYSSTW SSDLPHETPT 480
RALSTASGGG LGTGINTATS RASSRSTSGS TSRTSSTSGS ASGTSSGASS ATQNDTSTD 540
LGAPASLSLA TPCLFAILL ML 562

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SEQ ID NO: 13      moltype = AA  length = 1800
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VARIANT            1..1800
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
VARIANT            7..11
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            15..19
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            23..27
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            31..35
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            39..43
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            47..51
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            55..59
                   note = MISC_FEATURE - This region may encompass "SGGQQ,"
                   "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
                   may be absent
VARIANT            63..67

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	4..67
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	71..90
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	97..101
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	105..109
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	113..117
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	121..125
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	129..133
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	137..141
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	145..149
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	153..157
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	94..157
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	161..180
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	187..191
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	195..199
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	203..207
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	211..215
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	219..223
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	227..231
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	235..239

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	243..247
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	184..247
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	251..270
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	277..281
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	285..289
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	293..297
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	301..305
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	309..313
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	317..321
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	325..329
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	333..337
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	274..337
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	341..360
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	367..371
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	375..379
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	383..387
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	391..395
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	399..403
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	407..411

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	415..419
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	423..427
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	364..427
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	431..450
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	457..461
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	465..469
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	473..477
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	481..485
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	489..493
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	497..501
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	505..509
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	513..517
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	454..517
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	521..540
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	547..551
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	555..559
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	563..567
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	571..575
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	579..583

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	587..591
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	595..599
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	603..607
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	544..607
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	611..630
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	637..641
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	645..649
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	653..657
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	661..665
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	669..673
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	677..681
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	685..689
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	693..697
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	634..697
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	701..720
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	727..731
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	735..739
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	743..747
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	751..755

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	759..763
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	767..771
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	775..779
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	783..787
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	724..787
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	791..810
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	817..821
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	825..829
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	833..837
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	841..845
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	849..853
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	857..861
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	865..869
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	873..877
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	814..877
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	881..900
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	907..911
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	915..919
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	923..927

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	931..935
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	939..943
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	947..951
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	955..959
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	963..967
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	904..967
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	971..990
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	997..1001
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1005..1009
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1013..1017
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1021..1025
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1029..1033
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1037..1041
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1045..1049
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1053..1057
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	994..1057
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1061..1080
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1087..1091
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1095..1099

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1103..1107
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1111..1115
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1119..1123
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1127..1131
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1135..1139
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1143..1147
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1084..1147
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1151..1170
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1177..1181
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1185..1189
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1193..1197
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1201..1205
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1209..1213
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1217..1221
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1225..1229
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1233..1237
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1174..1237
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1241..1260
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1267..1271

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	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1275..1279
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1283..1287
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1291..1295
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1299..1303
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1307..1311
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1315..1319
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1323..1327
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1264..1327
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1331..1350
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1357..1361
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1365..1369
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1373..1377
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1381..1385
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1389..1393
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1397..1401
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1405..1409
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1413..1417
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1354..1417
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent

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VARIANT	1421..1440 note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1447..1451 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1455..1459 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1463..1467 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1471..1475 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1479..1483 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1487..1491 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1495..1499 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1503..1507 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1444..1507 note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1511..1530 note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1537..1541 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1545..1549 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1553..1557 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1561..1565 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1569..1573 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1577..1581 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1585..1589 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1593..1597 note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1534..1597

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	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1601..1620
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1627..1631
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1635..1639
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1643..1647
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1651..1655
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1659..1663
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1667..1671
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1675..1679
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1683..1687
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1624..1687
	note = MISC_FEATURE - This region may encompass 4-8 repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent
VARIANT	1691..1710
	note = MISC_FEATURE - This region may encompass 6-20 residues
VARIANT	1717..1721
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1725..1729
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1733..1737
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1741..1745
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1749..1753
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1757..1761
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1765..1769
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent
VARIANT	1773..1777

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VARIANT
note = MISC_FEATURE - This region may encompass "SGGQQ,"
"GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions
may be absent
1714..1777
note = MISC_FEATURE - This region may encompass 4-8
repeating "GPG-X1" repeating units, wherein X1 is "SGGQQ,"
"GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may
be absent
1781..1800
note = MISC_FEATURE - This region may encompass 6-20
residues
VARIANT
1..1800
note = MISC_FEATURE - This sequence may encompass 2-20
"GGY-[GPG-X1]n1-GPS-(A)n2" repeating units, wherein X1 is
"SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," n1 is 4-8 and
n2 is 6-20 and some positions may be absent
source
1..1800
mol_type = protein
organism = synthetic construct

SEQUENCE: 13
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 60
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 120
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 180
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 240
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 300
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 360
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 420
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 480
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 540
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 600
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 660
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 720
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 780
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 840
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 900
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 960
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 1020
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 1080
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 1140
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 1200
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 1260
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 1320
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 1380
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 1440
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 1500
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 1560
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 1620
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG 1680
PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG 1740
XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG PGXXXXXGPGS AAAAAAAAAA AAAAAAAAAA 1800

SEQ ID NO: 14
FEATURE
VARIANT
1..5
note = Description of Artificial Sequence: Synthetic peptide
source
1..5
mol_type = protein
organism = synthetic construct

SEQUENCE: 14
SGGQQ 5

SEQ ID NO: 15
FEATURE
VARIANT
1..5
note = Description of Artificial Sequence: Synthetic peptide
source
1..5
mol_type = protein
organism = synthetic construct

SEQUENCE: 15
GAGQQ 5

SEQ ID NO: 16
FEATURE
VARIANT
1..5
note = Description of Artificial Sequence: Synthetic peptide
source
1..5

-continued

	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 16		
GQGPY		5
SEQ ID NO: 17	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
VARIANT	1..4	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..4	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 17		
AGQQ		4
SEQ ID NO: 18	moltype = AA length = 70	
FEATURE	Location/Qualifiers	
VARIANT	1..70	
	note = Description of Artificial Sequence: Synthetic polypeptide	
VARIANT	7..11	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	15..19	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	23..27	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	31..35	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	39..43	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	47..51	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	55..59	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	63..67	
	note = MISC_FEATURE - This region may encompass "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," wherein some positions may be absent	
VARIANT	4..67	
	note = MISC_FEATURE - This region may encompass 4-8 "GPG-X1" repeating units, wherein X1 is "SGGQQ," "GAGQQ," "GQGPY," "AGQQ" or "SQ," and some positions may be absent	
source	1..70	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 18		
GGYGPXXXXX XGPGXXXXXG PGXXXXXGPG XXXXXGPGXX XXXGPGXXXXX XGPGXXXXXG		60
PGXXXXXGPS		70
SEQ ID NO: 19	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
VARIANT	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
VARIANT	1..20	
	note = MISC_FEATURE - This sequence may encompass 6-20 residues	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 19		
AAAAAAAAAA AAAAAAAAAA		20

-continued

SEQ ID NO: 20 moltype = DNA length = 2835
 FEATURE Location/Qualifiers
 misc_feature 1..2835
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..2835
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 20

ggtggttacg	gtccaggcgc	tggtaacaa	ggtccaggaa	gtggtggtca	acaaggacct	60
ggcgggtcaag	gacctacgg	tagtggccaa	caagggtccag	gtggagcagg	acagcagggt	120
ccgggaggcc	aaggacctta	cggaccagg	gctgctgctg	ccgcccgtgc	cgctgccgga	180
ggttacggtc	caggagccgg	acaacagggt	ccagggtggag	ctggacaaca	aggtccaggga	240
tcacaaggtc	ctggtggaca	aggtccatac	ggtcctggtg	ctggtcaaca	gggaccagg	300
agtcaaggac	ctggttcagg	tggtcagcag	ggtccaggag	gacagggtcc	ttacggccct	360
tctgccgtg	cagcagcagc	cgctgccgca	ggaggatacg	gacctggtgc	tggacaacga	420
tctcaaggac	caggaggaca	aggtccttat	ggacctggcg	ctggccaaca	aggacctggt	480
tctcagggtc	caggttcagg	aggccaacaa	ggcccaggag	gtcaaggacc	atacggacca	540
tccgctgcgg	cagctgcagg	tgctgcagg	ggatatggcc	caggagccgg	acaacagggt	600
cctggttcac	aagggtccag	atctggtggt	caacagggac	caggccggcca	gggaccttat	660
ggtccaggag	ccgctgcagc	agcagcagct	gttgagggtt	acggccctgg	tgcgggtcaa	720
caaggcccg	gatctcagg	tctggatct	ggaggacaac	aaggtcctgg	aggtcagggt	780
ccatacggac	cttcaggagc	agctgctgct	gcagccgctg	gtggttatgg	acctggtgct	840
ggtcaacaag	gaccgggttc	tcagggtccg	ggttcaggag	gtcagcagg	ccctggtgga	900
caaggacctt	atggacctag	tgcggctgca	gcagctgccg	ccgcagggtg	ttacgggtcca	960
ggcgtggttc	aacaagggtc	aggaagtgg	ggtcaacaag	gacctggcgg	tcaaggacct	1020
tacggtagt	gccacaagg	tccagggtga	gcaggacagc	agggcccggg	aggccaagga	1080
ccctacggac	caggtgctgc	tgtgccgccc	gctgccgctg	ccggagggtta	cggtccaggga	1140
gcccggacaac	aggttcagg	tggagctgga	caacaaggtc	caggatcaca	aggtcctggt	1200
ggacaaggtc	catacggctc	tgtgtctggt	caacaggggac	caggtagtca	aggacctggt	1260
tcagggtggt	agcagggttc	atgaggacag	ggtccttacg	gcccttctgc	cgctgcagca	1320
gcagccgctg	ccgacaggag	atacggacct	ggtgctggac	aacgatctca	aggaccaggga	1380
ggacaaggtc	cttatggacc	tggcgctggc	caacaaggac	ctggttctca	gggtccagg	1440
tcaggaggcc	aacaaggccc	aggaggtcaa	ggaccatacg	gacctccgc	tgcggcagct	1500
gcagctgctg	caggttgata	tggcccaggga	gcccggacaac	agggctcctg	ttcacaagg	1560
ccaggatctg	gtggtcaaca	gggaccaggc	ggccaggggac	cttatggctc	aggagccgct	1620
gcagcagcag	cagctgttgg	aggttacggc	cctggtgccg	gtcaacaagg	cccaggatct	1680
cagggtcctg	gatctggagg	acaacaagg	cctggaggtc	aggggtccata	cggaccttca	1740
gcagcagctg	ctgctgcagc	cgctggtggt	tatggacctg	gtgctggtca	acaaggaccg	1800
ggttctcagg	gtccgggttc	aggaggtcag	cagggccctg	gtggacaagg	accttatgga	1860
ccatagtgcg	ctgcagcagc	tgcgcgccga	ggtggttacg	gtccaggcgc	tggtaacaa	1920
ggtccaggaa	gtggtggtca	acaaggacct	ggcgggtcaag	gacctacagg	tagtggccaa	1980
caagggtccag	gtccgacagg	acagcagggt	ccgggaggcc	aaggacctta	cggaccagg	2040
gctgctgctg	ccgcgcgtgc	cgctgcggga	ggttacggtc	caggagccgg	acaacagggt	2100
ccagggtggag	ctggacaaca	aggtccaggga	tcacaaggtc	ctggtggaca	aggtccatac	2160
ggctcctggtg	ctggtcaaca	gggaccagg	agtcaggac	ctggttcagg	tggtcagcag	2220
ggtccaggag	gacagggttc	ttacggccct	tctgccgctg	cagcagcagc	cgctgccgca	2280
ggaggatacg	gacctggtgc	tggacaacga	tctcaaggac	caggaggaca	aggtccttat	2340
ggacctggcg	ctggccaaca	aggacctggt	tctcagggtc	caggttcagg	aggccaacaa	2400
ggcccaggag	gtcaaggacc	atacggacca	tccgctgcgg	cagctgcagc	tgctgcagg	2460
ggatatggcc	caggagccgg	acaacagggt	cctggttcac	aaggtccagg	atctggtggt	2520
caacagggg	caggccggca	gggaccttat	ggtccaggag	ccgctgcagc	agcagcagct	2580
gttggagggt	acggccctgg	tgcgggtcaa	caaggcccag	gatctcagg	tctggtgct	2640
ggaggacaac	aaggtcctgg	aggtcagggt	ccatacggac	cttcagcagc	agctgctgct	2700
gcagccgctg	gtggttatgg	acctggtgct	ggtcaacaag	gaccgggttc	tcagggtccg	2760
ggttcaggag	gtcagcagg	ccctggtgga	caaggacctt	atggacctag	tgcgggtgca	2820
gcagctgccg	ccgca					2835

SEQ ID NO: 21 moltype = AA length = 945
 FEATURE Location/Qualifiers
 VARIANT 1..945
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 source 1..945
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 21

GGYGPAGQ	Q	GGPGSGQ	Q	QPGGAGQ	Q	PGGQPGYGP	AAAAA	60
GYGPAGQ	Q	PGGAGQ	Q	SGPGGQ	Q	PGGAGQ	Q	120
SAAAAA	Q	GGYGPAGQ	R	SGPGGQ	Q	PGGAGQ	Q	180
SAAAAA	Q	GGYGPAGQ	Q	PGGAGQ	Q	PGGAGQ	Q	240
QPGGSG	Q	GGGQ	Q	PGGQ	Q	PGGQ	Q	300
QPGYPSA	A	AAAAAGY	G	GAGQ	Q	PGGQ	Q	360
PYPGAAAA	Q	AAAAAGY	G	AGQ	Q	PGGQ	Q	420
SGGQ	Q	PGGQ	Q	PGGQ	Q	PGGQ	Q	480
SGGQ	Q	PGGQ	Q	PGGQ	Q	PGGQ	Q	540

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AAAAAVGGYG	PGAGQQGP	SGSGGQQG	PGGQGPYGPS	AAAAAAAGG	YGPAGQQGP	600
GSQGPSSGGQ	QGPGGQGPY	PSAAAAAAA	GGYGPAGQQ	PGSGGQQGP	GGQGPYSSGQ	660
QGPAGAGQQG	PGGQGPYGP	AAAAAAAAG	GYGPAGQQG	PGAGQQGPG	SQGPGGQGPY	720
GPAGQQGPG	SQGPSSGGQ	GPGGQGPYGP	SAAAAAAA	GGYGPAGQR	SQGPGGQGPY	780
GPAGQQGPG	SQGPSSGGQ	GPGGQGPYGP	SAAAAAAA	GYGPAGQQG	PGSQGPSSG	840
QGPGGQGPY	GPAAAAAA	VGGYGPAGQ	QGPSSQGP	GGQGPGGQ	PYGPSAAAA	900
AAAGYGP	GQQGPSSGP	SGSGQGP	QGPYGPSAA	AAAAA		945

SEQ ID NO: 22	moltype = AA	length = 315
FEATURE	Location/Qualifiers	
VARIANT	1..315	
	note = Description of Artificial Sequence: Synthetic polypeptide	
source	1..315	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 22		
GGYGPAGQQ	GPSSGGQGP	GGQGPYSSGQ
GYGPAGQQG	PGAGQQGPG	SQGPGGQGPY
SAAAAAAA	GGYGPAGQR	SQGPGGQGPY
SAAAAAAG	GYGPAGQQG	PGSQGPSSG
QGPSSQGP	GGQGPSSG	PYGPSAAAA
QGPYGPSAA	AAAAA	

SEQ ID NO: 23	moltype = AA	length = 5
FEATURE	Location/Qualifiers	
VARIANT	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 23		
SGAGG		5

SEQ ID NO: 24	moltype = AA	length = 5
FEATURE	Location/Qualifiers	
VARIANT	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
GSAG		5

SEQ ID NO: 25	moltype = AA	length = 5
FEATURE	Location/Qualifiers	
VARIANT	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 25		
GGGA		5

SEQ ID NO: 26	moltype = AA	length = 181
FEATURE	Location/Qualifiers	
source	1..181	
	mol_type = protein	
	organism = Aliatypus gulosus	
SEQUENCE: 26		
GAASSSTII	TTKSASASAA	ADASAAATAS
IFGSSISSY	AAAIASASR	AAESNGYTT
DALSHALLYT	GRLNTANANS	LASAFAYAF
S		

SEQ ID NO: 27	moltype = AA	length = 126
FEATURE	Location/Qualifiers	
source	1..126	
	mol_type = protein	
	organism = Plectreureys tristis	
SEQUENCE: 27		
GAGAGAGAGA	GAGAGAGGA	STSVSTSSSS
AGAGFGSLG	LGYGVLSSA	QAQAQAQAAA
AAAAAA		

SEQ ID NO: 28	moltype = AA	length = 239
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FEATURE	Location/Qualifiers	
source	1..239	
	mol_type = protein	
	organism = Plectreureys tristis	
SEQUENCE: 28		
GAAQKQPSGE	SSVATASAAA	TSVTSGGAPV GKPVPAPIF YPQGFLQQGP APGPSNVQPG 60
TSQQGPIGGV	GGSNAFSSSF	ASALSLNRGF TEVISSASAT AVASAFQKGL APYGTAFALS 120
AASAAADAYN	SIGSGANAPA	YAQAFARVLY PLVQQYGLSS SAKASAFASA IASSFSSGTS 180
GQPSIGQQQ	PPVTISAASA	SAGASAAAVG GGQVGQGPYG GQQQSTAASA SAAAATATS 239
SEQ ID NO: 29	moltype = AA length = 182	
FEATURE	Location/Qualifiers	
source	1..182	
	mol_type = protein	
	organism = Araneus gemmoides	
SEQUENCE: 29		
GNVGYQLGLK	VANSLGLGNA	QALASSLSQA VSAVGVGASS NAYANAVSNA VGQVLAGQGI 60
LNAANAGSLA	SSFASALSSS	AASVASQSAS QSQAASQSQA AASAFRQAAS QSASQSDSRA 120
GSQSSTKTTT	TSTSGSQADS	RSASSSASQA SASAFAQQSS ASLSSSSSFS SAFSSATSIS 180
AV		182
SEQ ID NO: 30	moltype = AA length = 180	
FEATURE	Location/Qualifiers	
source	1..180	
	mol_type = protein	
	organism = Argiope aurantia	
SEQUENCE: 30		
GSLASSFASA	LSASAASVAS	SAAAQAASQS QAAASAFSRA ASQSASQSAA RSGAQSISTT 60
TTTSTAGSQA	ASQSASSAAS	QASASSFARA SSASLAASSS FSSAFSSANS LSALGNVGYQ 120
LGFNVANNLG	IGNAAGLGNA	LSQAVSSVGV GASSSTYANA VSNVAGQFLA GQGILNAANA 180
SEQ ID NO: 31	moltype = AA length = 199	
FEATURE	Location/Qualifiers	
source	1..199	
	mol_type = protein	
	organism = Deinopis spinosa	
SEQUENCE: 31		
GASASAYASA	ISNAVGPYLY	GLGLFNQANA ASFASSFASA VSSAVASASA SAASSAYAQS 60
AAAQAQAASS	AFSQAAQSA	AAASAGASAG AGASAGAGAV AGAGAVAGAG AVAGASAAAA 120
SQAAASSSAS	AVASAFQSA	SYALASSSAF ANAFASATSA GYLGSLAYQL GLTTAYNLGL 180
SNAQAFASTL	SQAVTGVGL	199
SEQ ID NO: 32	moltype = AA length = 171	
FEATURE	Location/Qualifiers	
source	1..171	
	mol_type = protein	
	organism = Nephila clavipes	
SEQUENCE: 32		
GATAASYGNA	LSTAAQFFA	TAGLLNAGNA SALASSFARA FSASAESQSF AQSQAFAQAS 60
AFQQAASRSA	SQSAAEAGST	SSSTTTTSTA ARSQAAQSQA SSSYSSAFQA AASSSLATSS 120
ALSRPFSSVS	SASAASSLAY	SIGLSAARSL GTADAAGLAG VLARAAGALG Q 171
SEQ ID NO: 33	moltype = AA length = 268	
FEATURE	Location/Qualifiers	
source	1..268	
	mol_type = protein	
	organism = Argiope trifasciata	
SEQUENCE: 33		
GGAPGGGPGG	AGPGGAGFGP	GGGAGFGPGG GAGFGPGGAA GGPGGPGGPG GPGGAGGYGP 60
GGAGGYGPGG	VGPGGAGGYG	PGGAGGYGPG GSGPGGAGPG GAGGEGPVTV DVDVTVGPEG 120
VGGPGGAGP	GGAGFGPGGG	AGFGPGGAPG APGGPGGPGG PGGPGGPGGV GPGGAGGYGP 180
GGAGGVGPAG	TGGFGPGGAG	GFGPGGAGGF GPGGAGGFPG AGAGGYGPGG VGPGGAGGFG 240
PGGVGPGGSG	PGGAGGEGPV	TVDVDVSV 268
SEQ ID NO: 34	moltype = AA length = 420	
FEATURE	Location/Qualifiers	
source	1..420	
	mol_type = protein	
	organism = Nephila clavipes	
SEQUENCE: 34		
GVSYGPGGAG	GPYGPGGPYG	PGGEGPGGAG GPYGPGGVGP GSGPGGGYGP GGAGPGGYGP 60
GGSGPGGYGP	GGSGPGGYGP	GGSGPGGYGP GSGPGGGYGP GGYGPGGSGP GGSGPGGSGP 120
GGYGPGGTGP	GGSGPGGYGP	GGSGPGGSGP GGYGPGGSGP GGFGPGGSGP GGYGPGGSGP 180
GGAGPGGVGP	GGFGPGGAGP	GGAAPGGAGP GGAGPGGAGP GGAGPGGAGP GGAGPGGAGP 240
AGGAGGSGGA	GGSGGTTIIE	DLDTIDGAD GPITISEELP ISGAGGSGPG GAGPGGVGPG 300
GSGPGGVGPG	GSGPGGVGPG	GSGPGGVGPG GAGGPGYGP 360

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SYGPGGSGGP GGAGGPYGP GEGPGGAGGP YGPGGAGGPY GPGGAGGPY GGEPPYGP 420

SEQ ID NO: 35 moltype = AA length = 376
 FEATURE Location/Qualifiers
 source 1..376
 mol_type = protein
 organism = *Latrodectus hesperus*

SEQUENCE: 35
 GINVDSDIGS VTSLLSGST LQMTIPAGGD DLSGGYPGGF PAGAQPSSGA PVDGPGPSAG 60
 GDVAKLARS LASTLASSGV FRAAFNSRVS TPVAVQLTDA LVQKIASNLG LDYATASKLR 120
 KASQAVSKVR MGSNTNAYAL AISSALAEVL SSSGKVADAN INQIAPQLAS GIVLGVSTTA 180
 PQFGVDLSSI NVNLDISNVA RNMQASIQGG PAPITAEGPD FGAGYPPGAP TDLSGLDMGA 240
 PSDGSRGGDA TAKLLQALVP ALLKSDVFRA IYKRGTRKQV VQYVNSALQ QAASSLGLDA 300
 STISQLQTKA TQALSSVSAD SDSTAYAKAF GLAIAQVLGT SGQVNDANVN QIGAKLATGI 360
 LRGSSAVAPR LGIDLS 376

SEQ ID NO: 36 moltype = AA length = 200
 FEATURE Location/Qualifiers
 source 1..200
 mol_type = protein
 organism = *Argiope trifasciata*

SEQUENCE: 36
 GAGYTGPSGP STGSPGYGP LGGGAPPGQS GFGGSAGPQG GFGATGGASA GLISRVANAL 60
 ANTSTLRTVL RTGVSQQIAS SVVQRAAQL ASTLGVDGNN LARFAVQAVS RLPAGSDTSA 120
 YAQAFSSALF NAGVLNASNI DTLGSRVLSA LLNGVSSAAQ GLGINVDSGS VQSDISSSS 180
 FLSTSSSSAS YSQASASST 200

SEQ ID NO: 37 moltype = AA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = protein
 organism = *Uloborus diversus*

SEQUENCE: 37
 GASAADIATA IAASVATSLQ SNGVLTASN SLSNQLASY VSSGLSSTAS SLGIQLGASL 60
 GAGFGASAGL SASTDISSSV EATSASTLSS SASSTSVVSS INAQLVPALA QTAVLNAAFS 120
 NINTQNAIRI AELLTQQVGR QYGLSGSDVA TASSQIRSA YSVQQGSASS AYVSAIVGPL 180
 ITALSSRGVV NASNSSQIAS SLATAILQFT ANVAPQFGIS IPTSAVQSDL STISQSLTAI 240
 SSQTSSSVDS STSAFGGISG PSGSPSPYGPQ PSGPTFGPGP SLSGLTGFTA TFASSFKSTL 300
 ASSTQFQLIA QSNLDVQTRS SLISKVLINA LSSLGISASV ASSIAASSSQ SLLSVSA 357

SEQ ID NO: 38 moltype = AA length = 32
 FEATURE Location/Qualifiers
 source 1..32
 mol_type = protein
 organism = *Euprostenops australis*

SEQUENCE: 38
 GGQGGGQGR YGQAGSSAA AAAAAAAAAA AA 32

SEQ ID NO: 39 moltype = AA length = 42
 FEATURE Location/Qualifiers
 source 1..42
 mol_type = protein
 organism = *Tetragnatha kauaiensis*

SEQUENCE: 39
 GGLGGGQAG QGGQAGAGG GYGSGLGGAG QGASAAAAAA AA 42

SEQ ID NO: 40 moltype = AA length = 42
 FEATURE Location/Qualifiers
 source 1..42
 mol_type = protein
 organism = *Argiope aurantia*

SEQUENCE: 40
 GYGPGAGQQ GPGSQGPGSG GQQGPGGLGP YGPSAAAAAA AA 42

SEQ ID NO: 41 moltype = AA length = 46
 FEATURE Location/Qualifiers
 source 1..46
 mol_type = protein
 organism = *Deinopis spinosa*

SEQUENCE: 41
 GPGGYGGPGQ QGPGQGYGP GTGQGGGQPS GQQGPAGAAA AAAAAA 46

SEQ ID NO: 42 moltype = AA length = 42
 FEATURE Location/Qualifiers
 source 1..42
 mol_type = protein

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organism = *Nephila clavata*

SEQUENCE: 42
 GPGGYGLGQQ GPGQQGPGQQ GPAGYGPSGL SGPGGAAAA AA 42

SEQ ID NO: 43 moltype = AA length = 3360
 FEATURE Location/Qualifiers
 source 1..3360
 mol_type = protein
 organism = *Argiope bruennichi*

SEQUENCE: 43

MNWSIRLALL	GFVVLSTQTV	FAVGQAATPW	ENSQLAEDFI	NSFLRFIAQS	GAFSPNQLDD	60
MSSIGDTLKT	AIEKMAQSRK	SSKSKLQALN	MAFASSMAEI	AVAEQGGLSL	EAKTNAIANA	120
LASAFLETTG	FVNQQFVSEI	KSIIYMQA	SSNEISGSAA	AAGGSGGGG	SGSQGGYGG	180
ASASASAAAA	YGSAPQAGG	PAPQGPSQQG	PVSQGPYPGP	AAAAAAAAGG	YGPAGQQGP	240
GSQGPSGGQG	QGPSQGPS	GGQQGPGQG	PYGPSAAAA	AAAAGGYGPG	AGQQGPSQG	300
PGSGGQGGPG	GGYPYGPAA	AAAAAVGGY	PGAGQQGPS	QGPSGGQG	PGQGPYPGS	360
AAAAAAAAGG	YGPAGQQGP	GSQGPSGGQ	QGPQGPGYPG	PSAAAAAA	GGYGPAGQQ	420
GPSSGGQQGP	GGQGPYSGQ	QGPAGAGQQG	PGQGPYPGP	AAAAAAAAG	GYGPAGQQG	480
PGAGQQGGPG	SGPGGQGPY	GPAGQQGPS	SGPGSGGQG	GPQGQGPYP	SAAAAAAG	540
GYGPAGQQG	PGSQGPSGG	QQGPSQGP	SGGQQGPGQ	GPYPGSA	AAAAGGYGPG	600
AGQQGPSQG	PGSGGQGP	GGQPYGPAA	AAAAAVGGY	PGAGQQGPS	QGPSGGQG	660
PGQGPYPGS	AAAAAAGG	YGPAGQQGP	GSQGPSGGQ	QPGGQGPYP	PSAAAAAA	720
GGYGPAGQQ	GPSSGGQGP	GGQPYSGQ	QGPAGAGQQG	PGQGPYPGP	AAAAAAAAG	780
GYGPAGQQG	PGGAGQGP	SGPGGQGPY	GPAGQQGPS	SGPGSGGQG	GPGGQGPYP	840
SAAAAAAG	GGYGPAGQ	SGPGGQGPY	GPAGQQGPS	SGPGSGGQG	GPGGQGPYP	900
SAAAAAAG	PGAGRQGPS	QGPSGGQG	PGQGPYPGS	AAAAAAGP	GARRQGPSQ	960
GPSSGGQGP	GGQPYSGQ	QGPAGAGQQG	PGQGPYPGP	AAAAAAAAG	GYGPAGQQG	1020
PGAGQQGGPG	SGPGGQGPY	GPAGQQGPS	SGPGSGGQG	GPQGQGPYP	SAAAAAAG	1080
GYGPAGQQG	PGSQGPSGG	QQGPGGQPY	GPAAAAAA	VGGYGPAGQ	QGPSQGPS	1140
GGQGPGGQG	PYGPSAAAA	AAAGGYPGA	GGQGPSQG	SGGQQGPG	QGPYPGSA	1200
AAAAAGYGP	GAGQGPSG	GGQGPSQG	YSGQGPSG	AGQGPSQG	PYGPAAAA	1260
AAAAGGYGP	AGQGPSG	QQGPEGPSQ	GPSSGGQGP	GGQPYPGA	AAAAAVGGY	1320
GPAGQQGGPG	SGPGSGGQG	GPQGQGPYP	SAAAAAAG	GYGPAGQQG	PGSQGPSGG	1380
QQGPGGQGPY	GPSAAAAA	AGGYGPAGQ	QGPSGGQG	PGQGPYSG	QQGPGAGQ	1440
GPQGQGPYP	GAAAAAAG	GGYGPAGQ	GPAGQQGP	SGQGPSQG	YGPAGQQGP	1500
GSQGPSGGQ	QPGGQGPYP	PSAAAAAA	GGYGPAGQ	GPSSQGPSG	GGQGPSQG	1560
SGGQQGPSG	GGYPGSA	AAAAAGGY	PGAGQQGPS	QGPSGGQG	PGQGPYPGP	1620
AAAAAAVGG	YGPAGQQGP	GSQGPSGGQ	QPGGQGPYP	PSAAAAAA	GGYGPAGQ	1680
GPSSQGPSG	QQGPGGQGP	YGPSAAAA	AAGGYGPAG	QQGPSGGQ	GPQGQGPYP	1740
GGQGPAGQ	QPGGQGPS	PAAAAAAG	AGGYGPAGQ	QPGGAGQQG	PGSQPGQG	1800
PYGPAGQQG	PGSQGPSGG	QQGPGGQPY	GPSAAAAA	AGGYGPAGQ	QGPSQGPS	1860
GGQGPGGQ	PYGPSAAAA	AAAGGYPGA	GGQGPSGGQ	QPGGQGPYP	SGQGPSG	1920
QQGPGGQGPY	GGYGPAGQ	QGPSQGPS	GGQGPSQG	PYGPSAAAA	AAAGPGARRQ	1980
GPSSQGPSG	QQGPGGQGP	YSGQGPSG	AGQGPSQG	PYGPAAAA	AAAAGGYGP	2040
AGQGPSG	QGPSQGP	GGQGPSQG	SGGQQPGQ	GPYPGSA	2100	
AAAAAGYGP	GAGQGPSG	PGSSGGQGP	GGQPYPGA	AAAAAVGGY	GPAGQQGP	2160
SGPGSGGQG	GPQGQGPYP	SAAAAAAG	GYGPAGQQG	PGSQGPSGG	QQGPGGQPY	2220
GPSAAAAA	AGGYGPAGQ	QGPSGGQG	PGQGPYSG	QQGPGAGQ	GPQGQGPYP	2280
GAAAAAAG	GGYGPAGQ	PGGAGQQGP	SGQPGGQGP	YGPAGQQGP	GSQGPSGGQ	2340
QPGGQGPYP	PSAAAAAA	GPAGRQGP	SGPGSGGQ	PGGQGPYP	SAAAAAAG	2400
PGARRQGPS	QGPSGGQG	PGQGPYSG	QQGPGAGQ	GPQGQGPYP	GAAAAAAG	2460
GGYGPAGQ	PGGAGQQGP	SGQPGGQGP	YGPAGQQGP	GSQGPSGGQ	QGPSQGPYP	2520
PSAAAAAA	GGYGPAGQ	GPSSQGPSG	GGQPGGQGP	YGPSAAAA	AAGGYGPAG	2580
QQGPSGGQ	PGGQGPYP	GGQPGGAGQ	QPGGQGPYP	PAAAAAAG	AGGYGPAGQ	2640
QGPSQGPS	GGQGPSG	PGSSGGQGP	GGPYGPSA	AAAAAAGGY	GPAGQQGP	2700
SGPGSGGQG	GPQGQGPYP	GAAAAAAG	GYGPAGQQG	PGSQGPSGG	QQGPGGQPY	2760
GPSAAAAA	AGGYGPAGQ	QGPSQGPS	GGQGPSQG	PYGPSAAAA	AAAGGYPGA	2820
GGQGPSGGQ	QPGGQGPYP	SGQGPSG	QQGPGGQPY	GPAAAAAA	AAGGYGPAG	2880
QQGPGAGQ	GPSSQPGQ	GPYPGAGQ	GPSSQGPSG	GGQPGGQGP	YGPSAAAA	2940
AAGGYGPAG	QGPSQGP	SGGQGPSG	GPYPGAAAA	AAAVGGYGP	AGQGPSQG	3000
PGSGGQGGPG	GGPYGPSA	AAAAAGGY	PGAGQQGPS	QGPSGGQG	PGQGPYPGS	3060
AAAAAAAAG	YGPAGQQGP	SGGQGPSG	QGPYSGQG	PGGAGQGP	GGQPYPGA	3120
AAAAAAGGY	GPAGQQGP	GAGQGPSQ	GPQGQGPYP	GAGQGPSG	GPSSGGQGP	3180
GGQGPYPGSA	AAAAAAGG	GPAGQQGP	SGPGSGGQ	GPQGQGPYP	SAAAAAAG	3240
GYGPAGQQG	PSSQAPVASA	AASRLSSQA	SARVSSAVST	LVSSGPTSPA	ALSNAISSV	3300
SVVSASNPGL	SGCDVLVQAL	LEIVSALVHI	LGSSSIGQIN	YAASSQYQM	VGNSVAQALG	3360

SEQ ID NO: 44 moltype = AA length = 3129
 FEATURE Location/Qualifiers
 source 1..3129
 mol_type = protein
 organism = *Latrodectus hesperus*

SEQUENCE: 44

MTWSTRLLS	FLFVLCTQSL	YALAQANTPW	SSKANADAFI	NSFISASANT	GSFSQDMED	60
MSLIGNTLMA	AMDNMGGRT	PSKLQALDMA	FASSVAEIAA	SEGDLGVTT	NAIADALTS	120
FYQTTGVVNS	RFISEIRSLI	GMFAQASAND	VYASAGSSGG	GGYGASSASA	ASASAAAPSG	180

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VAYQAPAAQ	ISFTLRGQ	VSQGGAGP	GGAGAAAAA	AAAGGAGQGG	QGGYGQGGYG	240
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GGTGQGGAGA	AAAAAAGGA	GQGGQGGYGQ	GGYGQGGYGQ	GGSGAAAAA	AAAGGAGQGG	360
QGGYGQGGYG	QGGAGQGGAG	AAAAAAAAG	GAGQGGYGRG	GAGQGGAAAA	AAAAAGAGQG	420
GYGGQGGAGQ	GSGAAAAAA	AGGAGQGGQ	GYGQGGYGQ	GSGAAAAAA	AGGAGQGGQ	480
GYGQGGYGQ	GAGQGGAGAA	AAAAAAGGAG	QGGQGGYGQ	GYGQGGAGQ	GAGAAAAAA	540
AGGAGQGGQ	GYGQGGYGQ	GAGQGGAGAA	AAAAAAGGAG	QGGQGGYGQ	GYGQGGAGQ	600
GAAAAAAA	GGAGQGGYGR	GGAGQGGAAA	AAGAGQGGYG	QGGAGQGGAG	AAAAAAAAG	660
AGQGGQGGYG	RGYGGQGGAG	QGGAGAAAA	AAAGGAGQGG	QGGYGQGGYG	QGGAGQGGAA	720
AAAAAAGGAG	QGGYGRGGAG	QGGAAAAAA	AAGAGQGGYG	QGGAGQGGAG	AAAAAAAAG	780
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AAASAAAAAG	AGQGGYGRGG	AGQGGAAAA	GAGQGGYGQ	GAGQGGAGAA	AAAAAAGGAG	900
QGGQGGYGRG	GYGRGGAGQ	GAGAAAAATA	AGGAGQGGQ	GYGQGGYGQ	GAGQGGAAAA	960
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AAAAAGGAGQ	GAGAAAAAA	AGGAGQGGQ	GGYGGQAGQ	GGAGAAAAA	RGAGQGGQGG	1140
YGRGGYGQGG	AGQGGAGAAA	AAAAAGGAGQ	GGQGGYGQGG	YGQGGAGQGG	AAAAAAAAG	1200
AGQGGYGRGG	AGQGGAAAA	GAGQGGYGQ	GAGQGGAGAA	AAAAAAGGAG	QGGQGGYGRG	1260
GYGQGGAGQ	GAGAAAAAA	AGGAGQGGQ	GYGQGGYGQ	GAGQGGAAAA	AAAAAGGAGQ	1320
GGYGRGGAGQ	GAAAAAAA	AGSGQGGYG	QGGAGQGGAG	AAAAAAGGA	GQGGQGGYGR	1380
GGYGGQGGAGQ	GGAGAAAAA	AAGGAGQGGQ	GGYGGQGGYG	GGAGQGGAAA	AAAAAAGGA	1440
GGQGGYGRGA	GAGAAAAAA	AGQGGAGQGG	AGQGGAGAAA	AAAAAGGAGQ	GGQGGYGRGG	1500
YGQGGAGQGG	AGTAAAAAA	GGAGQGGQGG	YGQGGYGQGG	AGQGGAAAA	AAAAGGAGQGG	1560
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GQGGAGQGG	AAAAAAGAG	GQGGYGQGG	GQGGAAAAA	AAAGGAGQGG	2040	
YGRGGAGQGG	AAAAAAGAG	SGQGGYGQGG	AGQGGAGAAA	AAAAAGGAGQ	GGQGGYGRGG	2100
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GGAGQGGYGR	GGAGQGGAA	AAGAGQGGYG	QGGAGQGGAG	AAAAAAGAG	AGQGGQGGYG	2220
RGYGGQGGAG	QGGAGAAAA	AAAGGAGQGG	QGGYGQGGNG	QGGAGQGGAA	AAAAAAGGAG	2280
QGGYGRGGAG	QGGAAAAAA	AAGAGQGGYG	GQGGAGQGGAG	AAAAAAGAG	AGQGGQGGYG	2340
RGYGGQGGAG	QGGAGAAAA	AAAGGAGQGG	QGGYGQGDY	QGGAGQGGAA	AAAAAAGGAG	2400
QGGYGRGGAG	QGGAAAAAG	GQGGYGQGA	GQGGAGAAAA	AAAGGAGRG	GQGGYGRGGY	2460
GQGGAGQGG	GAAAAAAG	GAGQGGQGGY	GQGGYGQGG	QGGAGAAAA	AAAGGAGQGG	2520
YRGGAGQGG	AAAAAAGAG	GYGGQAGQ	GAGAAAAA	AGGAGRGQ	GYRGGYGQ	2580
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AGQGGAAAA	AAAAAGAGQ	YGGQGGAGQ	AGAAAAAAA	GGAGQGGQGG	YRGGYGQGG	2940
AGQGGAGAAA	AGGAGQGGQ	GYGQGGYGQ	GAGQGGAAAA	AAAAAAGGAG	QGGYGGYGQ	3000
GGAGAAAAA	SGPQIYYG	QSVAAPAAA	ASALAAPATS	ARISSHASAL	LSNGPTNPAS	3060
ISNVISNAVS	QISSNPGAS	ACDVLVQALL	ELVTALLTII	GSSNIGSVNY	DSSGQYQV	3120
TQSVQNAFA						3129

SEQ ID NO: 45	moltype = DNA length = 20
FEATURE	Location/Qualifiers
misc_feature	1..20
	note = Description of Artificial Sequence: Synthetic primer
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 45	
tactacaggc tggctgttcc	20
SEQ ID NO: 46	moltype = DNA length = 46
FEATURE	Location/Qualifiers
misc_feature	1..46
	note = Description of Artificial Sequence: Synthetic primer
source	1..46
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 46	
ctcacttaat cttctgtact ctgaagaagt ccaactgttg aacgcc	46
SEQ ID NO: 47	moltype = DNA length = 46
FEATURE	Location/Qualifiers
misc_feature	1..46
	note = Description of Artificial Sequence: Synthetic primer
source	1..46

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	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 47		
agaagttgat tgagactttc aacgagggtc cccttcagct accttt		46
SEQ ID NO: 48	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 48		
tccctgctaa gccctaatacg		20
SEQ ID NO: 49	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 49		
ctctgattgc acgagaaggc		20
SEQ ID NO: 50	moltype = DNA length = 46	
FEATURE	Location/Qualifiers	
misc_feature	1..46	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..46	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 50		
ctcacttaat cttctgtact ctgaagtgaaggcgattgg agttgc		46
SEQ ID NO: 51	moltype = DNA length = 46	
FEATURE	Location/Qualifiers	
misc_feature	1..46	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..46	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 51		
agaagttgat tgagactttc aacgagctgg ctctgcttct ggtact		46
SEQ ID NO: 52	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 52		
gatgttgagg cgggcataag		20
SEQ ID NO: 53	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 53		
actgttcagg acgatacga		20
SEQ ID NO: 54	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Description of Artificial Sequence: Synthetic primer	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 54		
cgggtctccc tggaataga		20
SEQ ID NO: 55	moltype = DNA length = 20	

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FEATURE Location/Qualifiers
misc_feature 1..20
note = Description of Artificial Sequence: Synthetic primer
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 55
agttgtccgt cattagccct 20

SEQ ID NO: 56 moltype = DNA length = 20
FEATURE Location/Qualifiers
misc_feature 1..20
note = Description of Artificial Sequence: Synthetic primer
source 1..20
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 56
tgttcccttt cggctagaca 20

SEQ ID NO: 57 moltype = DNA length = 4324
FEATURE Location/Qualifiers
misc_feature 1..4324
note = Description of Artificial Sequence: Synthetic polynucleotide
source 1..4324
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 57

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taaacaagag	gaaggggtca	aaaacggaga	tatcttatac	ttggaaaaag	agatgacaat	120
cagtgatttc	atcaattttg	tatctagtgt	gccttctgtg	tttctgtgga	agcagcaacg	180
aggaaaggag	ggatccctag	atgattttta	caacgaactg	aacgactgct	ttgagggggg	240
taacatgaaa	gtaatatgga	actccgtcct	agtatttgcc	aggaggaagc	aaaggggtgt	300
ataggcttta	gtactttatag	aggaaacggg	gttacgtgca	agcgcgcgatg	cctgagcttt	360
gagggggggg	actttcacat	ctcttcttct	cacacttagc	cctaaccacag	agaataataa	420
aaagcattgc	aagatgagtg	ttgtcagcaa	gcaatacagc	atccacgaag	gcattatctt	480
tgtaattgaa	ttgaccccg	agcttcacgc	gccggcttca	gaagggaaat	ctcagctcca	540
gatcatctta	gagaatgtca	gtgaggttat	ttctgagcta	atcattacct	tgtcccggtac	600
aggaataggg	tgttacctta	ttaattacga	cgggtggtcaa	aacgacgaaa	tttaccctcat	660
ttttgagtta	caagacctga	atttggaaat	gatgaaacaa	ttgtaccaag	tcttgaggga	720
ccatgtaagt	gggcttaatc	ctctcgagaa	gcaattccca	attgaacaca	gtaaaccggt	780
atcagccact	ctgttcttct	acttaaggtc	tcttttttac	atggcggaaga	ctcataagcg	840
tactggaaga	cattacaact	tgaaaaagat	tttcttggtc	actaataacg	ataaacctta	900
caatggaaac	tctcagctga	gagttccctt	gaagaaaacc	ctggctgatt	acaatgacgt	960
agacattact	ttgattccgt	ttcttctgaa	caagccttca	gggtgcaagt	ttgacaagac	1020
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gaacgttatt	tataattcaa	atttttcttt	tttttctgta	cagacgcgtg	tacgcatgta	2760
acattatact	gaaaaccttg	cttgagaagg	ttttgggacg	ctcgaaggct	ttaatttgca	2820
agctgtatta	gtttcacttt	tcagcaacct	ggtcggaaag	atccacatca	agaatggata	2880

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ccaaccccaa gagtatgaaa atccttcctt acaatggcac ttcaaaatgt tacgtgacga 2940
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aaacagcctc gatgagacaa aaaccaagat catgaaacta cgggactatg tcaaggaaac 3060
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ccat 4324

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SEQ ID NO: 58      moltype = DNA length = 22
FEATURE
misc_feature       1..22
                    note = Description of Artificial Sequence: Synthetic
                    oligonucleotide
source             1..22
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 58
ggagttgaat cacatcttac tg 22

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SEQ ID NO: 59      moltype = DNA length = 1099
FEATURE
misc_feature       1..1099
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1099
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 59
gacaactaaa tgttctctgc gaatgtgtaa aatcaccata gttattaggg gataaaaacg 60
atgcttttgc aaactgatac tgtggcaacc aattagatat atccctgaaa cctagaattt 120
ccaaagaggg gatattgatg atatcattaa atcgatttat tgttttgact tgctctgagt 180
ttaaagggaat gtaagcatcg gcacctagag gatatgcttt gatcagctgt tcaactggaaa 240
actcttttcc ggatgttgga tcgacaaaact gagattttgt ctccacatct ttgaaagttg 300
aacctctcatt gacgacaaat ttgatcttcc ttggagtttc atgataaaac atagaataac 360
ctttcacaga aatgcacaag ttatttgcca ttttcaaaag acaggtgaag taaacccgct 420
tgatctcctt atgtctagaa attcgttggc ggatctcttc aattgacatc gaacaagcat 480
ctttatcata gaacaaaatt tctgagtatt ccgtcttgtc aaacttgaca cctgaaggct 540
tgttcagaag aaacggaatc aaagtaatgt ctacgtcatt gtaatcagcc aggggtttct 600
tcaaggggaa tctcagctga gagtttccat tgtaaggttt atcgttatta gtgaacaaga 660
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gcttctgac aacatcat 1099

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SEQ ID NO: 60      moltype = DNA length = 549
FEATURE
misc_feature       1..549
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..549
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 60
tcaggcctta gtctctttga acttctgctg aaagtacttg tcaatgttgg cgatgagatc 60

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tgcctttttg gaggccgtct ccagaaagt gtttcgtgac cttacgtatt ttcgtaattg 120
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aactaatac                                     549

SEQ ID NO: 61      moltype = DNA length = 552
FEATURE            Location/Qualifiers
misc_feature       1..552
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..552
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 61
ttcagtaatg tcttgtttct tttgttgtag tggtagacca ttttgacttc gtgaaagttt 60
ctttagaata gttgtttcca gagggcaaac attccacccg tagtaaatgt caagcgtagg 120
aagaccaaga ctggcataaa tcaggataaa gtgtcgagca ctggcagggt atcttctgaa 180
agtttctact agcagataag atccagtagt catgcataat gcaacaatgt accgtgtgga 240
tctaagaacg cgtcctacta accttcgcat tcgttggtcc agtttgtgt tatcgatcaa 300
cgtgacaagg ttgtcgattc cgcgtaagca tgcataccca aggaagcctg ttgcaattcc 360
aagttagcca gttccaacaa tctttgtaat attagagcac ttcattgtgt tgcgcttgaa 420
agtaaaatgc gaacaaatta agagataatc tcgaaaccgc gacttcaaac gccaatatga 480
tgtgcggcac acaataagcg ttcataatcc ctgggtgact ttctcgcttt aaaaaattat 540
cggaaaaaat tt                                     552

SEQ ID NO: 62      moltype = DNA length = 20
FEATURE            Location/Qualifiers
misc_feature       1..20
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source             1..20
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 62
cagaggccaa acattccacc                                     20

SEQ ID NO: 63      moltype = DNA length = 14
FEATURE            Location/Qualifiers
misc_feature       1..14
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source             1..14
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 63
ttaagagga gaaa                                     14

SEQ ID NO: 64      moltype = DNA length = 375
FEATURE            Location/Qualifiers
misc_feature       1..375
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..375
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 64
atggctaatac tgacctctgc tgttccggtt ctgaccgctc gtgacgttgc tgggtgctgtt 60
gagttctgga ccgaccgtct gggtttctct cgtgacttcg ttgaagacga ctccgctggt 120
gttgttcgtg acgacgttac cctgttcata tctgctgttc aggaccagggt tgttccggac 180
aacaccctgg cttgggtttg ggttcgtggt ctggacgaac tgtacgctga atggctctgaa 240
gttgtttcta ccaacttcgc tgacgcttct ggtccggcta tgaccgaaat cgggtgaacag 300
ccgtggggtc gtgagttcgc tctgctgtgc ccggttggtg actgcgttca cttcgttgct 360
gaagaacagg actaa                                     375

SEQ ID NO: 65      moltype = DNA length = 318
FEATURE            Location/Qualifiers
misc_feature       1..318
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..318
                   mol_type = other DNA

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-continued

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                                organism = synthetic construct
SEQUENCE: 65
cacgtccgac ggcgcccccac ggggtcccagg cctcggagat ccgtcccccct ttccctttgt 60
cgatatcatg taattagttta tgtcacgctt acattcacgc cctcccccca catecgctct 120
aaccgaaaaag gaaggagtta gacaacctga agtctaggtc cctattttatt tttttatagt 180
tatgttagta ttaagaacgt tatttatatt tcaaatTTTT cttttttttc tgtacagacg 240
cgtgtacgca tgtaacatta tactgaaaac cttgcttgag aaggttttgg gacgctcgaa 300
ggctttaatt tgcaagct                                     318

SEQ ID NO: 66      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source            1..20
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 66
aggagttaga caacctgaag                                     20

SEQ ID NO: 67      moltype = DNA length = 23
FEATURE           Location/Qualifiers
misc_feature       1..23
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source            1..23
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 67
gtaactagtc tctcggaatc cat                                     23

SEQ ID NO: 68      moltype = DNA length = 1623
FEATURE           Location/Qualifiers
misc_feature       1..1623
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source            1..1623
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 68
cttcagagta cagaagatta agtgagagaa ttctaccgtt cgtatagcat acattatacg 60
aagttatttc agtaatgtct tgtttctttt gttgcagtg tgagccattt tgacttcgtg 120
aaagtttctt tagaatagtt gtttccagag gccaaacatt ccaccgtag taaagtgcaa 180
gcgtaggaag accaagactg gcataaatca ggtataagtg tcgagcactg gcagggtgatc 240
ttctgaaagt ttctactagc agataagatc cagtagtcat gcatatggca acaatgtacc 300
gtgtggatct aagaacgcgt cctactaacc ttgcattctg ttggtccagt ttgttggtat 360
cgatcaacgt gacaagggtt tcgattccgc gtaagcatgc atacccaagg acgcctgttg 420
caattccaag tgagccagtt ccaacaatct ttgtaattt agagcacttc atttgtgtgc 480
gcttgaaagt aaaatgcgaa caattaaga gataatctcg aaaccgagc ttcaaacgcc 540
aatatgatgt gcggcacaca ataagcggtc atatccgctg ggtgactttc tcgctttaaa 600
aaattatccg aaaaaatttt tgacggctag ctacgtccta ggtacgctag cattaaagag 660
gagaaaaatga ctactcttga tgacacagcc tacagatata ggacatcagt tccgggtgac 720
gcagaggcta tcgaagcctt ggacgggttc ttactactg atacgggtgt tagagtcacc 780
gctacaggtg atggcttcac cttgagagag gttcctgtag acccaccctt aacgaaagtt 840
ttccctgatg acgaatcggg tgacgagctc gatgctgggt aggacgggtg ccctgattcc 900
agaacatttg tcgcatacgg agatgatggt gacctggctg gctttgttgt ggtgtcctac 960
agcggatgga atcgtagact cacagttgag gacatcgag ttgcacctga acatcggtgt 1020
cacggtgttg gtcgtgcact gatgggactg gcaacagagt ttgctagaga aagaggagcc 1080
ggacatttgt ggtagaagt gaccaatgtc aacgctcctg ctattcacgc atataggcga 1140
atgggtttca ctttgtgcgg tcttgatact gctttgtatg acggaactgc ttctgatggt 1200
gaacaagctc tttaactgtg tatgccatgt ccatagcacg tccgacggcg gccacgggt 1260
cccaggcctc ggagatccgt cccctttttc ctttgtcgat atcatgtaat tagttatgtc 1320
acgcttacat tcacgccttc cccccacatc cgctctaacc gaaaaggag gagttagaca 1380
acctgaagtc taggtcccta tttatttttt tatagttatg ttagtattaa gaacgttatt 1440
tatattttca atttttcttt tttttctgta cagacgcgtg tacgcatgta acattatact 1500
gaaaaccttg cttgagaagg ttttgggacg ctccaaggct ttaatttgca agctataact 1560
tcgtatagca tacattatac cttgttatgc ggccgcaaga agttgattga gactttcaac 1620
gag                                     1623

SEQ ID NO: 69      moltype = DNA length = 27
FEATURE           Location/Qualifiers
misc_feature       1..27
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source            1..27
                   mol_type = other DNA

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-continued

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                                organism = synthetic construct
SEQUENCE: 69
cttcagagta cagaagatta agtgaga                                27

SEQ ID NO: 70                moltype = DNA length = 34
FEATURE                      Location/Qualifiers
misc_feature                 1..34
                                note = Description of Artificial Sequence: Synthetic
                                oligonucleotide
source                      1..34
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 70
taccgttcgt atagcatata ttatacgaag ttat                                34

SEQ ID NO: 71                moltype = DNA length = 552
FEATURE                      Location/Qualifiers
misc_feature                 1..552
                                note = Description of Artificial Sequence: Synthetic
                                polynucleotide
source                      1..552
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 71
ttcagtaatg tcttggtttct tttgttgtag tggtagacca ttttgacttc gtgaaagtgt 60
ctttagaata gttgtttcca gaggccaaac attccaccgc tagtaaatg caagcgtagg 120
aagaccaaga ctggcataaa tcaggataaa gtgtcgagca ctggcagggtg atcttctgaa 180
agtttctact agcagataag atccagtagt catgcatatg gcaacaatgt accgtgtgga 240
tctaagaacg cgtcctacta accttcgcat tcgttggtcc agtttgtgt tatcgatcaa 300
cgtgacaagg ttgtcgattc cgcgtaagca tgcataccca aggacgcctg ttgcaattcc 360
aagttagcca gttccaacaa tctttgtaat attagagcac ttcatttgtg tgcgcttgaa 420
agtaaaatgc gaacaaatta agagataatc tcgaaaccgc gacttcaaac gccaatatga 480
tgtgcggcac acaataagcg ttcatatccg ctgggtgact ttctcgttt aaaaaattat 540
cggaaaaaat tt                                552

SEQ ID NO: 72                moltype = DNA length = 14
FEATURE                      Location/Qualifiers
misc_feature                 1..14
                                note = Description of Artificial Sequence: Synthetic
                                oligonucleotide
source                      1..14
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 72
ttaagagga gaaa                                14

SEQ ID NO: 73                moltype = DNA length = 570
FEATURE                      Location/Qualifiers
misc_feature                 1..570
                                note = Description of Artificial Sequence: Synthetic
                                polynucleotide
source                      1..570
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 73
atgactactc ttgatgacac agcctacaga tataggacat cagttccggg tgacgcagag 60
gctatcgaag ccttggaagg ttcattcact actgatacgg tgttttaggt caccgctaca 120
ggtagatggc tcaccttgag agagggttcct gtagaccac ccttaacgaa agttttccct 180
gatgacgaat cggatgacga gtctgatgct ggtgaggacg gtgaccctga ttccagaaca 240
tttgcgcac acggagatga tggtagacctg gctggctttg ttgtggtgtc ctacagcgga 300
tggaatcgta gactcacagt tgaggacatc gaagttgcac ctgaacatcg tggtcacggt 360
gttggtcgtg cactgatggg actggcaaca gagtttgcta gagaagagg agccggacat 420
ttgtgggtag aagtgaacaa tgtcaacgct cctgctattc acgcatatag gcgaatgggt 480
ttcactttgt gcggtcttga tactgctttg tatgacggaa ctgcttctga tggtagaaca 540
gctctttaca tgagtatgcc atgtccatag                                570

SEQ ID NO: 74                moltype = DNA length = 318
FEATURE                      Location/Qualifiers
misc_feature                 1..318
                                note = Description of Artificial Sequence: Synthetic
                                polynucleotide
source                      1..318
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 74
cacgtccgac ggcggccac ggggccagg cctcgagat cgcgtccct tttcctttgt 60

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cgatcatcatg taattagttta tgtcacgctt acattcacgc cctcccccca catccgctct 120
aaccgaaaaag gaaggagttta gacaacctga agtctaggtc cctattttatt tttttatagt 180
tatgttagta ttaagaacgt tatttatatt tcaaatTTTT cttttttttc tgtacagacg 240
cgtgtacgca tgtaacatta tactgaaaac cttgcttgag aaggttttgg gacgctcgaa 300
ggctttaatt tgcaagct 318

SEQ ID NO: 75      moltype = DNA length = 34
FEATURE           Location/Qualifiers
misc_feature       1..34
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source             1..34
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 75
ataacttcgt atagcataca ttataccttg ttat 34

SEQ ID NO: 76      moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature       1..35
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide
source             1..35
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 76
gcgccgcaa gaagttgatt gagactttca acgag 35

SEQ ID NO: 77      moltype = DNA length = 3869
FEATURE           Location/Qualifiers
misc_feature       1..3869
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..3869
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 77
tactacagcg tggtgtttcc tgcgatgggt tttaatgtcc tgaactgggt ttcgtttacc 60
ggtattacgc gagccacctt gactgtaagg gaacgatact ggactaagag agtaatgcga 120
aaggcaacag cgtttctggc gaacctaatc aatgacgggt acgagtttac tactcctaaa 180
gccagtctta ttgtcttaga gcgagtcacac gcttacttaa agggccaggg acctaattat 240
gacatcgatt ttgacgagca ggaggcggtt attaaagaaa tggaggaggt gaggacctct 300
ggtggatatg agaacagata ctcatattca ggaaccgatg aaacacccag agatccgggt 360
tgctgtttcc ttcccatgtc tttaaataaa tggcactttg atgtgctaga ctgcctgagg 420
atatacggtt ctacggaaga tctggaatct aaattattaa gtgttcagca attggtgtta 480
caatgttgca tgaagcacag tggcatgact ccagacatgg tctttgcaac ggaagtagct 540
cagaagccga ccttcgaaga cgacatagtt tgtgatgata ttgacgctta tgcccagggg 600
ggtgattgtc tagattatgt ttacacgccca agcaattact ccagaacttt agaaattcat 660
ggcaagattg ctacottaca acgagagctg gggctatgct ataatttct cggaattttg 720
gaccgttttt ccgattaagg tttttagctc cattgcgcga acccccgtcc tccagactcc 780
tcogttatcc agcattcagc atggacaggt tcaaaaaata aaatttcttg atatgggtcc 840
acttcaaaac tgccgctacc tgtaggaaaa aaaaagagaa cataaatatg ccgcgaacag 900
aaaacgtaat gtactgttct atataataac tgttcagatc aatcataaat tctcagtttc 960
aaactttccg ctacgcccaga ttattatcgt aaagaacgca tcattggctc tatgttgaag 1020
gatcagttct tgttatgggt tgccttgata gcgagcgta cggtttccgg cgtgatggca 1080
gtcctagcgc agtccggcca taacacgggt gaaaaaacgag atgccccaaa cgttgttggtc 1140
gttcaacagt tggactcttt cagagtacag aagattaaat gagagaattc taccgttcgt 1200
atagcataca ttatacgaag ttatttcagt aatgtcttgt tctttttgtt gcagtgggtg 1260
gccattttga cttcgtgaaa gtttctttag aatagtttgt tccagaggcc aaacattcca 1320
cccgtagtaa agtgcaagcg taggaagacc aagactggca taaatcaggt ataagtgtcg 1380
agcactggca ggtgatcttc tgaaagtttc tactagcaga taagatccag tagtcatgca 1440
tatggcaaca atgtaccgtg tggatctaag aacgcgtcct actaaccttc gcattcgttg 1500
gtccagtttg ttgttatcga tcaacgtgac aaggttgtcg attcccgcta agcatgcata 1560
cccaaggacg cctgttgcaa ttccaagtga gccagttcca acaatctttg taatattaga 1620
gcacttcatt gtgttgcgct tgaaagttaa atgcgaacaa attaagagat aatctcgaaa 1680
ccgcgacttc aaacgcgaat atgatgtgcg gcacacaata agcgttcata tccgctgggt 1740
gactttctcg ctttaaaaaa ttatccgaaa aaatttttga cggctagctc agtcctaggt 1800
acgctagcat taaagaggag aaaatgacta ctcttgatga cacagcctac agatatagga 1860
catcagttcc gggtagcga gaggtatcg aagccttggg cggttcattc actactgata 1920
cgggtgttag agtcacgcgt acaggtgatg gcttcacctt gagagaggtt cctgtagacc 1980
caccttaac gaaagttttc cctgatgacg aatcggatga cagtcctgat gctggtgagg 2040
acggtgaccc gatttccaga acatttgtcg catacggaga tgatgggtgac ctggctggct 2100
tgttgtgtgt gtccacagc ggaatgaaac gtgactcac agttgaggac atcgaagttg 2160
cacctgaaca tcgtgttcac ggtgttggtc gtgcactgat gggactggca acagagtttg 2220
ctagagaaag aggagccgga catttgttgt tagaagtgaac caatgtcaac gtcctcgcta 2280
ttcacgcata taggcgaatg ggtttcactt tgtgcggtct tgatactgct ttgatgacg 2340

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gaactgcttc	tgatgggtgaa	caagctcttt	acatgagtat	gccatgtcca	tagcaagctcc	2400
gacggcgcc	cacgggtccc	aggcctcgga	gatccgtccc	ccttttcctt	tgtcgatatc	2460
atgtaattag	ttatgtcacg	cttacattca	cgccctcccc	ccacatccgc	tctaaccgaa	2520
aaggaaggag	ttagacaacc	tgaagtctag	gtccctattt	atTTTTttat	agttatgtta	2580
gtattaagaa	cgttatttat	atttcaaatt	tttctttttt	ttctgtacag	acgctgttac	2640
gcatgtaaca	ttatactgaa	aaccttgctt	gagaagggtt	tggaagcctc	gaaggcttta	2700
atttgcaagc	tataacttcg	tatagcatac	attatacctt	gttatgcggc	cgcaagaagt	2760
tgattgagac	tttcaacgag	ggtcccttc	agctaccttt	ctctctgttt	ggtagttatt	2820
ctcgcgctgt	gtatagtata	gtataaaagg	gcctacattg	gataggcttc	aacattcctc	2880
aataaaca	catccaacat	cgcgcatctc	gcatttcgca	tttcacattt	cgcgctgcc	2940
ttccttttagg	ttctttgaat	catcatcaat	cgtcgccgtc	tacatcagag	caggacttat	3000
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tttgcctaaa	taggtttttg	ttagtttttc	ttcaaagccc	aaaagaaact	ctatttagat	3120
tcataccagaa	acaatctttt	ttccacccca	tttcgaagtg	ccgtggagca	cagacataaa	3180
aagatgacta	ccgttcaacc	tacagggccca	gacaggctca	ccctgccgca	tattctactg	3240
gaattcaacg	atggctcctc	gcagcatgca	gtgatcgagc	taagcatgaa	cgaggggatt	3300
aatatatcca	cccatgagtg	gaatccatcc	actaatgagc	aatcgccacg	ggaagagaga	3360
gcaccacccc	aacaatccaa	tcctatcgcat	catccagaat	catcgaaact	agctactcaa	3420
agtcccgctc	aggaaccgca	gactcagccc	ggcattccag	gactagatag	gctgccttt	3480
gatacctcgg	caacgggggtc	gtcagaacag	gttgacccag	tacaggggag	gatcctggat	3540
gatattatag	gccaatcatt	aaggacttcc	gaagaagacg	ataccgaatc	ccgccagaga	3600
ccacgagacc	agaagaacat	tatgatcacc	gtgaattact	tgtacgcaga	cgacacaaat	3660
tcagaagtg	ctaatacaaa	caaccagacg	cccaataaca	cttctagaac	ttccgacagt	3720
gaacgtgtgg	gctccttatc	gttgacaggt	ccggatctac	cagataatgc	cgacgattac	3780
tatatcgatg	tactcattaa	actaaccaca	agcattgccc	tcagcgtcat	cacgtccatg	3840
atcaagaaac	gattagggct	tagcagggga				3869

SEQ ID NO: 78 moltype = DNA length = 1157
 FEATURE Location/Qualifiers
 misc_feature 1..1157
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1157
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 78

tactacaggc	tggctgtttc	tgcgatgggt	tttaatgtcc	tgactggggt	ttcgtttatc	60
ggattatacc	gagccacctt	gactgtaagg	gaacgatact	ggactaagag	agtaatgcga	120
aaggcaacac	cgtttctggc	gaacctaatc	aatgacgggt	acgagtttac	tactcctaaa	180
gccagtctta	ttttgctaga	gcgagtcaac	gcttacttaa	agggccaggg	acctaattat	240
gacatcgatt	ttgacgagca	ggaggcgctc	attaaagaaa	tggaggagtt	gaggacctct	300
ggtgatagtg	agaacagata	ctcatattca	ggaaccgatg	aaacaccacg	agatccgggt	360
tgctgttttc	ttcccatttc	tttaataaaa	tggcactttg	atgtgctaga	ctgcctgagg	420
atatacggta	ctcaggaaga	tctggaatct	aaattattaa	gtgttcagca	attgggtgta	480
aatgttgca	tgaagcacag	tgccatgact	ccagacatgg	tctttgcaac	ggaagtagct	540
cagaagccga	ccttcgaaga	gcacatagtt	tgtgatgata	ttgacgctta	tgcccagggg	600
ggtgattgtc	tagattattg	ttacacgcca	agcaattact	ccagaacttt	agaaattcat	660
ggcaagattg	ctaccttaca	acgagagctg	gggctatgct	ataatattct	cggaattttg	720
gaccgttttt	ccgattaaag	tttttagctc	cattgcgcca	accccgctc	tccagactcc	780
ttcgttatcc	agcattcagc	atggacaggt	tcaaaaaata	aaatttcttg	atatgggtcc	840
acttcaaaac	tgcgccctacc	tgtaggaaaa	aaaaagagaa	cataaatatg	ccgcgaacag	900
aaaacgtaat	gtactgtttc	atatataaac	tgttcagatc	aatcataaat	tctcagtttc	960
aaactttccg	ctcagccaga	ttttattcgt	aaagaacgca	tcattggctc	tatgttgaag	1020
gatcagttct	tggtatgggt	tgctttgata	gcgagcgtac	cggtttccgg	cgtgatggca	1080
gctcctagcg	agtcggggca	taaacaggtt	gaaaaacgag	atgccaaaaa	cgttgttggc	1140
gttcaacagt	tggactt					1157

SEQ ID NO: 79 moltype = DNA length = 1089
 FEATURE Location/Qualifiers
 misc_feature 1..1089
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..1089
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 79

ggtcctcttc	agctaccttt	ctctctgttt	ggtagttatt	ctcggcgctg	gtatagtata	60
gtataaaagg	gcctacattg	gataggcttc	aacattcctc	aataaaca	catccaacat	120
cgcgcatctc	gcatttcgca	tttcacattt	cgcgccctgc	ttccttttagg	ttctttgaat	180
catcatcaat	cgtcgcgctc	tacatcagag	caggacttat	ctttgccttc	cccaaaaatt	240
gccactccgt	caaatagatt	cttttgaatc	cttgactatt	tttgccctaaa	taggtttttg	300
ttagtttttc	ttcaaagccc	aaaagaaact	ctatttagat	tcataccagaa	acaatctttt	360
ttcaccacca	tttcgaagtg	ccgtggagca	cagacataaa	aagatgacta	ccgttcaacc	420
tacagggcca	gacaggctca	cctgcgcgca	tattctactg	gaattcaacg	atggctcctc	480
gcagcatgca	gtgatcgagc	taagcatgaa	cgaggggatt	aatatatcca	cccatgagtg	540
gaatccatcc	actaatgagc	aatcgccacg	ggaagagaga	gcaccacccc	aacaatccaa	600

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tccatcgc	catccagaat	catcgaacat	agctactcaa	agtcctcgc	aggaaccga	660
gactcagccc	ggcattccag	gactagatag	gcctgccttt	gatacctcgg	caacggggtc	720
gtcagaacag	gttgaccacg	tacagggaag	gatcctggat	gatattatag	gccaatcatt	780
aaggacttcc	gaagaagacg	ataccgaatc	ccgccagaga	ccacgagacc	agaagaacat	840
tatgatcacc	gtgaattact	tgtacgcaga	cgacacaaat	tccagaagtg	ctaatacaaa	900
caaccagacg	cccataaaca	cttctagaac	ttccgacagt	gaacgtgtgg	gtcctctatc	960
gttgacggtt	ccggatctac	cgataaatgc	cgacgattac	tatatcgatg	tactcattaa	1020
actaaccaca	agcattgccc	tcagcgctcat	cacgtccatg	atcaagaaac	gattagggct	1080
tagcaggga						1089

SEQ ID NO: 80 moltype = DNA length = 3964
 FEATURE Location/Qualifiers
 misc_feature 1..3964
 note = Description of Artificial Sequence: Synthetic
 polynucleotide
 source 1..3964
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 80

gccttctcgt	gcaatcagag	ctgttgaaag	agagaagagg	gcacacggaa	gctgctgttc	60
aattgtgtga	attgaccgga	ttacaacctg	ctggagtgat	aggagagctg	gttcgtgacg	120
aggacggctc	tatgatcgca	ttagacgact	gtgttcagtt	tggtctccgc	cacaacgtaa	180
aaattatcaa	ccttgaccag	atcattgaa	acatggattc	caagaacagc	tagataccg	240
ggataggaat	acagagatat	catgattgag	gaacgtaaga	gctttttcga	aatgtgtgag	300
ttgtggtgag	ggccaggcgg	tggggaggtg	tgggggagcc	tccttggtcg	aatgtagata	360
tagtaagcaa	gacacaagag	cgcggaagt	cttcaacgag	cgccgcttgg	gtcttgtacg	420
caacgtaaat	actacacagt	tgagcttctc	gcgaaccggt	cgacattttg	atcatgcata	480
ctatgttgag	acacocatct	gtactattgc	ggcaaccagc	tgtaaatttg	actaattaaa	540
gctgatgaag	gatgcagggc	gtcgtcaatt	ttttgattga	ttgcatttaa	ttgtttgagc	600
cattcaaggc	tgaatgcccg	gcacccctga	ccctctctgt	gagtactata	aaccgcgagg	660
cagggtaccc	ttggccttct	gcgagactac	cagtcataac	gtatatccac	aatgtactag	720
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tcctttgac	ccagttcttc	gtctgagaat	aaagggtgct	aatcctaag	atttagccga	3360
acaatgggct	gctgcattcc	caaagggtgg	atatcttaaa	gacgttattg	cgcaaggacc	3420

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tttcttgacg ttctttttta acacatcggt tctgtacaag ttggtgatat ctgatgctct 3480
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tgaccaggca agagcttttt tcaagaaaat ggaaaatggc gacgaatcgg ctgtaagctt 3900
gtggaaaaga ttccgtgagt tatccattga gaagatcatt gatacttatg cccgcctcaa 3960
catc 3964

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SEQ ID NO: 81      moltype = DNA length = 1019
FEATURE           Location/Qualifiers
misc_feature       1..1019
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1019
                   mol_type = other DNA
                   organism = synthetic construct

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```

SEQUENCE: 81
gccttctcgt gcaatcagag ctgttgaaag agagaagagg gcacacggaa gctgctgttc 60
aattgtgtga attgaccgga ttacaacctg ctggagtgat aggagagctg gttcgtgacg 120
aggacggctc tatgatcgga ttacacgact gtgttcagtt tggctctcgc cacaacgtaa 180
aaattatcaa ccttgaccag atcattgaat acatggattc caagaacagc tagatacgat 240
ggataggaat acagagatat catgattgag gaacgtaaga gctttttcga aagtgtgagt 300
ttgtgggtgag ggcagagcgg tggggaggtg gtggggagcc tccttgggtc aatgtagata 360
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caacgtaatg actacacagt tgagcttgtc gcgaaccggt cgacattttg atcatgcata 480
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gctgatgaag gatgcagggc gtgcgtcaatt ttttgattga ttgcatttaa ttgtttgagc 600
cattcaaggc tgaatgcccg gcaccctaga cccttcttgt gactactata aaccgcagc 660
cagggtagcc ttggccttct gcgagactac cagtcataac gtatatccac aatgtactag 720
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SEQ ID NO: 82      moltype = DNA length = 1322
FEATURE           Location/Qualifiers
misc_feature       1..1322
                   note = Description of Artificial Sequence: Synthetic
                   polynucleotide
source             1..1322
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 82
ctggctctgc ttctgggtact tcttcagggtg catcttctgc tactcaaaat gacgaaacat 60
ccactgatct tggagctcca gtgcactctt taagtgaac gccatgtctt ttgcatctc 120
tgctgctcat gttgtagtag actttttttt tcaactgagt tttatgtact actgattaca 180
ttgtgtaggt gtaatgatgt gcactataat actaatatag tcaaaatgct acagaggaaa 240
gtgcaggttg cctgtgggtg tttttcttat tagcaccctc tgaacactct ttacctataa 300
catcctcagc catgctaata gcgcataaaa taaatcttcg aacttttttc cattttatgc 360
tcataaagct tccttactgt caccttatca aaagagcttt tgccactaaa gtatgcacac 420
ccagaattgc tcccgaatat cgtccaacaa tgctaggatc tgtggaaaagt ttgacaaata 480
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gtaacccagt ggaccttttg agaaactaca tcaactcaaga acttagtaaa atttctggag 600
ttgacaaaaga attgattttc ccagccttgg aatggggtac cacactggaa aaagggtgatc 660
ttttgatccc agtctctcgt ctgagaataa aggggtgctaa tcctaaaagt ttagccgaac 720
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accaggcaag agcttttttc aagaaaatgg aaaaatggcg cgaatcggct gtaagcttgt 1260
ggaaaagatt ccgtgagtta tccattgaga agtacattga tacttatgcc cgcctcaaca 1320
tc 1322

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SEQ ID NO: 83      moltype = DNA length = 66
FEATURE           Location/Qualifiers
misc_feature       1..66
                   note = Description of Artificial Sequence: Synthetic
                   oligonucleotide

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source                1..66
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 83
atgttcagct tgaaagcatt attgccattg gccttggtgt tggtcagcgc caaccaagtt 60
gctgca                                     66

SEQ ID NO: 84         moltype = AA length = 22
FEATURE              Location/Qualifiers
VARIANT              1..22
                      note = Description of Artificial Sequence: Synthetic peptide
source               1..22
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 84
MPSLKALLPL ALLLVSANQV AA                                     22

SEQ ID NO: 85         moltype = DNA length = 60
FEATURE              Location/Qualifiers
misc_feature         1..60
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source               1..60
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 85
atgaaagcat tcctgtgtgtt actactttta ctaggcctgt ccactacact cgctaaggca 60

SEQ ID NO: 86         moltype = AA length = 20
FEATURE              Location/Qualifiers
VARIANT              1..20
                      note = Description of Artificial Sequence: Synthetic peptide
source               1..20
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 86
MKAFLLLLLL LGLSTTLAKA                                     20

SEQ ID NO: 87         moltype = DNA length = 144
FEATURE              Location/Qualifiers
misc_feature         1..144
                      note = Description of Artificial Sequence: Synthetic
                      polynucleotide
source               1..144
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 87
atggaagggtg gcgaagaaga agttgagcgc attcctgatg aacttttcga tacaaaaaag 60
aagcatttgt tagataagct cataagggtc ggaataatcc ttgtactcct gatatggggc 120
actgttttgt tgctaaaaag tatt                                     144

SEQ ID NO: 88         moltype = AA length = 48
FEATURE              Location/Qualifiers
VARIANT              1..48
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
source               1..48
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 88
MEGGEEEVER IPDELFDTKK KHLLDKLIRV GIILVLLIWG TVLLKSI      48

SEQ ID NO: 89         moltype = DNA length = 57
FEATURE              Location/Qualifiers
misc_feature         1..57
                      note = Description of Artificial Sequence: Synthetic
                      oligonucleotide
source               1..57
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 89
atgagatttc cttcaatttt tactgctgtt ttattgcag catcctccgc attagct 57

SEQ ID NO: 90         moltype = AA length = 19
FEATURE              Location/Qualifiers
VARIANT              1..19

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source          note = Description of Artificial Sequence: Synthetic peptide
                1..19
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 90
MRFPSIIFTAV LFAASSALA                                     19

SEQ ID NO: 91      moltype = DNA length = 5679
FEATURE           Location/Qualifiers
misc_feature       1..5679
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source            1..5679
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 91
ggcgcgccgt ttaaaccctc caccagccat ataccactac aacaccacag aagagaaaga 60
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tgcaagcatt acgtaataat agcaacttta tgacaaatca ttccattttt ttccactgga 180
gcgtgcactg cgtaaatcat tctcttttga aggcaaggga agaacaacaa aatttttcc 240
tccgttatca aaacattgaa tcatgtctac tgaacccact tttaaattgg tccttgtcgg 300
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gttacatcga gaattactta caagaacgtt ccaaaactggc accgtgactt ggtcagagt 660
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gaaacgacct ctccaactat attcagatga tgaagacgaa gatttgcaaa ttgtaggatc 3540
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actccatttt gacgggttcaa accagaatga acaagccaga tggctggact tgcggataaa 3660
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tatcaggggtt attgtctcat gacgggatac atatttgaat gtatttagaa aaataaaca 5640
ataggggttc cgcgcacatt tccccgaaaa gtgccacct 5679

SEQ ID NO: 92      moltype = DNA length = 1157
FEATURE            Location/Qualifiers
misc_feature       1..1157
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1157
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 92
aaaccctcca ccagccatat accactacaa caccacagaa gagaaagagc tcatatcatc 60
cgtcatgaga gtagtaccagc acagaaatac agtcaagtaa aactagtagt caagcattac 120
taataaatag caactttatg acaaatcatt ccattttttt ccaactggagc gtgcaactgcg 180
taaatcattc tctttggaag gcaagggaag aacaacaaaa ttttctcttc cgttatacaa 240
acattgaatc atgtctactg aacccacttt taaattgggtc cttgtcgggtg atgggtggtac 300
cggtaaaagta agtgcaaat atttgatgag tccgataatg ttttcgcccc cttagttccc 360
ctcatgatta ctaacaattc atagaccacc ttcgttaaga gacaccttac tggagagttc 420
cgtaagaagt acattgctac ttggggagtc gaagttcatc ccttgctatt ccacactaac 480
tgtggtccta tcacattcaa cgtttgggac actgctggag aagagaagtt tgggtgactg 540
agagatgggtt attacattaa cgggtgactgt ggtatcatca tgttcgacgt tacatcgaga 600
attacttaca agaagcttcc aaactggcac cgtgacttgg tcagagtggtg tgagaacatt 660
ccaattgtgc tttgtggtaa caaggttgat gtcaaggaaa gaaaggtcaa ggctaagacc 720
atcactttcc acagaaagaa gaacttgcaa tactttgaca tttctgcccc gtccaaactac 780
aactttgaga agccattctt gtggttagct agaaagttgt ctggtgagcc ccaattagag 840
ttcgttgctg ctcccagctt gcaagcccca gaggttcaaa ttgatgccga tttataaaag 900
aagtaacgagc aagagaacgc cgaggctgcc gctatgccat tgcctgatga agatgatgcc 960
gacttgtaag cttttactta cagtacattg agaaccatac atagggcacg tatcgtaagt 1020
ttagttggtt gctgatgtaa gctagtttgt tctgttagtg tttcagaggtc gcagagggat 1080
ctctctagcc ttagacaaaa aaaaaaaggt tgacacgttg atacactctc tgtttcatcc 1140
gatctttcac ctacgag 1157

SEQ ID NO: 93      moltype = DNA length = 1125
FEATURE            Location/Qualifiers
misc_feature       1..1125
                    note = Description of Artificial Sequence: Synthetic
                    polynucleotide
source             1..1125
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 93
ggagagttat agagctgcta tgcgtgtaaa aatagtttaa atcttcgtaa agtatgttag 60

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tccatgtaat ttgctatgaa tcgatacgt aatctggatg ctgaacggat gcttactggc 120
atgcattatt cattacccat ctaagctgcg ccacaacca gtaaatgca gtgagggaag 180
cttcctctgta accgtctctgt ccttttaggg accatcgatc cccaacgatc aaatcgcgat 240
acatctatca actgtccctt tccatctatc tatgcaaggt aatgacagac tctgttaact 300
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caataaggag aaaacttgat gaccaatcca actccaattc ttttgccaga ctggaatcgt 480
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agataaatca atctgtcccc gcattctacat catcttctac cccagttccc tccagcacac 780
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tcaatagggtt catttccact ttggggcatt taacacagaga agttt 1125

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SEQ ID NO: 94      moltype = DNA length = 1292
FEATURE           Location/Qualifiers
misc_feature       1..1292
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source             1..1292
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 94
ttcagtaatg tcttgtttct tttgttgacg ttgtgagcca ttttgacttc gtgaaagtgt 60
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aagttccaaga ctggcataaa tcaggataaa gtgtcgagca ctggcagggt atcttctgaa 180
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tctaagaacg cgtcctacta accctcgcat tcgttgggtc agtttgtgt tatcgatcaa 300
cgtgacaagg ttgtcgattc cgcgtaagca tgcataccca aggaagcctg ttgcaattcc 360
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tgtgcgccac acaataagcg ttcataatcg ctgggtgact ttctcgcttt aaaaaattat 540
ccgaaaaaat ttttgacggc tagctcagtc ctaggtagcg tagcattaaa gaggagaaaa 600
tggttaaaat gacctctgct gttccgggtc tgaccgctcg tgacgttgct ggtgctgttg 660
agttctggac cgaccgctcg ggtttctctc gtgacttcgt tgaggacgac ttcgctgggt 720
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SEQ ID NO: 95      moltype = AA length = 124
FEATURE           Location/Qualifiers
VARIANT           1..124
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
source             1..124
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 95
MAKLTSAPV LTARDVAGV EFWTDLRGFS RDFVEDDFAG VVRDDVTLEFI SAVQDQVVPD 60
NTLAWVWVRG LDELYAEWSE VVSTNFRDAS GPAMTEIGEIQ PWGREFALRD PAGNCVHFVA 120
EEQD 124

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SEQ ID NO: 96      moltype = DNA length = 11322
FEATURE           Location/Qualifiers
misc_feature       1..11322
                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source             1..11322
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 96
ccagccagga cagaaatgcc tcgacttcgc tgcgtcccaa ggttgccggg tgacgcacac 60
cgtggaaacg gatgaaggca cgaaccagc ggacataagc ctgttcgggt cgtaagctgt 120
aatgcaagta gcgtatgcgc tcacgcaact ggtccagaac cttgaccgaa cgcagcgggt 180

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gtaacggcgc	agtggcggtt	tteatggctt	gttatgactg	tttttttggg	gtacagtcta	240
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gcagcaacga	tgttacgcag	cagggcagtc	gccctaaaac	aaagttaaac	atcatgaggg	360
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SEQ ID NO: 97      moltype = DNA length = 888
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                   note = Description of Artificial Sequence: Synthetic
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source             1..888
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 97
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tatggcaaaa gggatgttct ttatcacaaat agcagtgagg atgacatctt catcatcatt 240
cgtatcatca ataccatcga gttctgaatc caaagggttc tgttgttggg ataaggcatc 300
agattctttg agcaggtcgt ctttggcgtc ctcctctaga tgaaatttgc catcgttagg 360
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source             1..953
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misc_feature       1..991
                   note = Description of Artificial Sequence: Synthetic
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source            1..991
                   mol_type = other DNA
                   organism = synthetic construct

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SEQ ID NO: 100     moltype = DNA length = 460
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source            1..460
                   mol_type = other DNA
                   organism = synthetic construct

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ctttgcttca acttatgtcc cactatgaac tttccgtcac ggttgtaaac ttgaacgtcg 360
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agtagcgtga tttgatattg tgcgataatt tgtggccagg 460

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source            1..492
                   mol_type = other DNA
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SEQ ID NO: 102     moltype = DNA length = 371
FEATURE           Location/Qualifiers

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                  organism = synthetic construct

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tatttgaaag tcgaactggt ttctctggac ttttagcttg atgatcgatt gatataattc 300
atccttggag ttatacagta gtggctttcc tcccaggtgg tatcgtaaaa cctgggaagg 360
atttggtcca a                                     371

SEQ ID NO: 103      moltype = DNA length = 1487
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                  note = Description of Artificial Sequence: Synthetic
                  polynucleotide
source              1..1487
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 103
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SEQ ID NO: 104      moltype = AA length = 189
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source              1..189
                  mol_type = protein
                  organism = synthetic construct

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VGRALMGLAT EFARERAGAGH LWLEVTNVNA PAIHAYRRMG FTLCGLDIAL YDGTASDGEQ 180
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SEQ ID NO: 105      moltype = DNA length = 207
FEATURE            Location/Qualifiers
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                  polynucleotide
source              1..207
                  mol_type = other DNA
                  organism = synthetic construct

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aataacgggt tattgtttat aaatactact attgccagca ttgctgctaa agaagaaggg 180

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VARIANT                 1..69
                        note = Description of Artificial Sequence: Synthetic
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source                  1..69
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 106
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VSLEKREAE                                     69

SEQ ID NO: 107          moltype = AA length = 8
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VARIANT                 1..8
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source                  1..8
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 107
HHHHHHHH                                     8

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1.-66. (canceled)

67. A *Pichia pastoris* microorganism, in which the activity of a SEC72 protein has been eliminated or in which a sec72 gene has been deleted, and wherein said microorganism expresses a recombinant silk-like protein comprising a secretion signal peptide derived from yeast.

68. The microorganism of claim **67**, wherein said SEC72 protein comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 2.

69. The microorganism of claim **67**, further comprising a recombinantly expressed SSH1 translocon complex.

70. The microorganism of claim **69**, wherein said SSH1 translocon complex comprises a first polypeptide sequence at least 95% identical to SEQ ID NO: 4, a second polypeptide sequence at least 95% identical to SEQ ID NO: 6, and a third polypeptide sequence at least 95% identical to SEQ ID NO: 8.

71. The microorganism of claim **69**, wherein said SSH1 translocon complex comprises a first polypeptide comprising SEQ ID NO: 4, a second polypeptide comprising SEQ ID NO: 6, and a third polypeptide comprising SEQ ID NO: 8.

72. The microorganism of claim **67**, further comprising a recombinantly expressed translocon complex.

73. The microorganism of claim **72**, wherein said translocon complex is expressed from a recombinant SSH1 gene, a recombinant SSS1 gene, and a recombinant SBH2 gene.

74. The microorganism of claim **73**, wherein said SSH1 gene comprises SEQ ID NO: 3.

75. The microorganism of claim **73**, wherein said SSH1 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 3, wherein said SSS1 gene comprises a polynucleotide sequence at least 95% identical to

SEQ ID NO: 5, or wherein said SBH2 gene comprises a polynucleotide sequence at least 95% identical to SEQ ID NO: 7.

76. The microorganism of claim **72**, wherein said translocon complex comprises an SSH1 protein comprising a polypeptide sequence at least 95% identical to SEQ ID NO: 4, an SSS1 protein comprising a polypeptide sequence at least 95% identical to SEQ ID NO: 6, and an SBH2 protein comprising a polypeptide sequence at least 95% identical to SEQ ID NO: 8.

77. The microorganism of claim **67**, wherein activity of a YPS1-1 protease and a YPS1-2 protease have been attenuated or eliminated.

78. The microorganism of claim **77**, wherein said YPS1-1 protease comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 10 or wherein said YPS1-2 protease comprises a polypeptide sequence at least 95% identical to SEQ ID NO: 12.

79. The microorganism of claim **67**, wherein said secretion signal peptide is selected from the group consisting of: a PEP4 signal sequence, a CPY+4 signal sequence, a DAP2 signal sequence, and a MFa1 signal sequence.

80. The microorganism of claim **67**, wherein said secretion signal peptide is selected from the group consisting of: SEQ ID NO: 84, SEQ ID NO: 86, SEQ ID NO: 88, and SEQ ID NO: 90.

81. A method of producing a recombinant protein, comprising:

culturing the microorganism of claim **67** in a culture medium under conditions suitable for expression of the recombinantly expressed protein; and
isolating the recombinant protein from the microorganism or the culture medium.

* * * * *